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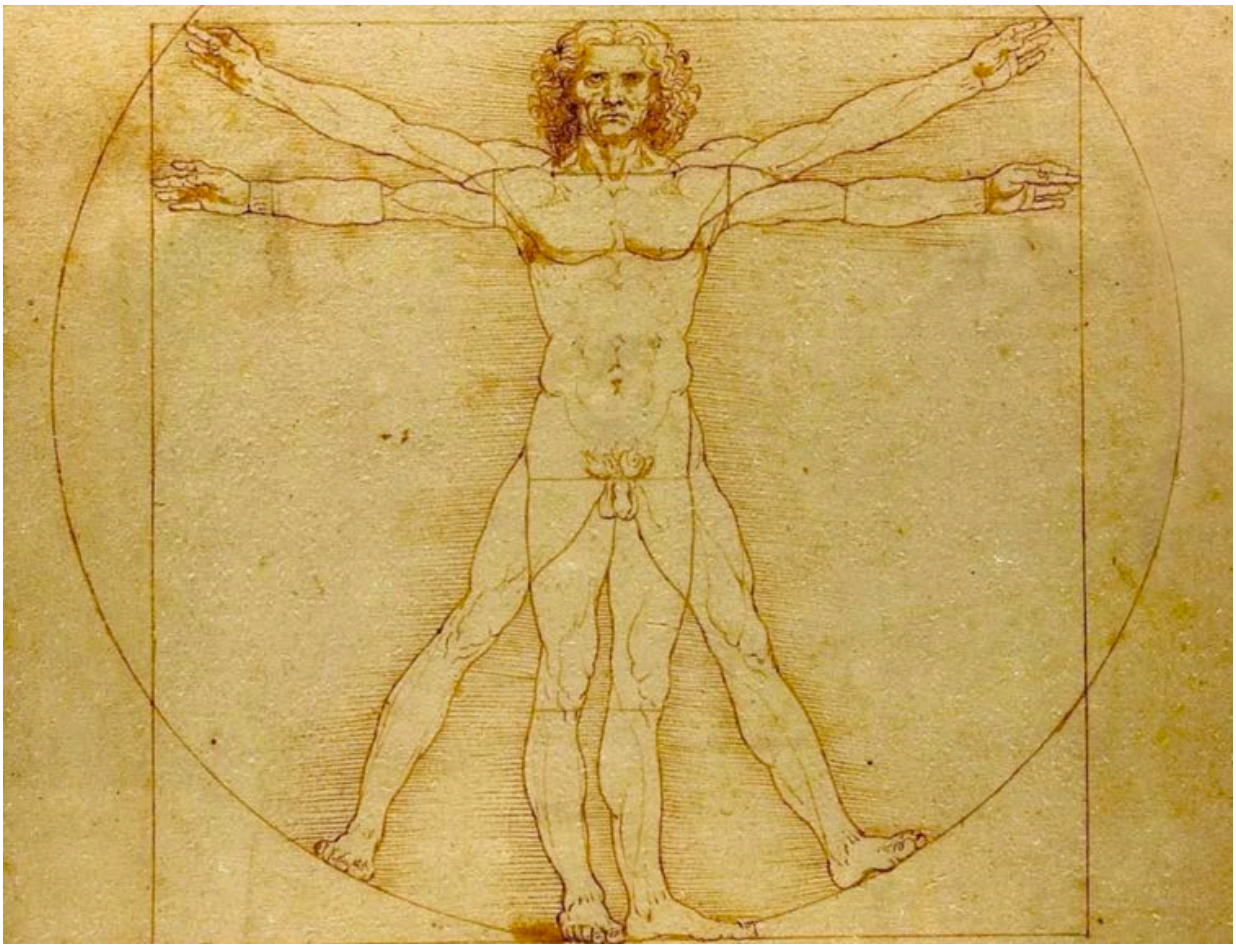

RHEUMNOTES© is a collection of clinical mind-maps designed to facilitate a fundamental understanding of Rheumatology and Clinical Immunology. Each individual **RHEUMNOTES©** page is comprised of a brief, concise evidenced-based review of a particular rheumatic disease process or condition encountered in clinical practice. **RHEUMNOTES©** has been successfully utilized in the continuing medical education programs at Walter Reed Army Medical Center and Brooke Army Medical Center in the instruction of medical students, interns, residents and staff physicians with academic applications in medical education, board review and clinical practice. This collection of **RHEUMNOTES©** is dedicated to the Department of Medicine Housestaff and their patients through whom we learn and teach the venerable art of medicine.

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F4. Fibromyalgia Syndrome
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A. ARTHRITIS

"Disease often reveals itself in a casual parenthesis."
- Dr. William Trotter

OUTLINE:

A1. Approach to the patient with musculoskeletal symptoms
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A13. Rubella Vaccine associated arthropathy

Initial Evaluation of adult patient with Musculoskeletal Symptoms

- History of significant trauma
  - Soft tissue injury, internal derangement, or fracture.
- Hot,swollen joint
  - Infection,systemic rheumatic disease,gout,pseudogout
- Constitutional signs&symptoms
  - Infection,sepsis,systemic rheumatic disease
- Weakness
  - Focal
    - Focal nerve lesion (compartment syndrome,entrapment neuropathy, mononeuritis multiplex,motor neuron disease,radiculopathy.
  - Diffuse
    - Myositis,metabolic myopathy,paraneoplastic syndrome, degenerative neuromuscular disorder, toxin myopathy, transverse myopathy
- Neurogenic pain
  - Asymmetric
    - Radiculopathy, RSD, entrapment neuropathy.
  - Symmetric
    - Myelopathy, peripheral neuropathy
- Claudication pain pattern
  - Peripheral vascular disease,GCA, lumbar spinal stenosis

Musculoskeletal Examination

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SCREENING EXAMINATION

⇩

FOCUSED EXAMINATION

- INSPECTION
- ACTIVE RANGE OF MOTION
- PASSIVE RANGE OF MOTION
- PALPATION
- SPECIAL MANEUVERS
- NEUROLOGICAL EXAM

	SYNOVITIS	BURSITIS	TENDINITIS	ENTHESITIS
INSPECTION	S++	S - / +	S + / Attitude	S --
ACTIVE ROM	T++ all planes / LROM	T+/- LROM (1 plane)	T+ LROM (1 plane)	T+ tendon insertion
PASSIVE ROM With stress:	T++ all planes / LROM	PROM >AROM T +/- (1 plane)	PROM >AROM T + tendon stretched	T + tendon stretched T+ when tendon stressed
Resisted Isometric movement	Ø pain	Ø pain	pain when tendon resisted	pain when tendon stressed
PALPATION	W++ T++ articular	T++ nonarticular	T++ along tendon	T+ along tendon insertion

S = swelling, W = warmth, T= tenderness, L = limited ROM

MUSCULOSKELETAL EXAM

DEFINITIONS

Synovitis: Inflammation of a synovial membrane associated with swelling, pain and limitation in ROM.

Tendinitis: Clinical & pathological disorder of a tendon associated with common features of local pain, dysfunction,inflammation and degeneration often from overuse.
- Triggering: unique disturbance associated with tendinitis m/b spontaneous digit locking requiring passive "snapping"release.

Bursitis: inflammation of the sac-like viscous fluid filled structures that form in utero to protect soft tissues from underlying bony prominences. Clinically, exquisite local tenderness at site of bursae.

Enthesitis: Inflammation of the muscular or tendinous attachment to bone.

ARTHRITIS

⇩

ACUTE
(< 6 weeks)

MONOARTICULAR
OLIGOARTICULAR (2-3 JOINTS)
POLYARTICULAR (4+ JOINTS)

CHRONIC
(> 6 weeks)

Nonarticular Rheumatic Pain

Inflammatory Rheumatic Diseases	Nonarticular Disorders
◆ Pain during movement	◆ Pain after rest
◆ Palpable joint swelling	◆ Swelling is sensational only
◆ Abnormal laboratory results	◆ All laboratory tests normal
◆ Aggravating factors possible	◆ Aggravating factors likely
◆ Diagnosis: Clinical exam & test results	◆ Diagnosis: Physical exam & outcomes

Sheon RP. Resident and Staff Physician 1989;35:65-70

SYNOVITIS

ACUTE (< 6 weeks)

CHRONIC (> 6 weeks)

MONOARTICULAR

MONOARTICULAR ARTHRITIS

INFLAMMATORY CONDITIONS

- Crystal Induced Arthritis
  - Monosodium urate (GOUT)
  - Calcium Pyrophosphate dihydrate (CPPD)
  - Apatite
  - Calcium oxalate
- Infectious Arthritis
  - Bacteria
  - Fungi
  - Lyme disease
  - Mycobacteria
  - Virus (IIIV,IIBV,etc)
- Systemic Diseases
  - Psoriatic arthritis
  - Reactive arthritis
  - Rheumatoid Arthritis
  - SLE

NONINFLAMMATORY CONDITIONS

- Amyloidosis
- Osteonecrosis
- Benign Tumor
  - Osteochondroma
  - Osteoid osteoma
  - Pigmented villonodular synovitis (PVS)
- Fracture
- Hemarthrosis
- Internal derangement
- Malignancy
- Osteoarthritis

SYNOVITIS

ACUTE (< 6 weeks)

CHRONIC (> 6 weeks)

POLYARTICULAR (4+JOINTS)

POLYARTICULAR ARTHRITIS

INFLAMMATORY CONDITIONS

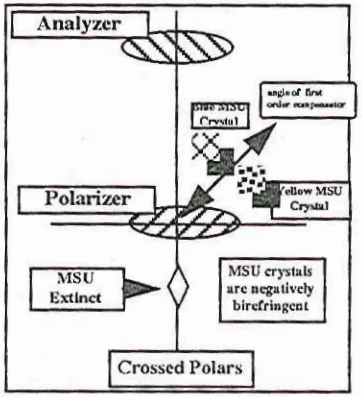
- Crystal Induced Arthritis
  - Monosodium urate (GOUT)
  - Calcium Pyrophosphate dihydrate (CPPD)
- Infectious Arthritis
  - Gonorrheal - Tuberculosis
  - Lyme disease - Spirochetal
  - Virus (HIV,HBV,etc)
  - SBE - Rheumatic Fever
- Systemic Diseases
  - Psoriatic arthritis - Inflammatory Bowel Disease
  - Reactive arthritis - RA
  - SLE - Malignancy
  - Systemic Sclerosis - PMR

NONINFLAMMATORY CONDITIONS

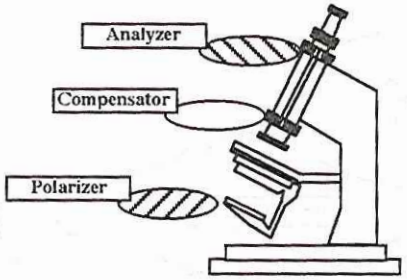
- Osteoarthritis
- Amyloidosis
- Osteonecrosis
- Benign Tumor
  - Osteochondroma
  - Osteoid osteoma
  - Pigmented villonodular synovitis (PVS)
- Malignancy

Gout

(Synovial Fluid)



Arthritis Rheum 1977;20:895-900



- Monosodium Urate Crystals (MSU)
  - Needle Shaped
  - Negatively birefringent
  - 5-10 u in length [PMN diameter]
- Acute Gouty Arthritis
  - Intracellular Phagocytized MSU crystals:
    - Sensitivity :84.4%
    - Specificity :100%
  - Extracellular MSU Crystals:
    - Common in intercritical gout

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Synovial Fluid Analysis

	Normal	GroupI (Noninflammatory)	GroupII (Inflammatory)	GroupIII (Purulent)
WBC/mm3	< 200	<2000	2500-50,000	50,000-300,000
Color	Clear	Xanthochromic	------>I	White
Clarity	transparent	transparent	translucent	opaque
Viscosity	High++	High	Low	Very low
Mucin clot	good	fair	fair	poor
Spontaneous clot	none	often	often	often

SYNOVIAL FLUID

DIFFERENTIAL DIAGNOSIS

Group I (Noninflammatory)

- Osteoarthritis
- SLE
- Mechanical derangement
- Neuropathic joint
- AVN
- Hemochromatosis
- Sickle Cell Disease
- Hypertrophic Osteoarthropathy
- Villonodular synovitis
- Wilson's Disease
- Acromegaly
- Hyperparathyroidism
- Paget's Disease
- Ochronosis
- Amyloidosis
- Sarcoidosis
- Erythema nodosum
- Osteochondritis
- Wilson's Disease
- Gaucher's Disease

Group II (Inflammatory)

- SLE
- Scleroderma
- Inflammatory Bowel Disease
- Vasculitis
- Villonodular synovitis
- Psoriatic Arthritis
- Juvenile chronic arthritis
- Polychondritis
- Whipple's Disease
- Hyperlipoproteinemias
- Behcet's syndrome
- Muticentric reticulohistiocytosis
- Erythema multiforme
- Carcinoma
- SBE
- Serum Sickness
- Agammaglobulinemia
- Leukemia
- Palindromic rheumatism
- Reactive Arthritis

Group III (Purulent)

- Bacterial Infections
- Tuberculosis
- Pseudosepsis

Group IV Hemarthrosis

- Trauma
- Coagulopathy
- Anticoagulation
- Tumors
- Pigmented Villonodular Synovitis
- Multicentric reticulohistiocytosis
- Vascular malformations
- Kasabach-Merritt Syndrome
- Chondrocalcinosis
- Neuropathic Joint

RHEUMATOID ARTHRITIS

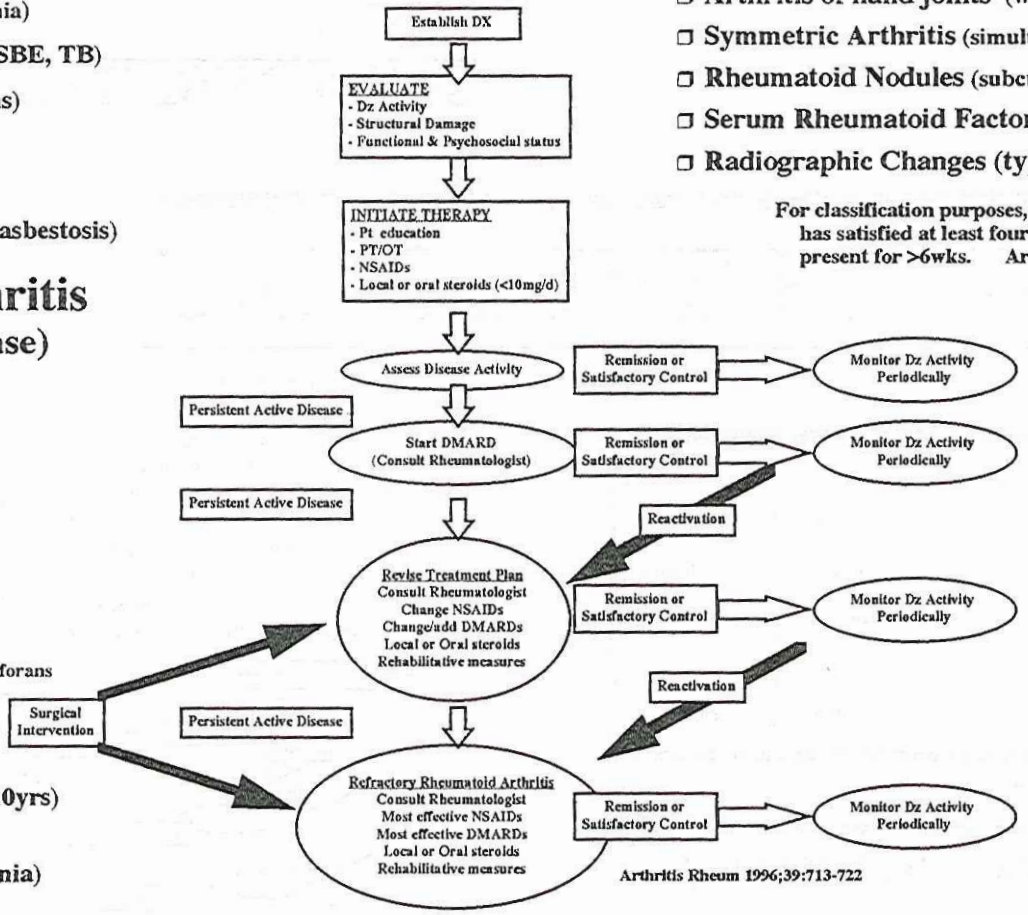
Rheumatoid Factor
(Differential Diagnosis)

- Normal people (1-4 %+, 10-25% age >70yrs)
- Systemic Autoimmune Disease
  (Sjogren's, cryoglobulinemia)
- Infections: (Acute viral, SBE, TB)
- Malignancy (Lymphomas)
- Chronic Liver Disease
- Pulmonary Disease
  (IPF, Sarcoid, Silicosis, asbestosis)

Rheumatoid Arthritis
(Extra-articular Disease)

- Skin:
  - Rheumatoid nodules
  - Cutaneous vasculitis
  - Ulcers
- Lung:
  - Nodules - BOOP
  - Pleurisy - Caplan's Syndrome
  - Fibrosis - Pulm HTN
- Ocular:
  - Xerophthalmia - Scleromalacia perforans
  - Episcleritis - Scleritis
- Neurological:
  - Peripheral neuropathy
- Vasculitis (@duration of dz > 10yrs)
- Felty's Syndrome:
  (RA+Splenomegaly+Leukopenia)
- Amyloidosis

Management



Arthritis Rheum 1996;39:713-722

Rheumatoid Arthritis
(Classification Criteria)

- ☐ Morning Stiffness (≥1hr)
- ☐ Arthritis of 3 or more joint areas simultaneously
- ☐ Arthritis of hand joints (wrist/MCP/PIP)
- ☐ Symmetric Arthritis (simultaneous involvement of both sides of body)
- ☐ Rheumatoid Nodules (subcutaneous nodules observed by physician)
- ☐ Serum Rheumatoid Factor
- ☐ Radiographic Changes (typical erosions, osteopenia)

For classification purposes, a pt shall be said to have rheumatoid arthritis if he/she has satisfied at least four of the these seven criteria. Criteria 1-4 must be present for >6wks. Arthritis Rheum 1988;31:315-324

DMARDS

- Hydroxychloroquine
- Sulfasalazine
- Methotrexate
- Gold (IM / PO)
- D-penicillamine
- Azathiaprine (Imuran)
- Cyclosporine A
- Cyclophosphamide
- Chlorambucil

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RHEUMATOID LUNG

Rheumatoid arthritis (RA) is a systemic inflammatory disease characterized by diarthroidial joint synovitis, synovial hyperplasia and eventual distruction if left untreated. The prevalence of RA in the general population ranges from 0.5% to 2% with woman older than 40yrs of age most commonly affected. The extra-articular manifestations of RA are protean and may be life threatening. A recent necropsy study in 1,246 patients with RA revealed that lung involvement was the second most common cause of death (18%) following infection (27%). *Semin Arthritis Rheum 1995;24: 242-54.*

I. Pulmonary Manifestations of Rheumatoid Arthritis.

- Pleural Involvement
  - Pleurisy
  - Pleural effusion
  - Pneumothorax
  - Empyema
- Rheumatoid nodulosis
- Interstitial Lung Disease
  - Interstitial fibrosis
  - BOOP
- Caplan's Syndrome
- Airway Involvement
  - Obstructive disease
  - Obliterative bronchiolitis
  - Bronchiectasis
- Pulmonary Vasculitis
- Drug-Induced Lung Diease
  - MTX
  - Gold
  - D-Penacillamine
  - SSZ

II. Pleural Disease

A. Pleural Effusion.
- Most common pulmonary manifestation. (20% symptomatic/ Frequency 40-75%)
- Usually occur in active disease and in patients with rheumatoid nodules

B. Characterisitics of Pleural Fluid in RA
- Aseptic Exudative effusion / high LDH
- pH: <7.2 — Cell type: Lymphocytes > PMNs
- Glucose level: Very Very Low: <20mg/dl. — Protein level: High > 4gm/l
- RF+ [nonspecific and may be seen in SLE/TB/CA]

III. Rheumatoid Nodules:

A. Central zone of fibrnoid necrosis, surrounding by layer of palisading mononuclear cells within an outer layer zone of vascular granlulation tissue, lymphocytes,plasma cells and fibroblasts.
- More common in Men > Women — May be single or multiple
- Usually asymptomatic. — Central cavitation @ 50%
- SPRA with rheumatoid nodules ++ — RML or BUL more common sites

B. Caplan's Syndrome (pulmonary nodulosis and pneumoconiosis in RA)
- Coal miners, foundry workers, roof tile makers, silica and asbestos exposure.

IV. Interstitial Lung Disease:

A. Pulmonary Fibrosis (1-5% of RA patients)

- Joint disease precedes lung involvement [>90%]
- More common in seropositive men (50 - 60yrs of age)
- clinical features very similar to Idiopathic Pulmonary Fibrosis (IPF)
- HRCT: Reticular, Reticulonodular or honeycomb pattern in lung bases
- PFTs: restrictive pattern [decreased lung volume,increased FEV1/VC ratio/decreased DLco]

B. BAL

- Both lymphocytic alveolitis and neutophilic inflammation +/- pulmonary fibrosis
-Lymphocytic alveolitis: predominant T-cells with increased CD4 + CD4:CD8 ratio.
-Neutrophilic alveolitis is associated with decreased CD4:CD8 ratio.

C. Risk Factors:

- RA independent risk factors: + Tobacco use & certain phenotypes of Anti-1antitrypsin.
- RA dependent risk factors: HLA-DRB1, HLA-B40, TNF

D. Indications for pulmonary Biopsy:

1. When clinical or radiographic manifestations persist despite appropriate therapy
2. When new findings appear in pt with chronic pulmonary disease
3. When the etiology of pulmonary nodule[s] are in doubt.

E. Nodulosis and BOOP have a better prognosis than Interstital pneumonitis

F. Pulmonary HTN may occur as a consequence of ILD

G. BOOP:

- Affects men and women equally usually preceded by febrile flu-like illness
- disease duration usually < 3 months---> Infection should be ruled out.
- BAL: lymphocytic alveolitis
- PFTs c/w restrictive pattern

V. MTX Pneumonitis:

- Uncommon but potentially life-threatening complication of MTX therapy
- The presence of ILD is not a contraindication to the use of MTX in RA.
- Heterogeneous clinical presentation

Acute or subacute onset of nonproductive cough,dyspnea, fever and rales.
CXR may be WNL or show diffuse ILD or acinar disease
Hypoxemia, eosinophilia and elevated LDH have been observed.
PFTs: WNL or Restrictive or Obstructive

- BAL:

cytology is nonspecific.
Lymphocytic interstitial infiltration and granuloma formation

VI. References:

Semin Arthritis Rheum 1995;24:242-254
Br J Rheumatol 1993;32:556-561
J Rheumatol 1996;23:933-6

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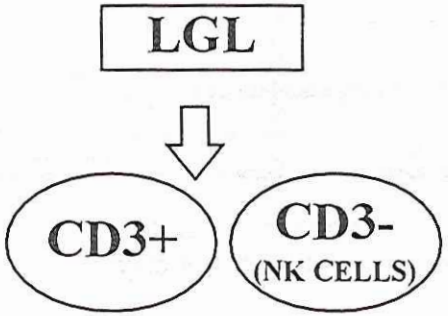
LARGE GRANULAR LYMPHOCYTE SYNDROME

FELTY'S SYNDROME

- Rheumatoid Arthritis
- Neutropenia
- Splenomegaly



Clinical Features of LGL patients

	OBSERVED	PERCENTAGE
Median age	57 (15-88)	
Male / Female	57/71	45/55
Recurrent Infections	50/129	39
Rheumatoid Arthritis	36/129	28

Blood 1993;82:1-14

Hematological Manifestations

	Observed	Percentage
Neutropenia (<2,000)	109/129	84
Anemia (Hct <36%)	48/97	49
Thrombocytopenia (<150,000)	13/69	19
Lymphocytosis (>4000)	93/126	74

Blood 1993;82:1-14

Serological profile

	Percentage
RF	57%
ANA	38%
Polyclonal hypergammaglobulinemia	45%
Hypogammaglobulinemia	6%
Antineutrophil AB	41%

DR4 Genotyping in LGL

	Felty's Syndrome	LGL/RA	LGL/øRA	Normal controls
n	22	10	12	191
DR4+	19(86%)	9(90%)	4(33%)	70
DR4-	3	1	8	121

Arthritis Rheum 1997;40:624

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GOUT

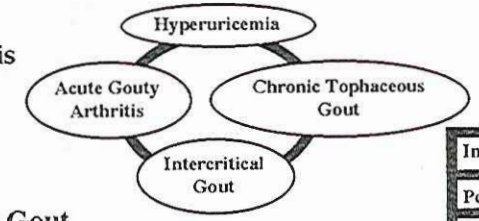
(Overview)

Epidemiology
(Gout)

Peak age (yrs)	Males: 40-50 Females: > 60
Gender	M 2-7: F1
Prevalence (per 1000)	Males 5-28; Females 1-6
Annual Incidence (per 1000)	Males 1-3; Females 0.2
Geography	Worldwide: Regional differences
Genetic Associations	Inherited Enz def / underexcretion
Environmental associations	Diet,Rx,toxins

Natural History
(Gout)

Asymptomatic Hyperuricemia
↓
Acute Gouty Arthritis
↓
Intercritical Gout
↓
Chronic Tophaceous Gout



Acute Gouty Arthritis
(Triad)

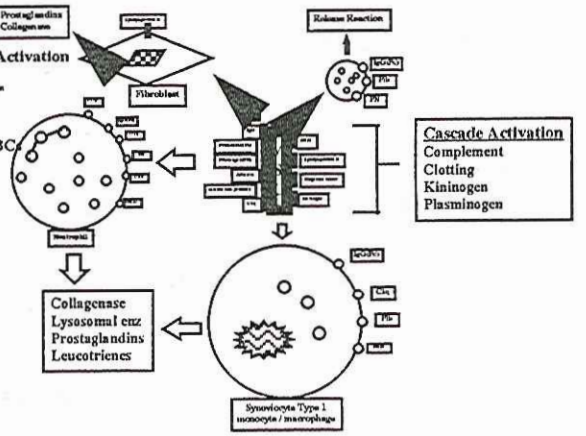
Criteria for Clinical Diagnosis

Acute monoarticular arthritis
Hyperuricemia
Dramatic response to colchicine

JAMA 1967;199:525
J Rheumatol 1988;15:495

Inflammatory Response
(Gout)

- Complement & Coagulation System Activation
  - Activation of classic complement pathway
  - Alternate complement pathway at higher MSU concentration
  - Activation of Hageman factor -> kallikrein,plasmin
- Platelet Activation
- Direct activation of Mononuclear WBC
  - Transcription & Elaboration of Cytokines
  - (IL-1, IL-6, IL-8, TNF, PLA2, PLAP)
- Neutrophil influx and activation
  - Superoxide
  - Activated phospholipase A2 & PLAP
  - Arachidonic acid, LTB4, IL-1, IL-8, PAF
- Synovial Mast Cell degranulation
  - C3a, C5a, IL-1, Histamine



Acute Gouty Arthritis
Associated Disorders

Obesity
HTN
Hyperlipidemia
ETOH

→ H.O.L.E.

Int J Obesity 1986;10:255-64
Ann Intern Med 1980;93:817-21

Incidence of gout in relation to serum urate concentrations

Incidence (per 1000)		Serum urate concentration (mg/dl)
Per year	5 yr	cummulative
0.8	5.0	<7.0 (0.42mmol/l)
0.9	6.0	7.0-7.9 (0.42-0.47mmol/l)
4.1	9.8	8.0-8.9(0.48-0.53mmol/l)
49	220	> 9.0 (0.54mmol/l)

Amer J Med 1987;82:421-6

PROVACTIVE FACTORS:

Medications
Trauma
ETOH
Ingestion
Surgery
Dietary excess
Hemorrhage
Venisection
Foreign protein therapy
Infections
Radiographic therapy

J Rheumatol 1985;12:376
Ann Intern Med 1985;136:582
Clin Exp Rheumatol 1985;3:105

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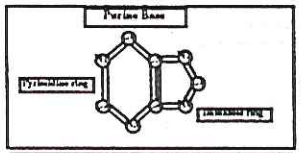
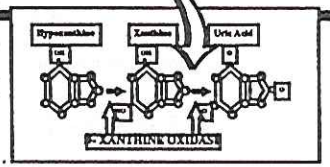
URIC ACID
(Metabolism)

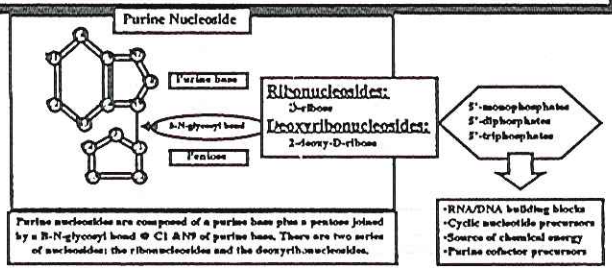
Purine Metabolism



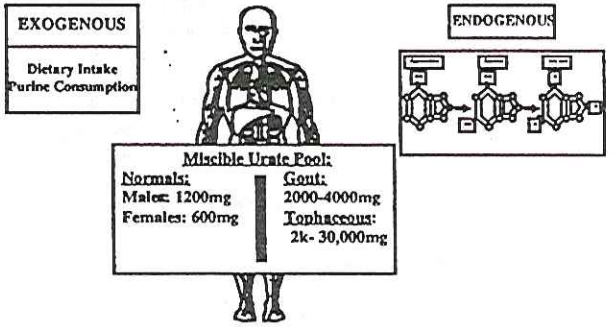
Free Purine Bases:
- Adenine
- Guanine
- Hypoxanthine
- Xanthine



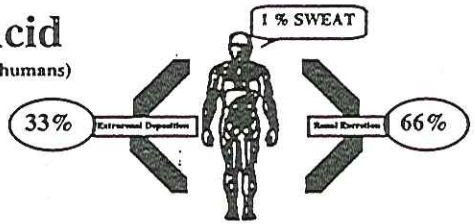
Purine Metabolism

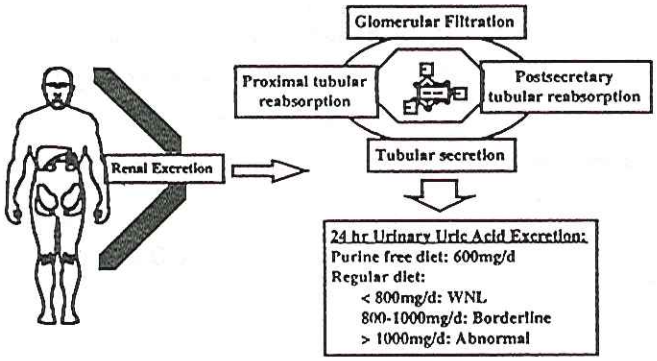


Uric Acid
(Humans)

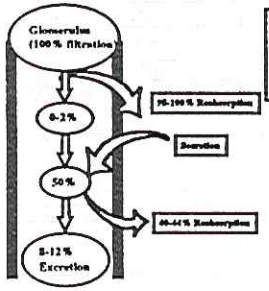


Uric Acid
(Elimination in humans)





UricAcid
(Excretion in the human kidney)



Four-compartment model illustrating bidirectional urate transport by the kidney. Tubular reabsorption and secretion are shown as a percentage of filtered urate.

- [Urate] glomerular fluid = [Urate] p
- Protein bound [Urate] p < 5%
- Proximal Tubule : 90-95%
- Mechanism: Active transport
- Fractional urate excretion: (FE urate)
  - [C urate / C creatinine x 100]

Uric Acid
(Factors controlling renal clearance)

- Urine Flow
  - Increase GFR = Increase Uric acid clearance
- Estrogens:
  - Women/children -> lower sUA
  - Exogenous estrogens -> Increase Urate clearance
- Surgery:
  - Abdominal (anaesthesia/IV fluids/vagotomy)
- Autonomic Nervous System
  - Anticholinergic agents -> Increased Urate clearance

Treatment
(Goals)

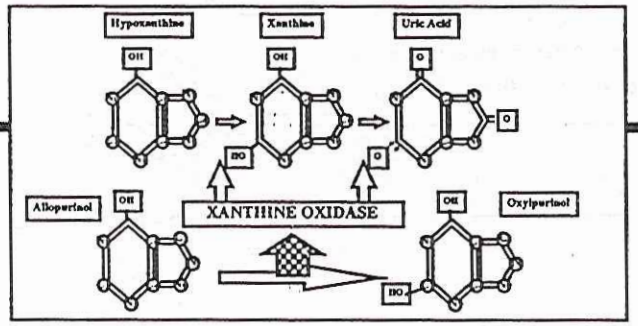
(1) Terminate acute attack promptly & gently

(2) Prevent recurrance of Acute Gout

(3) Prevent or reverse complications
- MSU / Uric acid deposition in Joints / Kidneys etc.

(4) Prevent or reverse associated features:
- Obesity
- HTN
- Hyperlipidemia

Allopurinol
(Mechanism of action)



Indications for Allopurinol

Hyperuricemia associated with increased uric acid production
- Urinary uric acid excretion ≥ 1gm / 24hrs
- Hyperuricemia associated with HPRT deficiency
- Hyperuricemia associated with PP-ribose-P synthetase overactivity
- Uric Acid nephropathy
- Nephrolithiasis
- Prophylaxis before cytolytic therapy

Intolerance or reduced efficacy of uricosuric agents
- Gout with renal insufficiency (GER < 60mg/min)
- Allergy to uricosurics

Treatment
(Hyperuricemia & Gout)

Allopurinol
(Dosing schedule)

GFR (ml/min)	DOSE (mg/d)
100	300
80	250
60	200
40	150
20	100
10	100 q2d
0	100 q3d

Treatment
Benemid (Probenecid)

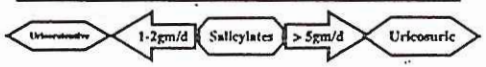
- Indications:
  - Underexcretor (24hr urine uric acid < 700mg)
  - GFR > 30-50 cc/min
  - Ø Acute Gout Ø Nephrolithiasis
- Dosage:
  - 250mg BID (x4d) -> 500mg BID (x2wks) -> Maintenance dose
  - Avg maintenance dose = 1000mg/d
- Drug Interactions:

Salicylates cancel benemid's uricosuric affect

Treatment
Anturane (Sulfinpyrazone)

- Indications:
  - Underexcretor (24hr urine uric acid < 700mg)
  - GFR > 30-50 cc/min
  - Ø Acute gout Ø Nephrolithiasis
- Dosage:
  - 50mg BID (x4d) -> 100mg BID (x1wk) -> Maintenance dose
  - Avg maintenance dose = 400mg/d
  - Maximun dose = 800mg/d
  - Potency 3-6x > probenecid
- Drug Interactions:

Salicylates cancel anturane's uricosuric affect



ADULT ONSET STILL'S DISEASE

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Adult Onset Still's Disease [ASD] is a systemic inflammatory disease of unknown etiology. First described in a case series of 14 patients by Bywaters in 1971 it has now become a recognized autoimmune disorder manifested by high spiking fevers, a salmon-colored evanescent rash, lymphadenopathy and arthritis. It occurs more commonly in females in the second and third decades of life. There is often a clinical history of antecedent culture negative pharyngitis and laboratory parameters usually reveal neutrophilic leukocytosis and the absence of both rheumatoid factor and antinuclear antibodies.

I. Clinical Presentation: [J Rheumatol 1987;14:1139-1146]

TABLE: 1

Clinical Manifestation	Frequency [%]
Female	53
Childhood episode	20*
Onset 16-35yrs	75
Arthralgias	100
Arthritis	94
Fever [>39 degrees C]	96
Weight Loss [> 10%]	36*
JRA rash	89
Sore Throat	56*
Myalgias	84
Lymphadenopathy	60*
Hepatomegaly	41
Splenomegaly	51
Pleuritis	23*
Pericarditis	27
Pneumonitis	12*
Abdominal Pain	12*

* denotes $p<0.05$

II. Diagnosis:

The diagnosis of ASD is based solely on the clinical findings as serological markers and other diagnostic tests do not exist.

A. DDX:

1. SBE
2. Meningococcemia
3. Gonococcemia
4. Rubella
5. HBV
6. HIV
7. Parvovirus B19
8. Syphilis
9. Lyme Disease
10. Toxic shock Syndrome
11. Reactive Arthritis
12. Serum Sickness
13. E. Multiforme
14. SLE
15. Vasculitis
16. Myeloproliferative Dz
17. Lymphoproliferataive

B. Definite Diagnosis of ASD

1. High spiking Fever
2. Evanescent Rash
3. Leukocytosis with neutrophilia.
4. Arthritis > 6wks

C. Probable ASD

1. Fever
2. Arthritis
3. Rash
4. Neutrophilic Leukocytosis
5. At least one other criteria [see table above]

III Treatment:

1. NSAIDs
2. Glucocorticoids
3. Inununosuppressive Medications
   - Hydroxychloroquine
   - Cyclophosphamide
   - Azathiaprine
   - LD MTX

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PSEUDOSEPTIC ARTHRITIS

Pseudoseptic arthritis is an acute arthritis, superimposed on a chronic inflammatory polyarticular arthritis, associated with fever, chills and sterile synovial fluid leukocytosis. The common systemic features and predilection to large joints may make the clinical presentation indiscernable from septic arthritis.

I. Clinical Characteristics described in the medical literature:

A. Pre-existing inflammatory polyarticular arthritis
- RA [most common]
- Gout
- Pseudogout
- Reiter's Syndrome
- Psoriatic Arthritis
- Sickle cell disease
- JRA

B. Fever / Chills
C. Large joint involvement [Mono or polyarticular]
D. Synovial Fluid Leukocyte counts > 50,000/mm3
E. Sterile synovial fluid
F. DMARD discontinuation a/o Steroid Taper

II. Clinical characteristics in Patients with Rheumatoid Arthritis.

- *J Rheumatol 1991;18:1319-22*

Table 1. *Demographic characteristics of 6 patients with RA with pseudoseptic arthritis*

Patient	Age/Sex	Disease Duration (years)	RF titer	ANA	Radiographic Erosions	Prednisone (mg/day)	Functional Class
1	63/F	5	—	—	+	5	III
2	59/F	4	1:5120	—	+	10	II
3	53/F	5	1:5120	+	+	—	II
4	39/F	10	1:5120	—	+	3	II
5	70/M	12	1:320	—	+	7	II
6	69/F	44	1:5120	+	+	10	III

Table 2. *Clinical characteristics of 6 patients with RA with pseudoseptic arthritis**

Patient	Temp °C	PB WBC/mm³	Affected Joints	SF WBC/mm³	% PMN	ESR mm/h	CRP mg/dl (nl < 0.8)	Antibiotic Therapy (days)	Comments
1	38.8	10.200	L Wrist	PUS**	ND	108	15	31	
2	37.7	8.000	R Knee	180.000	87	95	17	5	GSTM stopped 3 months before
			L Knee	258.000	93				
			L Ankle	54.000	89				
3	37.7	5.000	R Elbow	204.000	93	132	17	3	DP stopped 3 months before
			L Elbow	158.000	93				
4	37.7	13.500	R Elbow	78.000	86	78	ND	4	Poor control with DP
5	39.4	11.000	L Hip	PUS**	ND	111	36	5	Poor control with AZA
			R Hip	PUS**	ND				
			L Ankle	PUS**	ND				
			R Ankle	PUS**	ND				
			R Elbow	121.000	96				
6	40.0	9.500	R Elbow	267.000	93	65	5.8	5	Active disease
			R Knee	200.000	93				
			L Knee	78.000	87				

II. Clinical characteristics in Patients with Rheumatoid Arthritis.

- *West J Med 1985;143:471-473*

TABLE 1.—*Selected Clinical and Laboratory Variables in Patients With Rheumatoid Arthritis Who Were Admitted With the Diagnosis of Bacterial Infection of Joints*

Patient	Age yr	Sex	Episode	Admission Temperature °C	Peripheral Leukocyte Count per μl	ESR mm/hr	Rheumatoid Factor Titer	Synovial Fluid Findings: Nucleated Cell Count per μl	Synovial Fluid Findings: % PMNs	Synovial Fluid Findings: Glucose mg/dl
1	29	♂	1	38.3	11,400	53	1:320	35,000	89	10
			2	38.5	11,100	39		50,000	91	6
2	65	♂	3	37.0	19,300	ND	1:320	395,000	94	13
			4	38.4	11,600	ND		96,400	73	60
			5	38.5	20,000	70		60,000	90	< 10
			6	35.6	18,000	123		113,000	88	ND
3	37	♂	7	37.7	13,000	33	1:320	98,000	88	6
			8	36.7	8,600	ND		129,000	90	ND
4	64	♂	9	37.3	24,700	ND	1:1280	89,000	97	ND
5	69	♀	10	37.3	7,900	56	1:40	52,000	92	ND

ESR = erythrocyte sedimentation rate, ND = no data, PMNs = polymorphonuclear leukocytes

III. References

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2. Singleton JD, West SG. et al. Pseudoseptic Arthritis complicating rheumatoid arthritis: a report of six cases. J Rheumatol 1991;18:1319-22.
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COMMON CUTANEOUS MANIFESTATIONS OF RHEUMATIC DISEASE

I. Juvenile Rheumatoid Arthritis

A. Evanescent Rash:
- Erythematous and macular or papular
- Distribution: Trunks, neck and extremities
- Duration: transient, periods of fever.

B. Koebner's Phenonenon
- Provacation of JRA evanescent rash by making scratch marks at susceptible sites resulting in linear chains of maculopapules.
- Nonspecific for JRA

II. Spondyloarthopathies:

A. Ankylosing Spondylitis:

1. Iridocyclitis +/- synechiae:
   - Ciliary injection and irregularity of the pupil. Adhesions b/w iris and lens [synechiae] appear at the 1:00 & 4:00 positions. Synechiae are a result of repeated episodes of iritis and may lead to glaucoma and blindess.

B. Psoriatic Arthritis

1. Psoriasis :
   Type: papules and plaques, sharply marginated with silvery-white scale
   Distribution: single or localized lesion, elbows,knees,face, scalp
   Duration:months to years.
2. Nail dystrophy & nail pitting:
   - punctate pitting of the nail is common
   - onycholysis: uplifting of distal portions of nails.

C. Reiter's Syndrome / Reactive Arthritis

1. Balantis circinata
   -painless erythema->well circumscribed lesion->dry,scaly,crusted
2. Muco-cutaneous ulcerations:
   - hard palate more commonly affected
3. Ocular:
   -Conjuctivitis
   -Iritis
   - Episcleritis
4. Nail dystrophy
5. Keratoderma blennorrhagica:
   -Discrete, circinate, scally and plaque-like lesions on feet

III. Dermatomyositis

1. Periorbitial edema
2. Heliotrope discoloration of eyelids
3. Gottron's Nodes/papules: flat-topped, violaceous papules

IV. Systemic Sclerosis & Scleroderma like conditions: (Appendix A)

A. Edema --> Thickening --> Atrophy / hypo-hyper-pimentation
B. Acrosclerosis (Sclerodacytyly)
C. Calcinosis cutis
D. Mauskopf : Facial changes
- drawn pursed lips,shinny skin and atrophy of muscles
E. Morphea
F. Linear scleroderma
G. Eosinophilic fasciitis:
- "orange peel" appearance and "Groove Sign"

V. Vasculitis:

A. Cutaneous Vasculitis (Appendix B)
B. Henoch-Schonlein Syndrome
- Palpable purpura usually located on lower extremities.
C. Polyarteritis nodosa
- subcutaneous nodules --> necrotizing arteritis
D. Giant Cell Arteritis
-scalp necrosis
E. Wegener's Granulomatosis
- saddle nose
F. Behcet's Disease
- Mucocutaneous uclerations
- Hypopyon
G. Kawasaki Disease
- Polymorphous macular exanthem
- Muco-cutaneous uclerations
- Skin desquamation: palms,soles, extremities

VI. Relapsing polychondritis

A. Otological Changes:
-Pinna thickening and collapse secondary to progressive cartilage destruction.
B. Ocular manifestations:
- episcleritis, scleritis, conjunctivitis, iritis and scleromalacia

VII: Infectious Arthritis

A. Rheumatic Fever: Erythema marginatum
- circinate erupton extends centrifugally someimes leaving residual hyperpigmentation. Individual lesions appear principally as open or closed rings with sharp outer edges.
B. Lyme Disease : Erythema chronicum migrans

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HUMAN PARVOVIRUS B19 ARTHROPATHY

Parvoviruses are small, non-enveloped, single stranded, icosahedral DNA vertebrate viruses capable of automonous replication. These viruses have been associated with serious disease in animals. Parvoviruses preferentially select rapidly turning over cells, such as activated lymphoid cells and gut epithelium for replication. As of yet, only human parvovirus B19 [HPV B19] has been firmly associated with human disease. The most marked manifestation of HPV B19 in adults is an acute, usually self limited, arthropathy that is both more frequent and more persistent in women.

I. Human Disease associated with Parvovirus B19.
   a. Erythema infectiosum [Fifth Disease]
   b. Aplastic Crisis associated with chronic hemolytic anemia.
   c. Hydrops Fetalis
   d. Arthropathy.

II. Clinical Manifestations:

Joint Involvment

Pain [%]		Swelling [%]
8	head	0
31	neck	0
42	shoulders	4
8	chest	0
50	elbows	12
58	wrists	19
69	hands	58
8	hips	0
75	knees	19
69	ankles	35

Clinical Features of HPV B19 Infection in Adults:

	IgM & IgG		IgG alone		Neither IgM/IgG	
	M	F	M	F	M	F
	[20]	[34]	[156]	[162]	[37]	[44]
Symptoms [%]						
Any	70	76	23	34	30	34
Influenza	50	65	17	23	19	30
Joint	30	59	9	8	11	5
Rash	35	50	2	8	5	9
Joint&Rash	15	47	0.6	3	3	0

Arch Intern Med 1989;149:1153-1156

III. DIAGNOSIS:

A. Antibody Capture Enzyme Linked Immunosorbent assay [ELISA]

1. IgM Antibody to HPV B19: represents acute infection.
   - present early in illness begins to decrease in 1-2months, absent 2-3months.
2. IgG Antibody to HPV B19: represents prior exposure.
   - Nondiagnostic: common in general population.

B. Electron Microscopy

C. Polymerase Chain reaction amplification of B19 target DNA sequences.

IV. THERAPY:

A. Majority of individuals experience a self limited illness lasting < 2 months. . The arthropathy may be persistent or recurrent for greater than two months in 20 percent of patients.

B. Symptomatic therapy:

1. NSAIDs
2. Analgesics
3. Corticosteroids

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INFECTIOUS ARTHRITIS

I.Pathogenesis

1. Route of entry:
   a. Hematogenous: most common
   b. Dissemenation from osteomylitis
   c. Spread from adjacent soft tissue infection
   d. Diagnostic or therapeutic measures
   e. Penetrating damage via trauma
2. Synovial anatomical modifications:
   a. microvascular dilatation
   b. subsynovial edema
   c. synovial fluid volume increases ---> Increased intra-articular pressure resulting in microvascular thrombosis,ischemia, microabcess formation
   d. Immunoglobulin/complement concentrations approach those in plasma
   e. Immune complex deposition, metabolic end products and joint destruction

II. Clinical Manifestations

1. Monoarticular: 80%: Knee > Hip > Ankle > Shouder > Wrist
2. Local signs of inflammation: Pain = cardinal manifestation. S++ W++T++L++
3. Constitutional symptoms: Fever / chills / malaise
4. Associated Systemic Diseases:
   - DM - Cirrhosis -CRI
   - RA -CA -Age [>60]
   -Immunosuppression

III. Infectious Etiologies:

1. Bacterial Infection: most common*
   - Staphlococcus aureus* - Streptococcus pneumoniae/ pyogenes
   - Staph epidermidis - Bacteroides fragilis
   -Enterococcus faecium - Enterococcus faecalis
   - Eschericia coli - Haemophilus influenza
   - Klebsiella pneumoniae - Myobacterium tuberculosis
   - Neisseria gonorrrhoeae - Proteus mirabilis
   - Salmonella species - Pseudomonas aeruginosa
2. Mycoplasmatales: Associated with hypogammablobulinemia
   - Mycoplasma - Ureoplasma
3. Fungal Infection:
   -Coccidiodes immitis - Blastomycosis dermatitidis
   -Cryptococcus neoformans - Candida species
   - Sporotrichosis - Aspergillosis species
   - Histoplasmosis - Scedospoiosis
4. Spirochetes:
   - Borrelia burgdorferi
5. Viral
   - HBV - ParvoB19 -Rubella
   - Alphaviruses -Mumps -Adenovirus
   - EBV -CMV - HSV

IV. Diagnosis:

1. Arthrocentesis: Single most important procedure--> ASAP
   - A. Gram stain & Culture
   - B. Fluid analysis: Yellow/Turbid, Purulent/ reduced viscosity / +/- Crystals
     ** The presence of crystals does not exclude infectious process**
   - C. Cell Count: WBC # 10,000 - > 100,000 [mm3] :
     - 50% of CX+ cell counts may be < 30,000 WBC mm3
     - PMN predominance [>80%]
2. Laboratory Studies:
   - A. CBC with differential, ESR, CRP
   - B. Cultures : All appropriate sites: [blood/skin/urethra/cervix/ throat etc.]
     Blood CX may be positive in @50% with nongonococcal bacterial arthritis.
3. Imaging:
   - A. Radiographs: Little diagnostic value in early infection.
   - B. Scintigraphy: Helpful in SI/sternoclavicular/facet joint involvement.
   - C. CT Scan: SI / sternoclavicular joint or abscesses
   - D. MRI: HIP & deep seated bone joint infections.

V. Treatment:

1. Antimicrobials:
   - A. Parenteral administration: 2 wks followed by 1-2 wks PO.
   - B. Bacterial Arthritis:
     - Staph aureus: Nafcillin [cefazolin / Vancomycin / Clindamycin]
     - MRSA: Vancomycin
     - Streptococcus: Penicillin G [Cefazolin / Vacomycin / Clindamycin]
     - Enterococcus : Penicillin or ampicillin + Gentamycin [Vanc + aminoglycoside]
     - Neisseria gonorrhea: Ceftriaxone 1gm IM/IV QD
     - Hemophilus influenza: Ampicillin
     - Enterobacteriaceae: Ceftriaxone [Imipenum]
     - Pseudomonas: Antipseudomonal PCN + aminoglycoside
     - Bacteroides fragilis: Metronidazole
2. Drainage: Close-needle aspiration Q daily until sterile.
   - Surgical drainage: hip
   - Joint lavage: poor surgical candidates
3. Joint immobilization and Mobilization: Passive & Active ROM daily

VI. Prognosis:

- Factors: Duration of infection/ Comorbidities / organism virulence / affected joint/ age
- Duration of infection less than 7 days bodes more favorable prognosis
- Mortality : 5-15% Disability: 25-60%
- Polyarticular septic arthritis Mortality = 23% / RA = 56%

VII. References:

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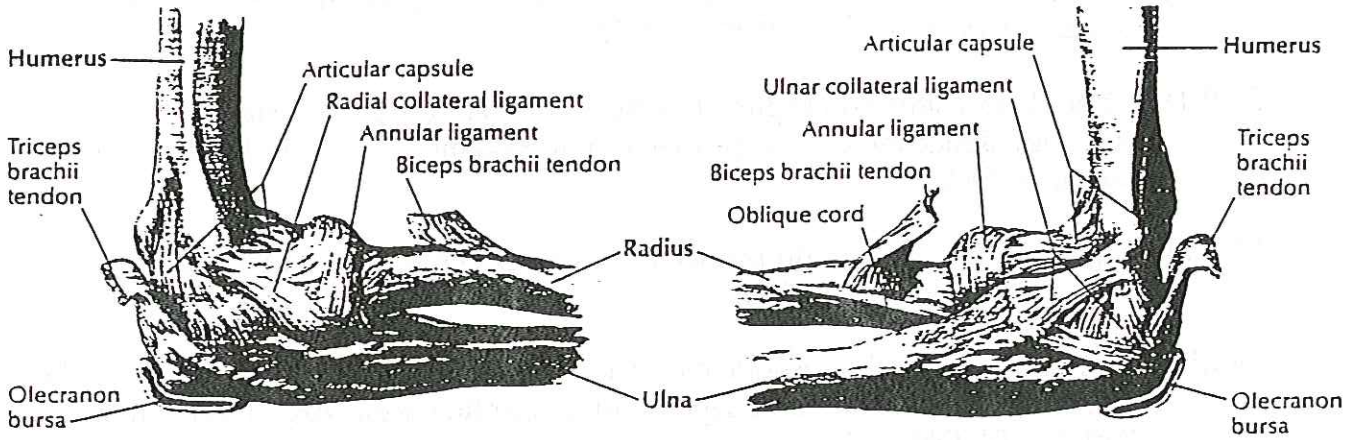
SEPTIC OLECRANON BURSITIS

The olecranon bursa is a subcutaneous synovial pouch located on the extensor aspect of the elbow. It serves to reduce friction between tissues overlying the olecranon process of the ulna. This area is vulnerable to trauma and infection through puncture wounds or cellulitis.

I. Anatomy



In 90° flexion: lateral view

In 90° flexion: medial view

II. Classification of Olecranon Bursitis

DIFFERENTIAL DIAGNOSIS OF OLECRANON BURSITIS

	Traumatic (idiopathic) noninflammatory	Aseptic inflammatory			Septic inflammatory
		Rheumatoid arthritis	Crystalline	Uremia	
Clinical symptoms					
Trauma	+++	+	+	0/+	+++
Tenderness	++	+	++	+	+++
Skin breakage	++	0	0	0	+++
Clinical signs					
Fever	0	0	0	0	++
Warmth	+	+	++	+	+++
Erythema	+	+	++	+	+++
Cellulitis	0	0	+	+	+++
Lymphadenopathy	0	0/+	0/+	0	++
Edema	0	0	0	0	++
Bursal fluid					
Color	C, S, H	C, S	S, H	S	S, H, P
White blood cells per mm^3	50–10,000 (mean 1,100)	1000–60,000 (mean 3000)	1000–50,000 (mean 2900)	1200	350–450,000 (mean 75,000)
Gram stain	Negative	Negative	Negative	Negative	Positive (70%)
Crystals	Negative	Rare cholesterol	Monosodium urate or pyrophosphate	Rare	Negative
Culture	Negative	Negative	Negative	Negative	Positive

Differential diagnosis: 0 usually absent; + rare; ++ occasional; +++ common

Bursal fluid color: C Clear; H Hemorrhagic; S Serosanguineous; P Purulent

SEPTIC OLECRANON BURSITIS

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RUBELLA VACCINE - ASSOCIATED ARTHROPATHY

Acute arthralgia and arthritis after rubella vaccination or infection has been recognized for many years. The onset of joint complaints usually occurs within a 2-to 6-week period after administration of rubella vaccine in susceptible individuals and within one week after the onset of rash in wild rubella infection. The most common clinical manifestations include arthralgias, myalgias, arthritis and parathesias. The majority of individuals who develop acute rubella-associated arthritis undergo a transient, self-limited clinical course. [Arch Intern Med 1993;153:2268-74]

I. Rubella Vaccine

A. Rubella Virus:
- TOGAVIRIDAE
- Spherical shapped 50-70nM [RNA]

B. VACCINE:
1. Cendehill Strain [rabbit kidney]
2. HP 77 Strain [Duck embryo culture]
3. **RA 27/3 Strain**
   - Grown in WI38 cells derived from human embryonic lung
   - Most immunogenic
   - Only strain currently licensed for use in United States

II. Rheumatic Manifestations:

A. Demographics:
- Females affected more than males
- Children [1-5%]
- 10 - 58% of rubella seronegative adults.

B. Symptoms: [Onset within 2 weeks/ duration 1 - 5 days]
1. Arthralgias = most common
2. Myalgias
3. Parathesias
   - Carpal Tunnel syndrome
   - Brachial Radiuloneuritis
   - no reported cases of permanent neurological deficits.

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RUBELLA VACCINE - ASSOCIATED ARTHROPATHY

C. Distribution of Joint Involvment:

1. Knee
2. PIP
3. MCP
4. Ankle
5. Wrist

III. DIAGNOSIS:

1. Diagnosis of exclusion
2. Serological Testing:
    a. RV IgG antibodies [ELIZA]
    b. RV Immunoblotting
    c. RV neutralization antibody titres
    d. RV detection in PBMNC [PCR]

IV. PROGNOSIS:

- Usually self limited illness.
- A small and as yet poorly defined proportion have been observed to progress to the development of persistent musculoskeletal symptoms lasting months to years.

References:
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B. BONE

"The whole of science is nothing more than a refinement of everyday thinking."
-Albert Einstein

OUTLINE:

B1. Osteonecrosis (Avascular necrosis)
B2. Hypertrophic Osteoarthropathy (HOA)
B3. Tuberculous Spondylitis (Pott's Disease)
B4. Osseous Sarcoidosis
B5. Diffuse Idiopathic Skeletal Hyperostosis (DISH)
B6. Metabolic Bone Disease

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OSTEONECROSIS

Osteonecrosis, commonly referred to as avascular necrosis, represents the in situ death of a segment of bone. There are two major forms of osteonecrosis. The first is infarction of medullary bone which is usually silent and affects only the marrow cavity and its trabecular architecture. In the second form, both the medullary bone and the cortex of the surrounding shaft are involved. This type frequently occurs in the immediate subchondral regions of the proximal femur, distal femur and proximal humerus. The most commonly affected sites are the the proximal and distal femoral heads, leading to hip or knee pain. Many patients with hip disease actually first present with referred knee pain.

I. Pathogenesis:

Interruption of the blood supply and resultant necrosis of the marrow, medullary bone or cortex may occur by one of at least four mechanisms.

A. Mechanical vascular interruption
   - Fracture, dislocation, stress fractures

B. Thrombosis and embolism:
   - Cholesterol emboli, Nitrogen bubbles, sickel cell crisis.

C. Injury to or pressure on a vessel wall
   - vasculitis, radiation injury, Gaucher's disease

D. Venous occlusion [Chandler's disease]
   - venous pressure exceeds arterial pressure in a closed system

II. Classification of Osteonecrosis:

Traumatic Osteonecrosis
Septic Osteonecrosis

Nontraumatic Osteonecrosis
- Stress or fatigue fractures
- ETOH abuse
- Dysbarism
- Gaucher's Disease
- Rheumatoid Arthritis
- Systemic Lupus Erythematosus
- Vasculitis
- Hemoglobinopathy
- Coagulopathy
- Radiation injury
- Corticosteroid administration
- Pregnancy
- Gout
- Pancreatitis
- Idiopathic

N Engl J Med 1992;326:1473-79

III. Diagnosis:

- History and physical examination reveals pain, localized tenderness, limited ROM.
- Radiographs: "crescent sign"
- Bone scan:
  - Early: photon deficiency at site ["doughnut sign"]
  - Later: uniform high level of activity
- MRI: gold standard
  - Reduced signal on both T1 and T2 weighted images.

IV. RADIOLOGICAL STAGING

Stage	Criteria
0	Normal X-ray, normal bone scan
I	Normal X-ray, abnormal bone scan
II	Sclerosis and or cyst formation in Femoral Head. a. mild [<20%] b. moderate [20-40%] c. severe [>40%]
III	Subchondral Collapse [crescent sign] w/o flattening a. mild [<15%] b. moderate [15-30%] c. severe [>30%]
IV	Flattening of head without joint narrowing or acetabular involvement. a. mild [<15% of surface &<2mm depression] b. moderate [15-30% of surface or 2-4mm depression] c. severe [>30% of surface or >4mm depression]
V	Flattening of head with joint narrowing and or acetabular involvement. a. mild* b. moderate* c. severe * * determined as above + acetabular involvement
VI	Advanced degenerative changes

V. Treatment:

Nonsurgical management

Non-weight bearing activity for several weeks using crutches and exercises.

Prophylactic surgery

- Core Decompression
- Bone grafting
- Osteotomy
- Electric stimulation
- Capacitance coupling

Arthroplasty

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HYPERTROPHIC OSTEOARTHROPATHY

Hypertrophic osteoarthropathy [HOA] is a syndrome characterized by [1] chronic proliferative periostitis of long bones; [2] clubbing of fingers a/o toes [3] oligosynovitis or polysynovitis. As a minimum, the diagnosis of HOA is based on the presence of digital clubbing and long bone periostitis. It is estimated that between 1 & 12 % of patients with bronchogenic CA develop HOA.

I. Classification of Hypertrophic Osteoarthropathy

I. Primary
- A. Pachydermoperiostosis
- B. Idiopathic
- C. Clubbing [familial, idiopathic]

II. Secondary
- A. Pulmonary
  1. Neoplasms [bronchogenic CA]
  2. Infections
  3. Other [cystic fibrosis, pneumoconiosis]
- B. Cardiovascular
  1. SBE
  2. Cyanotic congenital heart disease
  3. PDA
  4. Other [aneurysm, aortic prosthesis]
- C. Gastrointestinal
  1. Neoplasms [Esophageal CA, Liver CA]
  2. Infections [amebic dysentery]
  3. Hepatobiliary [biliary cirrhosis,cirrhosis]
  4. Other [inflammatory bowel disease]
- D. Neoplasms [lymphomas, intra-thoracic]
- E. Miscellaneous [CTD]
- F. Thyroid acropachy

II. Clinical Presentation:

A. Clubbing:
1. softening of nail bed
2. loss of 15 degree angle b/w nail & cuticle
3. accentuated convexity of the nail.
4. fingertip develops clubbed appearance with local warmth and sweating
5. Nail and adjacent skin develop shiny or glossy change

B. Articular symptoms [30-40%]
- + Pain/tenderness/swelling
- resembles RA
- Knees,ankles,wrists, elbows are affected.
- synovial effusions frequently found.

III. Radiographic Abnormalities:

A. Periostitis = predominant radiographic feature.

B. Osseous thickening is most pronounced in the tubular bones of the extremities:
tibia, fibula, radius, ulna

C. Thickening of calvarium and base of skull

D. Acro- osteolysis may be seen in youger patients

E. Periosteal changes appear as continuous, thin sclerotic line of new bone at the distal end of the diaphysis of the long bones.

F. Radionucleotide bone scan:

- pericortical linear concentration of nuclide along the radial, femoral and tibial shafts along with periarticular uptake c/w synovitis.
- "double strip" or " parallel track" sign and are found before radiographic changes.

IV. References:

1. Clin Radiol 1996;51: 694-7
2. Clin Oncol 1996;82: 259-61
3. J Nucl Med 1995;36:822-5.

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TUBERCULOUS SPONDYLITIS

[Pott's Disease]

Musculoskeletal infections caused by mycobacteria are often a diagnostic challenge. Characteristic pulmonary or extra-pulmonary findings are not always present leading to a delay in diagnosis. Patients typically present with chronic indolent localized involvement of the bones, spine, peripheral joints and soft tissue. Skeletal tuberculosis makes up about 1 -2 percent of all cases and about 10 percent of extra-pulmonary cases in the United States. In about 50 % of these cases, no pulmonary involvement can be detected. The spine is the dominant site of involvement in skeletal tuberculosis, accounting for 50-60 percent of cases. The suspected source of infection is hematogenous spread from reactivation of previous subclinical disease.

I. Location :

A. Lower thoracic and thoracolumbar spine account for 48-67% of lesions.

B. First lumbar vertebra most commonly affected and the frequency of involvement decreases equally on proceeding in either direction from this level.

C. Cervical spine rarely affected.

D. Infection usually begins in the anterior subchondral bone of a single vertebra adjacent to the intervetebral disc. Progression begins with extension, first from the cancellous to cortical bone and then across the disc spance to adjacent vertebrae leading to vertebral collapse.

E. Paraspinal abcesses [Psoas] / Pott's paraplegia [cord compression] / SI joint [10%]

F. Sites of Involvement:

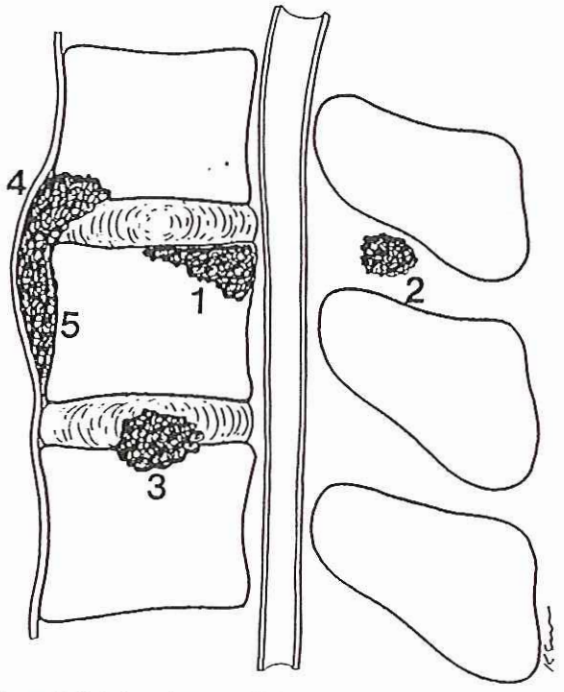


Figure 91–1. Tuberculous spondylitis: Sites of involvement. Tuberculous lesions can localize in the vertebral body (1) or, more rarely, the posterior osseous or ligamentous structures (2). Extension to the intervertebral disc (3) or prevertebral tissues (4) is not infrequent. Subligamentous spread (5) can lead to erosion of the anterior vertebral surface. (From Resnick D: Diagnosis of Bone and Joint Disorders, 3rd ed. Philadelphia, WB Saunders, 1995, p 2464.)

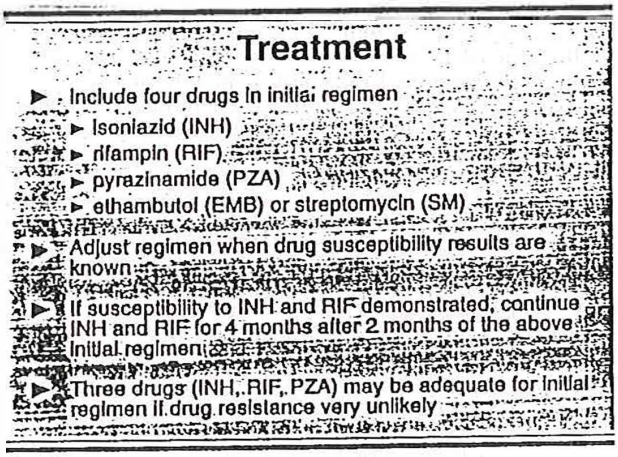
II. Diagnosis:

A. Radiographic studies: [nonspecific]
- Plain radiographs
- Scintigraphy
- CT scan
- MRI

B. CT Guided Bone BX
- AFB smear
- Mycobacterial colony counts in bone bx are relatively low.
- Histology c/w granulomatous inflammation

III. Treatment:

A. Chemotherapy
- Medical Research Council of Great Britain [MRC trial]
  INH + PAS x 18 months [+/- streptomycin months 1-3]
- CDC guidelines recommend treatment same as for pulmonary TB with duration of therapy for 12 months.



B. Indications for surgery
- Neurological deficits
- Spinal instability
- Spinal deformity with >50% vertebral collapse or deformity >5%
- Nonresponse or non-compliance with medical therapy
- Non-diagnostic needle bx

IV. References:

1. Turgut M. Pott's Disease. J Neurosurg 1997;87:136-138
2. Shaw BA. N Engl J Med 1996;334:958
3. Ventura G. Presumed Pott's Disease. Lancet 1996;347:476-477
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OSSEOUS SARCOIDOSIS

Sarcoidosis can affect the bones. A large international, multi-center review of more than 3000 cases revealed the prevalence of bone involvement to be about 5%[1 - 13%]. Bone involvement rarely occurs in the absence of cutaneous sarcoid. It has been estimated that 80-90 per cent of patients with sarcoidosis involving bone have radiographic evidence of pulmonary disease. Although bone changes are often asymptomatic, clinical manifestations may be more prominant. Soft tissue swelling and cutaneous lesions of the hands and feet can accompany osseous disease. These clinical findings are symmetric in distribution and are most prominant over the proximal and middle phalanges of the digits.

I. Radiographic Manifestations:

A. General findings:
- Osteoporosis/osteopenia
- Decreased cortical thickness
- Lacework trabecular pattern
- Lytic lesions
- Cystic lesions
- Osteosclerotic Lesions
- Acro-osteosclerosis
- Pathological fractures
- Looks like Osteomyelitis and neoplasm
- Typically lytic lesions in epiphyses progressively destroy cortical bone and bone medulla without periosteal reaction.

B. Hands & Feet:
1. Hands > Feet most commonly affected.
2. Unilateral or Bilateral without close symmetry between lesions.
3. Middle and Distal phalanges most common > Proximal phalanx.
4. Diffuse trabecular alterations are especially characteristic
5. Marginal scalloping of bone
6. Acro-osteosclerosis/ Jaccouds have been reported

C. Long Tubular Bones:
- Involvement is rare. Periostitis usually is not observed.

D. Skull and Face:
- Calvarial destruction is unusual.

E. Thorax:
- Rib involvement is infrequent.
- Lesions lytic a/o sclerotic

F. Spine and Spinal Cord:
- Vertebral involvement is uncommon.
- Granulomas w/in vertebral marrow can occur.
- Disk space is preserved.

II. Articular Involvement

Joint symptoms occur in 10-35% of patients with sarcoidosis and more frequently in females than in men. There are two fundamental patterns of articular disease: acute polyarthritis and chronic polyarthritis.

A. Acute Polyarthritis:
- Peripheral symmetric polyarticular affecting small and medium sized joints, especially the ankles, knees, elbows, wrists and articulations of hands.
- Soft tissue swelling, joint effusion, pain, tenderness, limitation in ROM and stiffness.
- Findings diappear in 4-6 weeks.

B. Chronic Polyarthritis:
- Persists for months to years, subsides and recurs and can lead to disability and irreversible joint damage.
- Ankles>Knees>shoulders>wrists and small joints of hands.
- Cutaneous and pulmonary manifestations are frequent.
- Polyarticular > Monoarticular pattern
- Synovial fluid analysis c/w inflammation: WBC 15,000-20,000 [lymphocytes predominant]
- Osseous Sarcoidosis may or may not be present.
- Periarticular osteoporosis
- Mild diffuse joint space loss and eccentric well defined erosions present.

III Radiographic Differential Diagnosis:

Sarcoidosis
Gout
Rheumatoid Arthritis
Xanthomatosis
Tuberous Sclerosis
Fibrous Dysplasia
Enchondromatosis [Oiler's Disease]
Tuberculosis
Fungal Disease
Metastasis
Plasma Cell Myeloma
Hyperparathyroidism
Basal Cell Nevus Syndrome
Hemangiomas

IV. Diagnostic Studies:

1. Histological diagnosis: non-caseating granulomas[skin, lung, bone]
2. Radiographs [hands & feet]
3. Bone Scan: extent of appendicular and axial involement and measure for progression of disease.
4. MRI:

V. Therapy

1. NSAIDs: First line therapy for musculoskeletal involvment. In absence of Neuro/Ophtho/progressive and or significant Pulmonary involvement.
2. Corticosteroids:
   - Reserved for progressive disease a/o NSAIDs failure.
   - Osteopenia increases with steroid use: Calcium/Vitamin D3 relatively contraindicated
     Consider Salmon Calcitonin in slowing steroid induced osteopenia.
3. Colchicine in treatment of acute arthritis.
4. Steroid sparring agents:
   - MTX
   - Cyclosporine
5. Synovectomy

VI. References:

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NIH ARB Rounds
30 January 1998

DIFFUSE IDIOPATHIC SKELETAL HYPEROSTOSIS
(Forestier's Disease)

Diffuse Idiopathic Skeletal Hyperostosis (D.I.S.H.) is a disease of new bone formation of unknown etiology which may occur anywhere in the axial skeleton; however, it primarily affects vertebrae of the thoracic spine and the thoracic-lumbar junction. DISH is a relatively common radiologic finding and is present in 6-12% of routine autopsies. Characteristic radiographic findings are flowing ossification of the anterior longitudinal ligament with relatively well preserved intervertebral disc spaces. Curr Opin Rheumatol 1995;7:356-358

I. Radiologic Features of DISH (Clin Exp Rheumatol 1993;11:603-608)

1. Ossification of anterior longitudinal ligament and presence of spurs (of at least four contiguous vertebrae).
2. Preserved intervertebral disc spaces
3. Radiological picture of syndesmophytes at some level.
4. Disc spaces commonly well maintained
5. Absence of vertebral body squaring
6. Absence of sacroillitis
7. Extra-spinal: Irregular new bone formation ("whiskering")

II. Associated Conditions

A. Diabetes Mellitus
B. Obesity
C. Acromegaly
D. Synthetic retinoid therapy

IV. Differential Diagnosis

A. Ankylosing Spondylitis
B. Seronegative Spondyloarthropathies
C. Synovitis-Acne-Pustulosis-Hyperostosis-Osteitis (SAPHO)

III. References:

Resnick D. Shapiro RF, Weisner KB, Niwayama G, Utsinger PD, Shaul SR. Diffuse idiopathic skeletal hyperostosis (DISH):ankylosing hyperostosis of Forestier and Toter Querol. Semin Arthritis Rheum 1978;7:153-87.

Ramos-Remus C, Gomez-Vargas A, LeClercq S, Russell AS. Radiologic features of DISH may mimic ankylosing spondylitis. Clin Exp Rheumatol 1993;11:603-608.

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METABOLIC BONE DISEASE

HYPERPARATHYROIDISM

Parathyroid hormone [PTH] is a polypeptide consisting of 84 amino acids. The hormone is secreted in response to decreased ionized calcium. PTH stimulates osteoclastic bone resorption and oteocytic oteolysis of bone, both of which release calcium from skeletal stores. PTH stimulates tubular resorption of calcium in the kidney and increased intestinal absorption of calcium in the gut. These activities result in increased serum calcium levels and periarticular calcium deposition.

I. Epidemiology:

- Female predominance: 2-3:1
- Postmenopausal women [age > 60]: 188 cases per 100,000
- Prevalence: 1 in 1000

II. Signs and Symptoms:

Systemic: Fatigue, weakness, malaise
Musculoskeletal: bone pain, arthralgia, joint effusion, myalgia, muscle weakness
Genitourniary: polyuria, nocturia, renal colic
Neuropsychiatric: impaired concentration, lethargy, confusion, depression,coma
Gastrointestinal: anorexia, nausea, vomiting, polydipsia
Cardiovascular: HTN
Ophthalmologic: conjuctival and corneal calcification

III. Clinical presentation:

A. Parathyroid adenoma: 85%
B. Multiple endocrine neoplasia [MEN]
   1. MEN I: [pituitary/pancreas/parathyroid]
   2. MEN IIa: [Medullary thyroid CA/hyperparathyroid/pheochromocytoma]
C. Metabolic derangements :
   1. Hypercalcemia
   2. Hyper PTH
   3. Hypophosphatemia [< 3.0] 50% of patients.
   4. Serum chloride >106 [serum CL/PO4 ratio > 33]
D. Radiographic Abnormalities:

IV. Articular Manifestations of Hyperparathyroidism:

Mechanism	Characteristic sites	Characteristic appearance
Subperiosteal bone resorption	Hands, wrists, feet	Marginal erosions + adjacent bone resorption & proliferation.
Subchondral bone resorption	SI, sternoclavicular,AC symphysis pubis,	Subchonral erosions, collapse
Subligamentous bone resorption	Trochanters, ischial&humeral tuberosities, calcanei, clavicles	Osseous erosion with reactive bone formation
CPPD	Knees, wrists, symphysis pubis	Chondrocalcinosis
Urate cyrstal dz	various sites	Soft tissue swelling,oss erosions
Tendon & Ligament laxity	SI, AC joints, spine	Joint instability, soft tissue swelling, osseous erosions,
Tendon avulsion/rupture	Quads, infrapatellar,triceps flexor & extensor tendons	Subluxation, dislocation
Brown Tumors	Face,pelvis,ribs,femur	Accum fibrous tissue, giant cells

V. Calcium Pyrophosphate Dihydrate Crystal disease [CPPD]:

Affects primarily articular cartilage, menisci, synovium and nearby tendinous or ligamentous structures, occurs mostly in the elderly, although it may be a marker for important underlying metabolic abnormalities.

A. Associated Conditions:

Hyperparathyroidism
Hemochromatosis
Hemosiderosis
Hypophosphatasia
Hypomagnesemia
Hypothyroidism
Gout
Neuropathic joint
Aging
Amyloidosis
Trauma / Surgery
Wilson's Disease

B. Clinical Features:

1. Acute synovitis [Pseudogout]
   - Knee>wrist>shoulder>ankle> elbow
   - Resembles Gout: Rapid onset, severe pain and stiffness
   - May be self-limited illness
   - CPPD crystals [rhomboids or rods, weak positive birefringence]
2. Chronic pyrophosphate arthropathy
   - Chronic pain, inactivity, stiffness/ Chondrocalcinosis on XRAY
   - Secondary Osteoarthritis [bony swelling, crepitus, restricted ROM]
   -Knees: Bi/tricompartmental dz / Patello-Femoral compartment***
   - Progressive, severely destructive arthropathy

C. Therapy:

- Aspiration and corticosteroid injection
- Analgesics
- NSAIDs
- Colchicine 0.6mg PO [seldom warranted]
- ACTH 40 IU IM
- Oral Magnesium

VI. Renal Osteodystrophy

In the presence of chronic renal insufficiency, the parathyroid glands undergo hyperplasia of the chief cells secondary to phosphate retention and consequent lowering of serum calcium. Renal osteodystrophy is a term applied to the bone disease that is apparent in patients with chronic renal failure.

A. Hyperparathyroidism:

- Subperiosteal, intracortical,endosteal, trabecular, subchondral, subligamentous bone resorption, Brown tumors, bone sclerosis & Chondrocalcinosis.
- Osteosclerosis: predominates in axial skeleton, pelvis, ribs and superior and inferior portions of vetebral bodies [Rugger-jersey spine].

B. Primary vs Secondary Hyperparathyroidism [Renal Osteodystrophy]

Finding	Primary	Secondary
Brown tumors	common	less common
Osteosclerosis	rare	common
Chondrocalcinosis	not infrequent	rare
Periostitis	rare	not infrequent

C. CONNECTIVE TISSUE DISEASE

"Plurality should not be assumed without necessity"
-William of Ockham

OUTLINE:

C1. Autoimmune Hepatitis
C2. Antiphospholipid Antibody Syndrome
  a. Overview
  b. Pulmonary embolus
  c. Deep vein thrombosis
C3. Systemic Lupus Erythematosus
  a. Overview
  b. Diagnosis
  c. SLE & Urticaria
  d. Pulmonary disease
  e. Autoantibodies/nephritis
  f. Hemophagocytic syndrome
C4. Behcet's Disease
C5. Mixed Cryoglobulinemia
C6. Systemic Sclerosis (Scleroderma)
  a. Overview
  b. Sine Scleroderma
  c. Interstitial Lung Disease
  d. Scleredema
C7. Myopathies
  a. Metabolic
  b. Infectious
  c. Inflammatory
  d. Antisynthetase syndrome
C8. Miscellaneous:
  a. Immune thrombocytopenic purpura
  b. Relapsing polychondritis
  c. Pulmonary Alveolar Hemorrhage
  d. Gastrointestional manifestations of Sjogren's syndrome

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AUTOIMMUNE HEPATITIS

CLINICAL FEATURES:

- FEMALES > MALES
- Alkaline phosphatase : Aminotransferase (< 3.0)
- Hyperglobulinemia (>1.5)
- Cirrhosis
- Autoantibodies (ANA,AMA,LKM)
- Negative viral serologies

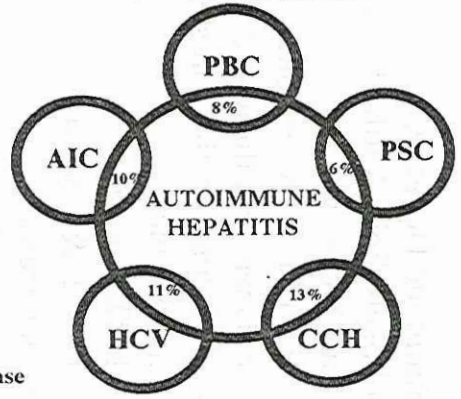
DIFFERENTIAL DIAGNOSIS

- Autoimmune liver disease
  - Autoimmune hepatitis
  - Primary biliary cirrhosis
  - Primary sclerosing cholangitis
  - Overlap syndrome
  - Autoimmune cholangiopathy
- Chronic viral hepatitis
  - HBV
  - HCV
  - Chronic hepatitis Delta
  - Chonic hepatitis @ other viruses
- Wilson's Disease
- Chronic Rx induced hepatitis
- Alpha 1-antitrypsin deficiency
- AIDS cholangiopathy
- Granulomatous hepatitis
- SLE
- Graft-versus-host disease
- ETOH steatohepatitis
- NonETOH steatohepatitis

AUTOANTIBODIES

TYPE	CHARARCTERISTIC AUTOANTIBODIES	OCCASIONALLY PRESENT
I. (Classic)	Antinuclear Anti-smooth muscle Antiactin Anti-asialoglycoprotein receptor	Antimitochondrial Anti-soluble liver antigen Anti-liver-pancreas protein ANCA
II.	Anti-LKM-1 Anti-liver cytosol 1	Antinuclear

OVERLAP SYNDROMES



- PBC: Primary Biliary Cirrhosis (8%)
- PSC: Primary Sclerosing Cholangitis (6%)
- CCH: Cryptogenic chronic hepatitis (13%)
- HCV: Chronic Hepatitis C Virus (11%)
- AIC: Autoimmune Cholangitis (10%)

ASSOCIATED CONDITIONS

- Chronic autoimmune thyroiditis
- Hyperthyroidism
- Ulcerative colitis
- Hemolytic anemia
- Idiopathic thrombocytopenia
- Diabetes mellitus
- Diabetes insipidus
- Celiac disease
- Polymyositis
- Pulmonary fibrosis
- Pericarditis
- Glomerulonephritis
- Acute Lichenoid Pityriasis
- Febrile panniculitis
- Hypereosinophilic syndrome
- Sjogren's Syndrome
- Mixed connective tissue disease
- Myasthenia gravis

TREATMENT

	Monotherapy	Combination therapy
Initial:	Prednisone 20-30mg	Prednisone 10-20mg/d + AZA 50-100mg
Maintenance:	Prednisone 5-15mg or AZA 100-200mg	Prednisone 5-10mg/d + AZA 50-150mg

Ann Intern Med 1996;125:588-598
New Engl J Med 1996;334:897-903

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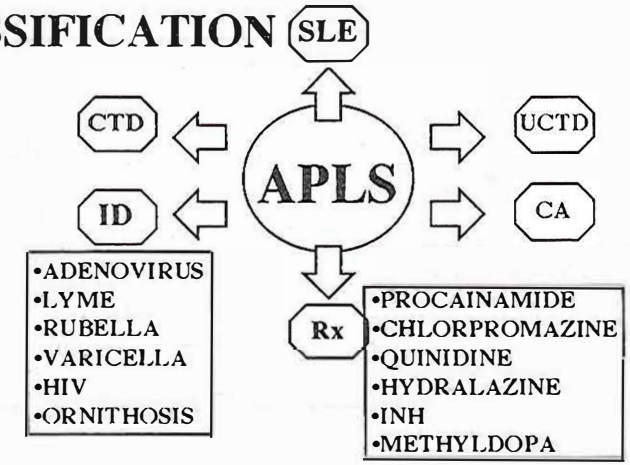
ANTIPHOSPHOLIPID ANTIBODIES

DEFINITIONS:

- **Antiphospholipid Antibodies [aPL]**
  - Antibodies [IgG or Ig M] directed against negatively charged phospholipids. High degree of cross-reactivity with cardiolipin and phosphotidylserine
- **Anticardiolipin Antibodies [aCL]**
  - Antiphospholipid antibodies measured by ELISA using cardiolipin as the phospholipid antigen.
- **Lupus Anticoagulant [LA]**
  - Antiphospholipid antibodies that prolong the clotting time of phospholipid dependent coagulation assays through their interaction with phospholipids.
- **Biologically False Positive Syphilis Serologies**
  - [BFPSTS] Antiphospholipid antibodies that cross react with RPR/VDRL reagin which contains cardiolipin.

CLASSIFICATION



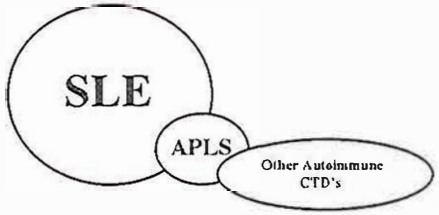
Prevalence of aPL and LAC

	aPL [%]	LAC [%]
Normal	<2	<1
Normal Pregnancy	<2	<2
SLE	30-40	10-20
RA	10	<5
HIV	60-90	20-40
Stroke*	10 - 20	
MI *	10 - 20	

* < 40 years of age

Seminars in Arthritis & Rheumatism 1990;20:81-96

Secondary Antiphospholipid Antibody Syndrome



Univariate Correlation between laboratory variables and the thromboses clinical outcome in SLE patients

Variable	Patients with thromboses [n=20]	Patients w/o thromboses [n = 370]	RR	P value
aCL+	13	171	1.4	0.114
Prolonged aPTT	10	55	3.38	0.001
Direct Coombs +	12	134	1.67	0.046
Indirect Coombs +	7	56	2.33	0.026
Thombocytopenia	0	18	0	0.327
BFP-STS [VDRL]	0	15	0	0.343

Proposed Mechanisms of Antibody-Mediated Thrombosis

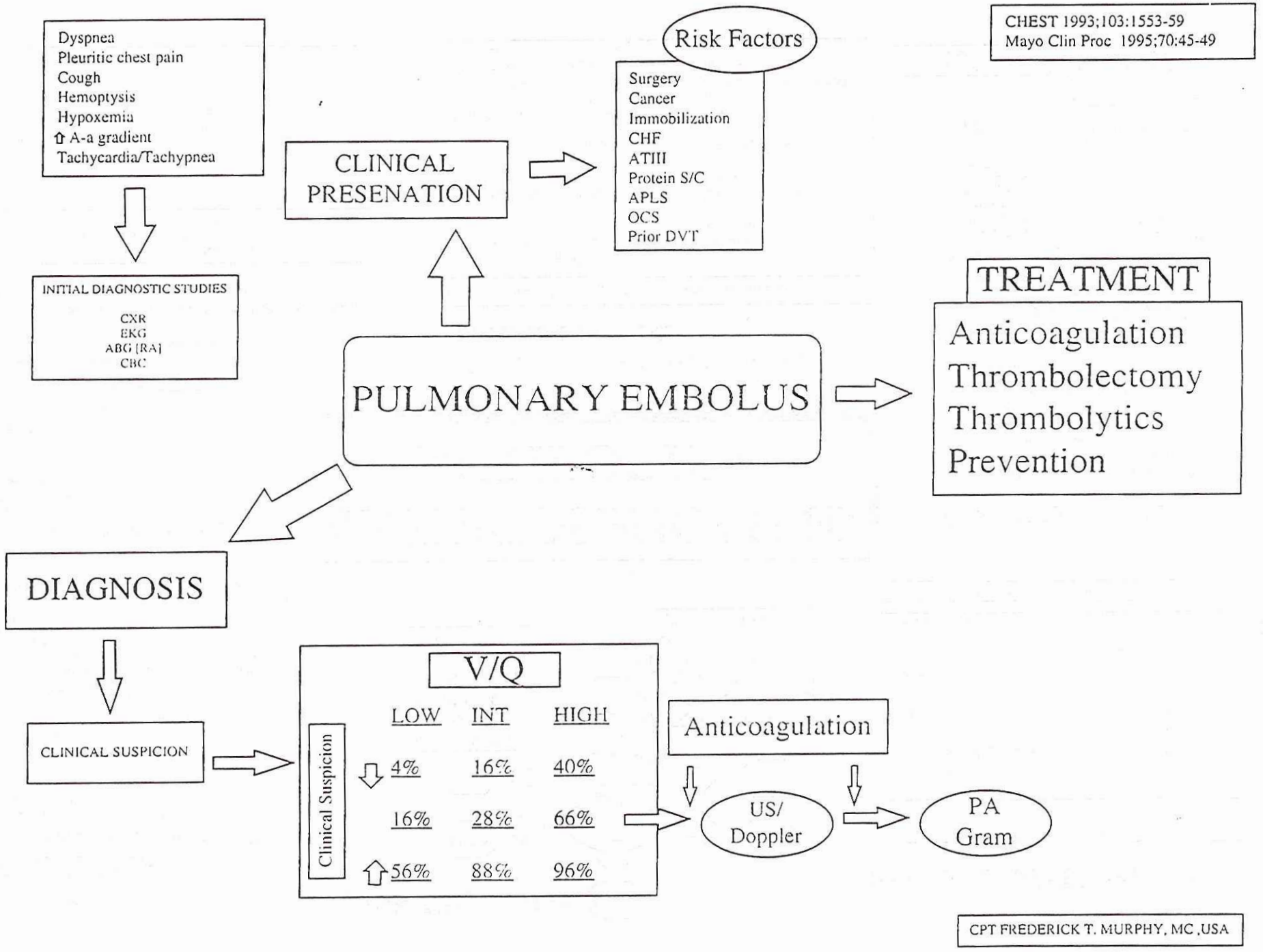
Proposed Mechanism	Autoantibodies implicated
Inhibition of Protein C Pathway	
- Protein C Activation	Antiprotein C Antithrombomodulin
- Inactivation of Factor Va / VIIIa	Antiprotein C & S Antifactor V
Inhibition of prostacyclin production	Antiphospholipase A2
Inhibition of ATIII activation	Antiheparin sulfate Anti-B2GPI
Impaired fibrinolysis	Anti-B2GPI, Antifactor XII
Enhanced PLT activation	Anti-B2GPI
Enhanced Factor Xa generation	Anti-B2GPI

Obstet Gynecol 1992;80:614
Lancet 1988;2:393-394

TREATMENT [APLS]

- **ANTICOAGULATION THERAPY**
  [Heparin followed by warfarin INR≥ 3]
  - Stroke and other Arterial Occlusion Syndromes
  - Venous Occlusion Syndromes
  - High titer aCL [IgG]
- **ANTIPLATELET THERAPY**
  - Not Standardized
  - Hydroxychloroquine
  - Pregnancy
- **NO [Ø] THERAPY INDICATED**
  - INFECTIOUS ETIOLOGY
  - DRUG INDUCED
- **PREGNANCY**
  - REGIMEN 1: ASA [81MG]
  - REGIMEN 2: [40mg with taper] ASA [81MG] + Prednisone
  - REGIMEN 3: Heparin 10K - 12K U/day returning to ASA ASA [0 -12 Weeks] then [13-32Weeks] then
  - Experimental: [IVIG] + Low molecular Intravenous Immunoglobulin weight heparin [LMWH]

Ann Intern Med 1992;117:303
Am J Hematol 1985;19:265-275
Am J Med 1994;96:254-9
Lupus 1993;2:S13-15


Dyspnea
Pleuritic chest pain
Cough
Hemoptysis
Hypoxemia
⇧ A-a gradient
Tachycardia/Tachypnea
INITIAL DIAGNOSTIC STUDIES
CXR
EKG
ABG [RA]
CBC
CLINICAL PRESENATION
Risk Factors
Surgery
Cancer
Immobilization
CHF
ATIII
Protein S/C
APLS
OCS
Prior DVT
CHEST 1993;103:1553-59
Mayo Clin Proc 1995;70:45-49
PULMONARY EMBOLUS
TREATMENT
Anticoagulation
Thrombolectomy
Thrombolytics
Prevention
DIAGNOSIS
CLINICAL SUSPICION
V/Q
LOW INT HIGH
Clinical Suspicion
4% 16% 40%
16% 28% 66%
56% 88% 96%
Anticoagulation
US/ Doppler
PA Gram
CPT FREDERICK T. MURPHY, MC ,USA

CPT. Frederick T. Murphy, MC, USA

DEEP VEIN THROMBOSIS

Risk Factors

- bed rest
- travel
- SX
- LOW CO

venous stasis

- Cancer
- pregnancy/OCS
- ATIII
- Protein C /S
- APLS

Hypercoagulation

- Thrombophlebitis
- Trauma
- prior DVT

Vascular Intima damage

Treatment

- **Goals of Therapy:**
- Prevention of PE
- Restoration of venous patency
- Prevention of postphlebitic synd.
- Proximal/Distal DVT = SAME

- ANTICOAGULATION
- THROMBECTOMY
- THROMBOLYTICS
- PREVENTION

DIAGNOSIS

- CLINICAL PRESENTATION
  - PAIN
  - EDEMA
  - ERYTHEMA
  - CORD PRESENT
  - HOMEN'S SIGN

- Diagnostic Testing*

[Symptomatic DVT]

	Sensitivity	Specificity
Impedence Plethysmography	92%	95%
Ultrasonography/** B-Mode	97%	97%
Doppler flow velocity	88%	88%
MRI	96%	100%

* Proximal thrombi [thigh/popliteal area]
**The sensitivity of distal thrombi is far less satisfactory

ANTICOAGULATION

1. heparin ----> warfarin
   [1.5-2.5x] [INR=2-3]
2. LMWH
3. Duration of therapy:
   1st event = 3 mos.
   2+ event = 6 mos.

Ref: N.Engl J.Med 1994;331:1630-41

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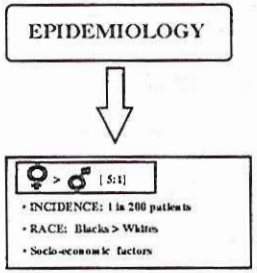
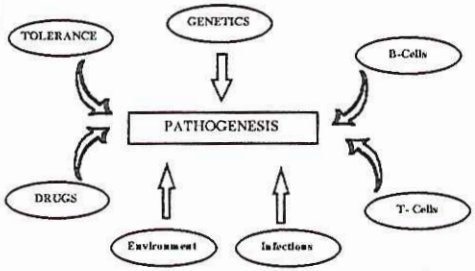
SYSTEMIC LUPUS ERYTHEMATOSUS
(Overview)

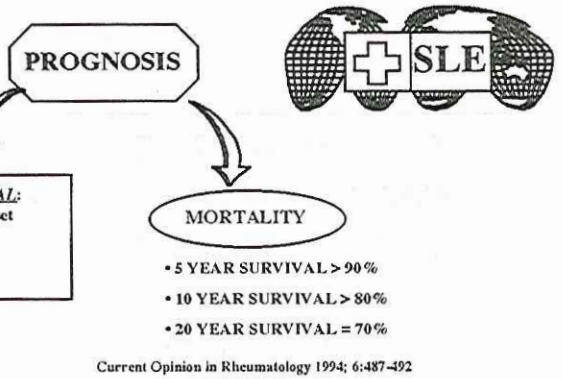
Classification Criteria

- ❑ Malar Rash
- ❑ Discoid Rash
- ❑ Photosensitivity
- ❑ Oral Ulcers
- ❑ Neurologic Disorder
- ❑ Arthritis
- ❑ Serositis
- ❑ Renal disorder
- ❑ Hematologic disorder
- ❑ Immunologic disorder
- ❑ Antinuclear Antibody

Arthritis Rheum 1982;25:1271-1277



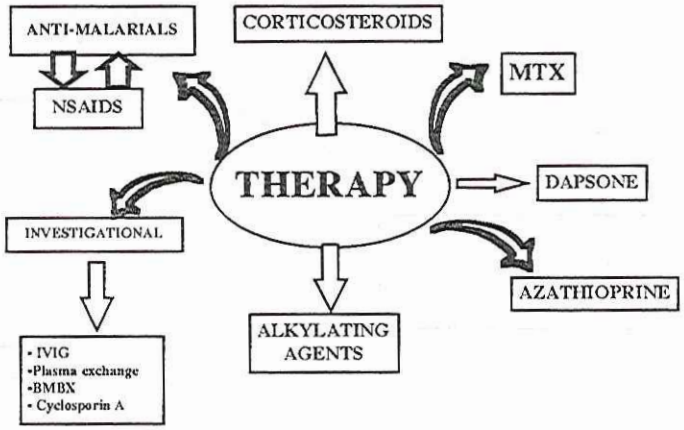




Current Opinion in Rheumatology 1994; 6:487-492

DRUG INDUCED LUPUS

- **Definite Association:**
  - Methyldopa - Hydralazine
  - Procainamide - INH
  - Chlorpromazine
- **Possible Association:**
  - PTN - Quinidine
  - Penicillamine - Lithium
  - Atenolol - Captorpril
- **Questionable Association:**
  - Griseofulvin - Gold Salts
  - Antibiotics:
    - Pen
    - Tetracycline
    - Streptomycin



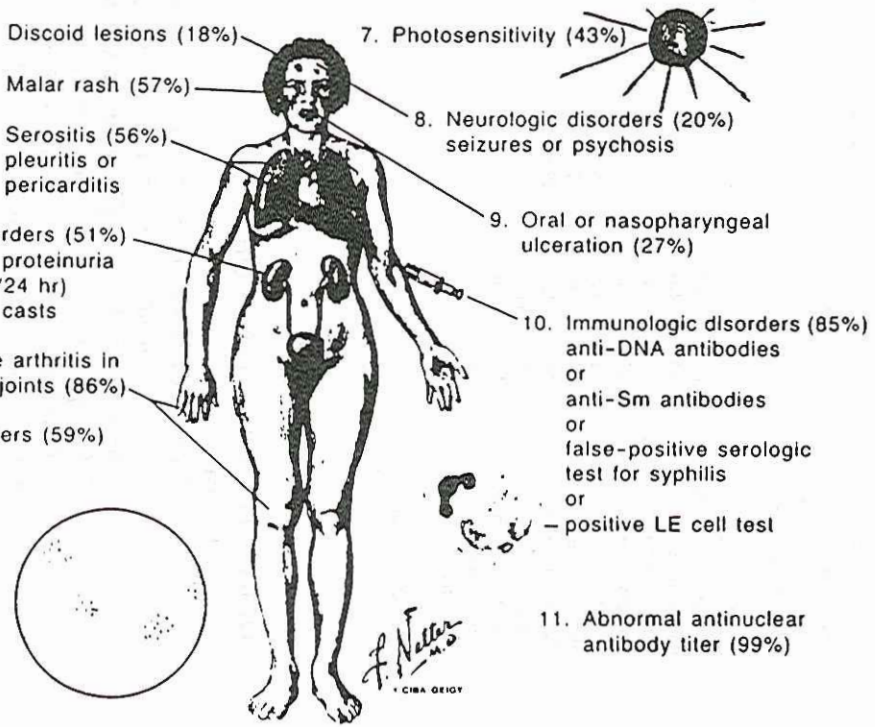
DIAGNOSIS OF SLE

Revised criteria for the classification of SLE.

Diagnostic markers: dsDNA and anti-Sm antibodies.

Major Diagnostic Criteria of Systemic Lupus Erythematosus (SLE)*

At least 4 of 11 should be present for definitive diagnosis



*American Rheumatism Association Criteria revised in 1982

Sensitivity and specificity of 1982 SLE criteria

1982 criteria	Sensitivity: SLE (177 patients)	Specificity: Control (162 patients)
Malar rash	101/177 (57)	156/162 (96)
Discoid rash	31/177 (18)	161/162 (99)
Photosensitivity	76/176 (43)	155/162 (96)
Oral ulcerations	47/177 (27)	155/162 (96)
Arthritis	152/177 (86)	60/162 (37)
Serositis	99/177 (56)	139/162 (86)
Renal disorder	91/177 (51)	147/157 (94)
Neurologic disorder	35/177 (20)	159/162 (98)
Hematologic disorder	105/177 (59)	144/162 (89)
Immunologic disorder	149/176 (85)	113/122 (93)
Antinuclear antibody	174/175 (99)	68/139 (49)
Application of 1982 criteria requirements	170/177 (96)	155/162 (96)

Analysis of false positive and false negative cases

	Current patient database (177 SLE and 162 controls)	
	1982 criteria	1971 criteria
False positive	7 (96%)*	8 (95%)*
False negative	7 (96%)†	22 (88%)†

* Percentages denote overall specificity of the 1982 and 1971 criteria.

† Percentages denote overall sensitivity of the 1982 and 1971 criteria.

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SLE & URTICARIA

Urticaria has been noted in 4 -13% of patients with SLE. It is unclear if this occurrence in SLE is more frequent than in allergic patients without SLE. The presence of urticaria alone is not suggestive of a SLE flare. On the other hand, urticarial vasculitis can be a specific skin finding in SLE and may be associated with elevated levels of circulating immune complexes and active systemic disease.

I. Cutaneous Manifestations in SLE [Br J Dermatol 1996;135:355-362]

	Our series		Dubois and Tuffanelli[3]	Estes and Christian[6]	Lee *et al.*[7]	Grigor *et al.*[4]	Weinstein *et al.*[8]		Worrall *et al.*[9]	Pisitiner *et al.*[10]
No. of cases	73		520	150	110	375	84		100	464
Year of study	1995		1964	1971	1977	1978	1987		1990	1991
Types of skin lesion	%	(number)	%	%	%	%	%	(number)	%	%
Skin lesion all types	100	(73)	72	81			82	(69)	90	55
Raynaud's phenomenon	60	(44)	18·4	21	46	32				25
Photosensitivity	63	(46)	32·7		50	28			48	37
Mucous membrane lesions	55	(40)	9·1	7		34	34·5	(29)	36	19
Butterfly eruption	45	(33)	20·9				21	(18)		34
Urticaria	44	(32)	6·9	13	5					4
Alopecia—non-scarring	40	(29)	21·3	37	38	64				
Chronic discoid lesions	25	(18)	28·6	14	28	22	11	(9)		23
Chilblains	20·5	(15)								
Alopecia—scarring	14	(10)	3·6							
Butterfly blush	14	(10)	36·7	39	36	68				
Vasculitis	11	(8)		21		70	18	(15)		
Bullae	8	(6)	0·4	2						
Psoriasiform rash	7	(5)		2				(18)		

II. Differential Diagnosis of Urticaria in SLE

A. Urticaria:

Transient wheals, edematous papules & plaques, usually pruritic and of larger edematous areas that involve the dermis and subcutaneous tissue.

1. Immune mediated: IgE / associated with atopy
2. Complement mediated: complement –>anaphylatoxins –>Mast Cell
3. Physical Urticaria: Dermographism
4. Cold Urticaria: children/young adults: "ice cube test"
5. Solar Urticaria: Action spectra 290-500nm, histamine mediated.
6. Urticaria/Angioedema: Mast cell releasing agents: ASA/azo dyes/benzoates
7. Hereditary Angioedema: Autosomal Dominant/ C1 esterase inhibitor deficiency

B. Urticaria Vasculitis [UV]:

Cutaneous lesions resembling urticaria which are slow to resolve sometimes greater than 96 hours and may leave residual pigmentation. UV may be assoicated with pruritis and/or a burning sensation of skin. It more commonly affects females 2.6:1 males with a mean age of 42 years.

1. Extra-cutaneous manifestations: Arthralgias [60-70%], Arthritis [28%], GI [25%], glomerulonephritis [15%], Chest pain [25%] and pulmonary disorders [30%]. Pseudotumor cerebri [8%], fever, angioedema, episcleritis and seizures.
2. Laboratory findings: elevated ESR [66%], Hypocomplementemia [38%]
3. Histological findings:
   - LCV or necrotizing vasculitis with perivascular infiltrate consisting primarily of abundent neutrophils in the interstices of collagen surrounding the blood vessels in the papillary and reticular dermis. Occasional eosinophils.
4. DIF: IgM and C3 in affected blood vessel.
   IgA & IgG occur in less than 10% / Fibrin [<20%]

C. Hypocomplementemic Urticaria Vasculitis Syndrome [HVUS]

Urticaria Vasculitis in the clinical setting of hypocomplementemia with antibodies directed against vascular endothelial cells.

1. IgG antibodies to C1q
2. The prevalence of antiribonucleoprotein [anti-RNP] is increased.
3. Acquired C1 inhibitor defeciency with angioneurotic edema has been observed.

III. Therapy:

A. Supportive measures
B. Corticosteroids: Prednisone 10-60mg/d
C. Histamine [H1/H2] Antagonists
D. NSAIDs: Indomethacin
E. Dapsone
F. Azathioprine
G. Cyclophosphamide
H. Plasmaphoresis

References:

1. Gibson LE et al. Cutaneous vasculitis. Rheum Dis Clin North Amer 1995;21:1097-1113
2. Yell JA et al. Cutaneous manifestations of Systemic Lupus Erythematosus. Br J Dermatol 1996;135:355
3. Brickman CM, Tsokas GC. et al. Immunoregulatory disorders associated with hereditary angioedema. J Allergy Clin Immunol 1986;77:749-57
4. Gammon WR et al. Urticarial Vasculitis: report of a case and review of the literature. Arch Dermatol 1979;115:76-80
5. O'Loughlin S et al. Chronic urticaria - like lesions in systemic lupus erythematosus. A review of 12 cases. Arch Dermatol 1978;114:879-883.
6. Wantanabe T. et al. Classification of lupus erythematosus based upon cutaneous manifestations. Dermatology 1995;190:277-283.
7. Laman SD et al. Cutaneous manifestations of lupus erythematosus. Rheum Dis Clin North Am 1994;20:195-212

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PULMONARY MANIFESTATIONS of SYSTEMIC LUPUS ERYTHEMATOSUS

Pleural Disease

Rheum Dis Clin North Am 1994;20:159-93

- Most common pulmonary manifestation
- Pleuritis: 45-60% +/- pleural effusion
- Effusion: 16-50%
  - May be presenting feature of disease
  - Unilateral or bilateral
  - Exudative !!! Transudative = CHF/Nephrosis/cirrhosis
  - LE cells/ ANA in pleural fluid = specific for SLE
- Therapy:
  - NSAIDS
  - Corticosteroids
  - Hydroxychloroquine

ACUTE LUPUS PNEUMONITIS

Semin Arthritis Rheum 1984;14:52-9
Curr Opin Rheumatol 1990;1992;4:314-21

- Incidence in SLE: 1%- 4%
- Symptoms:
  - dyspnea - Cough
  - Pleuritic pain - Hypoxemia
  - Fever - Hemoptysis
- CXR:
  - Lower lobe Infiltrates [Uni or Bilateral]
- Histopathological diagnosis:
  - Alveolar wall damage / Necrosis / hemorrhage / edema
  - Hyaline membranes: Hematoxylin bodies / LE cells
- Therapy:
  - Corticosteroids: 1mg/kg/day or "Pulse dose"
  - HD Cytoxan / Plasmaphoresis

Chronic Interstital Lung Disease

Ann Intern Med 1973;79:37-45
Clin Chest Med 1982;3:565-78

- Prevalence in SLE: 1% - 6%
- Clinical Features: CILD vs IPF
  - Progressive SOB- Cough
  - Rales - Bibasilar infiltrates
  - PFTs: reduced FVC,/TLC & DLco
  - Arterial desaturation with exercise
- Diagnosis: HRCT / BX = gold standard
  - "ground glass" --> Active Inflamm -> steroid responsive
  - "Macroscopic Honeycomb" --> end stage, irreversible
- Histopathology:
  - Lymphocytic infiltrates, peribronchial lymphoid hyperplasia
  - Interstitial fibrosis, Type II pneumocyte hyperplasia
- Therapy:
  - Corticosteroids -Cyclophosphamide
  - Imuran -Colchicine

BRONCHIOLITIS OBLITERANS & BOOP

- Small Airway disease
- Clinical Features:
  - Acute or Subacute
  - Fever/ Cough/ Constitutional features
  - CXR: focal alveolar infiltrates/ air bronchograms
- Histopathology:
  - Organizing inflammatory exudate -> bronchial obstruction
  - Polypoid masses of granulation tissue
  - Lung parenchema usually spared
- Therapy: prognosis excellent
  - Corticosteroids: 40-60mg/d x 4-6 wks --> slow Taper
  - CR 60%-80%

PULMONARY HEMORRHAGE

Chest 1996;110:1305-16
Rheum Dis Clin North Am 1994;20:159-93

- Mortality = 50%
- Clinical Features:
  - Diffuse alveolar infiltrates - Hypoxemia
  - Dyspnea - Anemia
- Associated with anti-DNA in SLE
- Overlap with Acute lupus pneumonitis
- Diagnosis: FOB & BAL
  - Gross Blood - Hemosiderin-ladin macrophages
- Therapy: No controlled studies
  - Pulse Methylprednisolone [1g QD x 3]
  - Day #4: IV Methylprednisolone 80 - 120mg/day -> 1mg/kg/d
  - HD Cytoxan: critically ill patients or pulse steroid failure.
  - Plasmaphoresis : not assoc with increased risk of infection
  - PRBC transfusion: HCT > 25%

PULMONARY VASCULAR DZ

J Rheumatol 1986;13:1-5
Clin Exp Rheumatol 1988;6:301-4

- Pulmonary Arterial HTN
- Demographic Features:
  - Young Females 18 - 49 yrs [10F:1M]
  - Raynaud's [75%] - Renal disase [63%]
  - anti-RNP [25%] - RF+ [80%] - LAC+
- Symptoms: insidious SOB/SSCP/Fatigue
- Clinical Features: Loud P2/RV heave/ R-CHF
- Diagnosis: TTE / R heart catheterization
- Pathogenesis:
  - Pulmonary vasculitis - PA thrombosis
  - Pulmonary artery vasoconstriction
- Therapy:
  - CA channel blockers - Prostacyclin
  - Anti-coagulation -Vasodilators

NEUROMUSCULAR DISEASE & DIAPHRAGMATIC DYSFXN

Chest 1988;93:352-358

- Low Lung Volumes
- Reduced Inspiration/Expiration forces
- Respiratory Muscle Weakness:
  - Transdiaphragmatic pressures:
    Transnasal esophageal & gastric balloons
  - Phrenic nerve stimulation
  - Diaphragmatic electromyography
- HRCT: not useful
- Controversial issue
- May play role in lung volume restriction

AUTOANTIBODIES IN LUPUS NEPHRITIS

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Walter Reed Army Medical Center

Glomerulonephritis is a common manifestation of Systemic Lupus Erythematosus [SLE]. Nearly all patients with SLE with have detectable renal abnormalities documented by EM or IF. Nephritis is a principal cause of morbidity and mortality in patients with SLE. The role of antibodies in the immunopathogenesis of lupus nephritis is poorly understood. The presence of ds-DNA antibodies has been associated with enhanced disease activity; however, other autoantibodies may also be implicated as potential mediators in the pathogenesis of lupus nephritis.

NEPHRITOGENIC AUTOANTIBODIES IN LUPUS*

Antibody	Evidence for Nephritogenicity [species studied]
Anti-DNA	Associated with nephritis [mouse,human]; association stronger for anti-ds-DNA Concentrated in glomeruli [mouse,human] Induce disease by adoptive transfer [mouse,human] DNA immunization induces nephritis [mouse]
Anti-histones/nucleosomes	Associated with nephritis[mouse, human] Concentrated in glomeruli [mouse,human] Induce disease by adoptive transfer [mouse]
Anti-C1q	Associated with nephritis [human]
Cryoglobulins	Associated with nephritis [mouse,human] Induce disease by adoptive transfer [mouse]
Antilaminin	Concentrated in glomeruli [mouse] Induce disease by adoptive transfer [mouse]
Antifibronectin	Present in lupus [human]
Anticollagen	Present in lupus [human]
Anti- RNA polymerase I	Associated with nephritis [mouse] Concentrated in glomeruli [mouse]
Anti-Ro	Associated with nephritis [human]
Antiribosomes	Associated with nephritis [human]

* Table 1: Arthritis Rheum 1996;39:904-913

Proposed immunopathogenic mechanisms in lupus nephritis*

Mechanism	Comments
Circulating immune complexes	Nucleosome/antinucleosome complexes are likely to participate in nephritis. Existence and role of DNA/anti-DNA complexes is controversial. Cryogolbulins may also contribute to nephritis Role of other immune complexes in nephritis is uncertain
Cross-reactive antibodies	Variant of molecular mimicry theme Recent technical and theoretical concerns warrant reappraisal of polyreactivity concept Probably less common than binding of autoantibodies to their cognate antigen
Planted antigen	Histones and nucleosomes bind avidly to glomerular basement membrane [GBM] and glomeruli Histones facilitate the binding of DNA to GBM Type IV collagen and heparin-sulfated proteoglycan are important for hsitone nucleosome binding to GBM

* Table 2: Arthritis Rheum 1996;39:904-913

Clinical and Laboratory features in which signficant differences between group I and group II patients were found.*

	Group I [n=8]	Group II [n=12]	Pvalue
SLE disease duration [mean yrs]	1.94	5.21	<0.02
Anti-dsDNA [mean reciprocal titre]	406.3	3.3	<0.001
Anti-ssDNA [# positive]	8	6	<0.04
CH50 [mean units/units in controls]	0.16	0.91	<0.001
Anti-Rp, anti-SM, and Anti-RNP [no. with all 3 precipitns, ø anti-LA]	6	0	<0.001

* Table 1: Arthritis Rheum 1993;36:1560-1565

APPENDIX 3

WHO Classification of lupus nephritis in 148 biopsies*

I. Normal glomeruli	12
a) Nil [by all techniques]	3
b) Normal by light but deposits on EM or IF	
II. Pure mesangial alterations [mesangiopathy]	62
a) Mesangial widening a/o mild hypercellularity	51
b) Moderate hypercellularity	11
III. A. Focal segmental glomerulonephritis	19
a) 'Active' Necrotizing lesions	14
b) 'Active and sclerosing lesions	5
B. Focal proliferative glomerulonephritis	3
a) 'Active' Necrotizing lesions'	1
b) 'Active and sclerosing lesions	2
IV. Diffuse glomerulonephritis	37
a) Without segmental lesions	9
b) 'Active' Necrotizing lesions'	13
c) 'Active and sclerosing lesions	14
d) With sclerosing lesions	1
V. Diffuse membranous glomerulonephritis	11
a) Pure membranous glomerulonephritis	2
b) Associated with lesions of category II	7
c) Associated with lesions of category III	ø
d) Associated with lesions of category IV	2
VI. Advanced sclerosing glomerulonephritis	4

*Gladman D D, Urowitz MB, Cole E, et al: Kidney biopsy in SLE. I. A clinical-morphologic evaluation. Quart J Med 1989;73:1125-1153

Lupus Nephritis

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HEMOPHAGOCYTIC SYNDROME

Acute hemophagocytic syndrome is most commonly a reactive process manifested by fever, hepato-splenomegaly, cytopenia, transaminitis, hyper-triglyceridemia and/or depressed detectable levels of fibrinogen. This syndrome may be associated with infectious processes, malignancy and autoimmune diseases. Treatment of underlying infection usually resolves the symptom complex. In the clinical setting of autoimmune disease, especially SLE, initiation of pulse dose steroid therapy has proven to be beneficial.

I. Diseases associated with Reactive Hemophagocytic Syndrome:

A. Infections

Viral:

EBV,CMV,HSV,Human herpesvirus-6, Rubella,HAV,HCV,

Bacterial:

Mycobacterium, Brucella, Serratia, Pneumococcus, Staphlococcus, Mycoplasma, Salmonella typhi, Enteric gram negative rods

Fungal:

Candida, Histoplasma, Cryptococcus

Parasitic:

Leishmania, Babesia

Rickettsial:

Coxiella

B. Neoplasms:

NHL
Hodgkin's Disease
Multiple Myeloma
Acute Leukemia [Lymphoblastic, myeloid]
Chronic Leukemia [Lymphocytic, hairy cell]
Carcinoma of stomach, Ovary

C. Miscellaneous and Autoimmune disorders:

Systemic Lupus Erythematosus
Kawasaki Disease
Adult Still's Disease
Immunosuppressive therapy
Drugs: PTN
The Myelodysplastic Syndrome
Congenital or Acquired Immunodeficiency.

II. Diagnosis

A. Blood smear/Bone Marrow Aspirate

- increase in mature looking histiocytes
- many histiocytes show phagocytosis or RBC,PLT, granulocytes & erythoblasts

B. Tissue Biopy:

-Lymph node/Liver/ Renal

III. Treatment

- Infection
- Steroids
- VP16

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BEHCET'S DISEASE

Behcet's disease is a chronic, multisystemic, inflammatory syndrome that includes mucocutaneous, ocular, cardiovascular, neurologic and gastrointestinal involvement. The simultaneous manifestation of recurrent choroiditis with oral and genital ulcerations was initially recognized by Hippocrates in the 5th century B.C. In 1937 a Turkish dermatologist, Hulusi Behcet, described the triad of recurrent oral ulceration, genital ulceration and relapsing inflammation of the eye in the disease which bares his name. This heterogeneous disease occurs more frequently in Japan and in Mediterranean countries [Greece, Turkey, Italy]. Ethnic aggregations, and associations with HLA-B5 and HLA-B51 support the theory of genetic predisposition.

I. Diagnostic Criteria:

1990 International Study Group for Behcet's Disease

Recurrent oral ulcerations	Minor aphthous, major aphthous, or herpetiform ulcers, observed by patient or physician which recur at lease three times in one 12 month period.
Plus 2 of:	
Recurrent genital ulceration	Aphthous-like ulceration or scarring, observed by patient or physician.
Eye Lesions	Anterior uveitis, posterior uveitis, or cells in vitreous on slit lamp examination or retinal vasculitis observed by ophthalmologist.
Skin Lesions	Erythema nodosum, pseudofolliculitis, or papulopustulare lesions or acneiform nodules observed by physician.
Pathergy Test	Positive pathergy test performed with oblique insertion of a 20gauge or smaller needle under sterile conditions and read by a physician.

Lancet 1990;335:1078-80

II. Organ System Involvement in Bechet's Disease:

Manifestation	Frequency [%]
Oral ulcerations	90-100
Genital ulcerations	64-88
Eye lesions	27-90
Skin lesions	48-88
Joint involvement	18-64
CNS	10-29
Gastrointestinal	0-59
Thrombophlebitis	10-37

Seminars in Neurology 1992:12:346-57

III. NEURO-BEHCET'S

Involvement of the central nervous system occurs in 10-30% of patients with Behcet's syndrome. It usually begins within 5 years from onset of disease and is associated with a poor prognosis. It occurs more frequently in males and in association with orogenital ulcers and uveitis. Neurologic involvement usually follows systemic manifestations in the course of disease. Ocular involvement is seen more frequently in patients with neurologic involvement compared to skin findings. The disease has an acute or gradual onset and its course is characterized by remissions and exacerbations. The brainstem is the most common site of involvement [85-95%].

A. NEUROLOGIC MANIFESTATIONS OF BEHCET'S DISEASE

Meningoencephalitis
Brainstem syndrome
Cerebrovascular insufficiency
Intracranial hypertension
Personality changes
Dementia
Peripheral Neuropathy
Autonomic nervous system dysfunction

B. Presenting Signs and Symptoms:

1. Pyramidal signs [82%]
2. Headache [65%]
3. Cerebellar signs [24%]
   Sensory impairment [24%]
4. Pseudobulbar signs [12%]
   Intracranial HTN [12%]
   Meningeal irritation [12%]

C. Patterns of Neurologic Manifestations:

1. Chronic meningoencephalitis : m/b fever,stiff neck,HA and CSF pleocytosis
2. Brainstem syndrome : m/b cranial nerve palsy or bulbar palsy resembling MS
3. Cerebral Infarction: m/b transient or persistent of brain dysfunction with pyramidal or extrapyramidal signs.
4. Cerebral vein thrombosis leading to intracranial HTN, papilledema & optic nerve atrophy
5. Confusional state leading to severe dementia and personality deterioration

D. Laboratory and Immunologic Studies:

Serum:

Elevated: ESR/CRP/Alpha2-globulins/ mild leukocytosis
Neopterin/Lysozyme and alpha1-acid glycoprotein

CSF:

-Marked Pleocytosis with predominance of neutrophils/lymphocytes
- Elevated protein/ low glucose
- IL 6 levels +++
-Oligoclonal bands [IgA/IgM]+++

IV. Treatment:

A. Corticosteroids:

Acute stage: Prednisone 1mg/kg day or more: ocular/CNS
Chronic stage: Prednisone 4 -6mg /day: mucocutaneous/articular

B. Immunosuppressive Medications:

Medication	Indication
1. Aziathiaprine 2.5mg/kg/day	Ocular
2. Chlorambucil 6-8mg QD	Ocular,cutaneous,arthritic,vascular
3. Cyclophosphamide 2-3 mg/kg/day	Mucocutaneous/arthritic/ocular
4. Cyclosporine A 5-10mg/kg/day	CNS/ocular/mucocutaneous
5. Colchicine 1mg /day	Ocular/GI
6. Thalidomide 200-400mg/day	Mucocutaneous/arthritic
7. Dapsone 100mg/day	Mucocutaneous
8. Plasmaphoresis	

V. References:

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MIXED CRYOGLOBULINEMIA

Mixed cryoglogulinemia is a systemic disorder characterized by purpura, arthralgia and weakness. The diagnosis is based upon the detection of cryoglobulins in serum which contain at least two immunoglobulin isotypes. Cryoglobulins are classified, on the basis of their immunoglobulin composition, into three subtypes. Type I consists of an isolated monoclonal component alone, type II of mixed cryoglobulins containing monoclonal IgM and polyclonal IgG immunoglobulins, and type III of polyclonal immunoglobulins only. Type II and III often have rheumatoid factor activity. Type I is frequently associated with immunoproliferative disorders such as Waldenstrom's macroglobulinemia. Most of type II and type III mixed cryoglobulinemia is classified as essential. Because of the high prevalence of coexisting essential mixed cryoglobulinemia and hepatocellular abnormalities it has been postulated that hepatotrophic viruses may be involved in the pathogenesis of this disease.

I. Clinical findings in studies of patients with essential mixed cryoglobulinemia*

	Study [number of patients]			
	Gorevic [50]	Tarantino [116]	Ferri [52]	Invernizzi [79]
Purpura	100	88	95	97
Arthralgia	70	78	90	51
Weakness	NR	67	100	NR
Peripheral Neuropathy	28	31	69	2.5
Hepatic involvement	70	88	62	NR
Splenomegaly	NR	50	NR	NR
Lymphadenopathy	16	rare	NR	NR
Lung involvement	NR	rare	NR	NR
Sjogren's syndrome	14	rare	40	NR
Raynaud's phenomenon	22	35	31	10
Skin ulcers	30	NR	NR	10
Female	66	61	75	66
Mean age [yrs]	NR	52	59	42

*except for age, numerical values are the percent with the characteristic. NR = not reported
Arthritis & Rheumatism 1993;36:1341-9

II. Prevalence of HCV in serum samples of patients with Essential Mixed Cryoglobulinemia

Study	Patients n	RIBA [%]	RNA [%]
Ferri et al.	26	96	91
Agnello et al.	19	42	84
Misiani et al.	51	66	81

Ann Intern Med 1995;123:615-20

III. Detection of HCV

The genome of HCV encodes 3 putative structural proteins: the nucleocapsid protein, and the envelope proteins gp35 and gp70 at the 5' end. The genome also encodes nonstructural proteins and has been divided into 5 regions [NS1-NS5]. HCV can be detected in both serum and cryoprecipitate.

A. First generation assays [ELISA,RIBA]:
   detect antibody to c100-3 antigen located on NS3 region.

B. Second generation assays [ELISA,RIBA]:
   combine c100-3 with another nonstructural protein, c33c, to form a new composite antigen c200. 2nd generation assays also detect antibodies against c22-3, a nucleocapsid protein.

C. RNA PCR: more direct indication of HCV infection.
   Nested primers from highly conserved 5' noncoding region [5'NC].

IV. Treatment

A. Interfon - alpha Therapy: highly effective
   - resolution of purpura,neuropathy,kidney function and normalization of LFTs/HGB.
   - reduction in cyroglobulin levels and viremia
   - 3 million units TIW

B. Plasmaphoresis

V. References:

1. Wong, VS et al. Incidence, character and clinical relevance of mixed cryoglobulinaemia in patients with chronic hepatitis C virus infection. Clin Exp Immunol 1996;104:25-31
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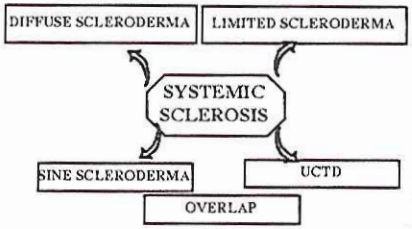
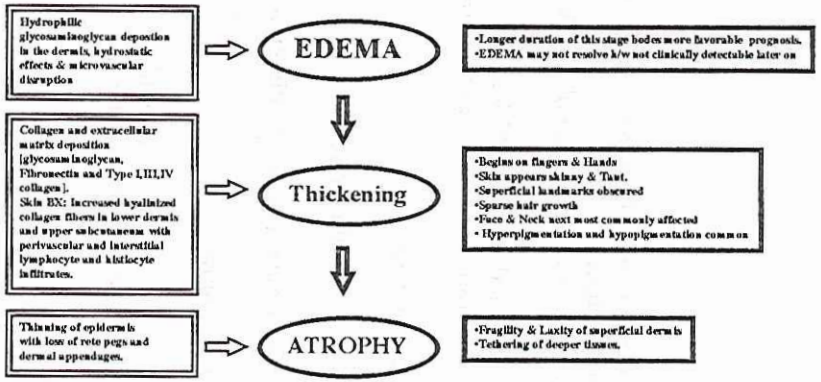
SCLERODERMA

CLASSIFICATION OF SYSTEMIC SCLEROSIS



Cutaneous Manifestations of Systemic Sclerosis



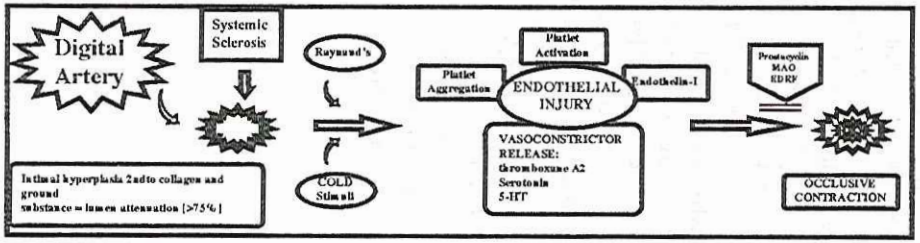
CLINICAL DIFFERENCES BETWEEN LIMITED AND DIFFUSE DISEASE

	DIFFUSE	LIMITED
Raynaud's	90%	99%
Finger swelling	95%	90%
Tendon Friction Rub	70%	5%
Arthralgia	98%	90%
Calcinosis	20%	40%
Telangiectasias	60%	90%
Esophageal dysmotility	80%	90%
Small bowel involvement	40%	60%
ILD	7%	35%
Pulmonary HTN	5%	25%
Myocardiopathy	15%	10%
Renal Crisis	20%	1%
Sicca Sydrome	15%	35%

Systemic Sclerosis & Raynaud's Phenomenon

- Typical Presenting manifestation of SS [@75%]
  - Nearly all CREST syndrome will present with Raynaud's
- 5% w/o Raynaud's Phenomenon
  - Male > Female
  - Increased mortality [myocardial and renal involvement].

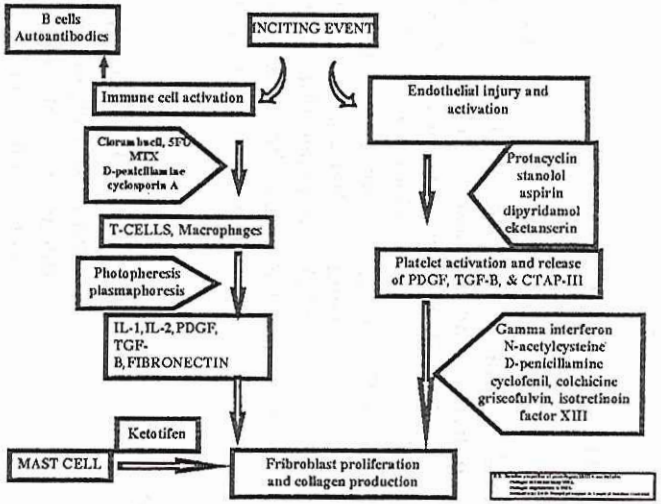
Pathogenesis of Raynaud's in Systemic Sclerosis:



COMPARISON OF LABORATORY FEATURES AND SURVIVAL IN SYSTEMIC SCLEROSIS

	DIFFUSE	LIMITED
ANA	90%	90&
ANTICENTROMERE	5%	50-90%
Anti-Scl-70	20-30%	10-15%
Cumulative Survival		
5 year	70%	90%
10 year	50%	70%

PATHOGENESIS OF SYSTEMIC SCLEROSIS AND POTENTIAL SITES OF DRUG ACTION



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NIH ARB Rounds
30 Jan 98

SINE SCLERODERMA

Progressive systemic sclerosis with sine scleroderma (SSSS) is characterized by internal organ manifestations, vascular and serologic abnormalities but without clinically detectable skin changes. SSSS was originally reported by Rodnan et al who described four patients who died of organ failure secondary to systemic sclerosis but with minimal or no cutaneous involvement.

I. Clinical Characteristics of Patients presenting with Systemic Sclerosis Sine Scleroderma (Amer J Med 1989;87:525-27).

	Lomeo et al.	Literature review	Total
ANA	5/6	2/2	87%
Esophageal dysfunction	5/6	3/3	89%
Restrictive PFTs	4/6	2/2	75%
DLco decreased	5/6	1/1	86%
Telangiectasias	4/6	1/1	71%
Raynaud's	5/6	0/2	63%
Nailfold capillary changes	6/6	ND	100%
Pulmonary Fibrosis	4/4	2/2	100%
Development of taut skin (w/in 7yrs)	3/6	4/4	70%

II. Clinical considerations:

A. The pulmonary system is the second most common organ system affected.
B. Pulmonary involvement is now the leading cause of mortality assoc with systemic sclerosis.
C. Occupational exposure may play a role in the development of systemic sclerosis presenting with pulmonary manifestations and may explain the male predominance in one study.
D. Pulmonary HTN should be excluded.
E. Esophagram, ANA and widefield microscopy should be performed.

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Rodman GP, Fennell RH. Progressive systemic sclerosis sine scleroderma. JAMA 1962;180:665

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INTERSTITIAL LUNG DISEASE IN SYSTEMIC SCLEROSIS [SSc]

Most patients with SSc have pulmonary involvement and approximately 13% of these patients will ultimately develop severe interstitial fibrosis. This condition is now the most frequent cause of death in scleroderma patients. The major clinical feature of SSc interstital lung disease is a restrictive ventilatory defect manifest by complaints of dyspnea on exertion and nonproductive cough with bibasal crackles on auscultation. Pulmonary function tests reveal reduced FVC, FEV1, residual volume [RV], total lung capacity [TLC] and DLco. Routine chest radiographs may show linear and reticular shadows and in more advanced cases a honeycomb appearance. [Rheum Dis Clin North America 1996;22:825-40]

I. Clinical Presentation

Clinical Features	Interstitial Pulmonary Fibrosis
Clinical subset	limited or diffuse SSc
Frequency	@75% post mortum
Symptoms	Dyspnea, cough
Signs	Crackles, digital pitting
Capillary microscopy	Capillary dropout,dilated loops
PFTs	Decreased: FVC/FEV1/DLCO/TLC Increased: FEV1/FVC
Antibodies:	+ Scl-70
HLA:	DR3/DR52a
CXR	WNL or Reticulonodular pattern
HRCT	Septal & subpleural lines, Ground-glass opacifications, subpleural cysts.
5 yr survival	@45% from initial PFTs

II. BAL Results

BAL Data	SSc patients	Normal controls	P Value
Total cells/ml	17.2	7.8	0.01
Alveolar macrophages	15.4	7.2	0.02
Lymphocytes	0.7	0.5	NS
Granulocytes [pmn/eos]	1.1	0.1	0.04
IgG [mg/ml]	1.44	0.51	0.01
Albumin [mg/ml]	3.34	1.99	0.10
Immune complexes [ug/ml HAGG equiv]	32.7	8.9	<0.001
Fibronectin [ng/10^6 alveolar macrophages/24hr]	306	18	0.04

III. Prognostic Factors

A. Prognostic factors for poor outcomes for lung dz in SSc
   1. Gender: Male
   2. Early lung involvment
   3. Severe Raynaud's phenomenon
   4. Low DLco
   5. Cigarette smoking
   6. Symptomatic cardiac involvment

B. Antibodies
   1. Anti-centromere antibody inversely correlated with severe restrictive disease.
   2. Anti-topo I antibody is associated with restrictive lung disease but no correlation between severe and moderate restrictive disease.

IV. Treatment:

A. Cyclophosphamide [CYC][4]
   1. Retrospective, uncontrolled study demonstrated significant improvement in FVC in SSc patients with FVC < 70%. Mean improvement in FVC of 435ml/yr [$p<0.005$]. No improvement in DLco demonstrated.
   2. Dose: Oral CYC ----> 1-2mg/kg/day x 6 months.
      IV CYC -----> 500-75mg/m2 for at least 6 months.

B. CYC & LD Prednisone[3]
   1. Open, prospective study in 14 patients with early disease & alveolitis.
      - CYC 1-2mg/kg/day & Prednisone < 10mg/day.
      - 6/14 patients demonstrated >10% increase in FVC at 6months
      - 12 patients treated w/CYC 18-24months had an FVC $\geq$ time of entry.
      - No significant improvement or decline in DLco

C. Ineffective Therapies [6] [No improvement FVC]
   - D-penicillamine, para-aminobenzoic acid, colchicine, griseofulvin, chlorambucil,
   - Extracorporeal photophoresis.

V. References:

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SCLEREDEMA

Originally described by Buschke, Scleredema is a rare cutaneous mucinosis characterized clinically by non-pitting induration of the skin. It typically affects the face, neck and upper portion of the trunk and proximal extremities. There has been a long-standing association between scleredema and the development of adult onset IRDM. In recent years, an association of monoclonal gammopathy with scleredema has been detected. The natural history of scleredema differs from that of Sclererderma in that patient's with scleredema rarely develop Raynauds phenomenon nor evidence of visceral organ involvement. *Clin Exp Dermatol 1988;13:271-274*

I. Classification of Scleredema (Arch Dermatol 1968;98:320)

1. ACUTE ONSET following acute respiratory tract infection, with resolution usually taking place within a few months.
2. INSIDIOUS ONSET and persistence of the lesions over a period of years.
3. LONG-STANDING Scleredema associated with severe complicated diabetes mellitus.

II. Proposed etiologies of Scleredema

Autoimmune:

Via sensitization of collagen by products of antecedent infection leading to abnomral accumulation of ground substance.

Adult Onset Diabetes Mellitus:

Via altered mucopolysaccharide mechanism, accelereated collagen aging, or increased susceptibility to infection.

Coccoid Forms:

Cell wall-deficient phase infection with gram positive, variably acid-fast microaerophilic Corynebacterium.

Adrenal Cortex Stimmulation:

Sudden stress may stimulate release of steroids similar to estrogen

Drug Reaction:

Associated with Beta-Blocker therapy.
Via reduction or inhibition of enzyme systems that break down hyaluronic acid.
Via physiologic dysfunction of Rouget cells in capillary walls, abnormal permeability.
Increase in Mast Cells producing exess hyaluronic acid.
Fibroblast stimulation->increased hyaluronic acid->increased procollagen & mRNA.

Chemical Mediated Injury, RX therapy, or Infection:

Prostaglandins, Adenosine 3'5'cyclic phosphate and endotomin mediated.
Restricted location of scleredema involvement suggests localized circulating stimulator.
Chronic immunostimulation leading to monoclonal gammopathy;antibodies directed against an unknown pathologic substance may stimulate synthesis of hyaluronic acid by fibroblasts.

J Am Acad Dermatol 1992;26:319-321

IV. Differential Diagnosis:

Sclerederma
Pseudosclerederma [Werner's Syndrome]
Vinyl Chloride Disease
Porphyria Cutane tarda
Scleromyedema

III. Histopathological Findings:

a. Increased thickness of dermis with dense fibrosis and fenestration between collagen bundles.
b. Direct Immunoflourescent is Negative
c. Toluidine Blue staining for mucin deposition is Negative
d. Alcian blue staining demonstrates scant deposits of fibrillar material.
d. Increased Acid mucopolysaccaride staining (J Cutan Pathol 1990;17:211-213)
e. Focal lymphocytic perivascular infiltrate in papillary dermis

IV. Scleredema and Monoclonal Gammopathy

The possible role of monoclonal immunoglobulins in the pathogenesis of scleredema is unclear but it seems as though both conditions may be related. It has been demonstrated that the serum from patients with scleredema and monoclonal gammopathy stimulates collagen production in dermal fibroblast cultures, which suggests that circulating serum factors, possibly related to paraproteins may enhance the synthesis of extracellular macromolecules within the dermis.

A. MGUS (IgG K light chains)
B. Multiple Myeloma (IgG K, IgG v & IgAK)

V. Therapy:

- No treatment is known to modify the course of scleredema.
- Some cases resolve spontaneously
- Electron- Beam Therapy has been reported in a patient with IgA gammopathy
- Improvement noted in patient's with Multple Myeloma following Chemotherapy.

VI. References:

Clin Exp Dermatol 1995;20:149-52
Clin Exp Dermatol 1988;13:271-274
J Rheumatol 1988;15:202-20
J Am Acad Dermatol 1992;26:319-321
J Cutan Pathol 1990;17:211-213
J Am Acad Dermatol 1984;11:134

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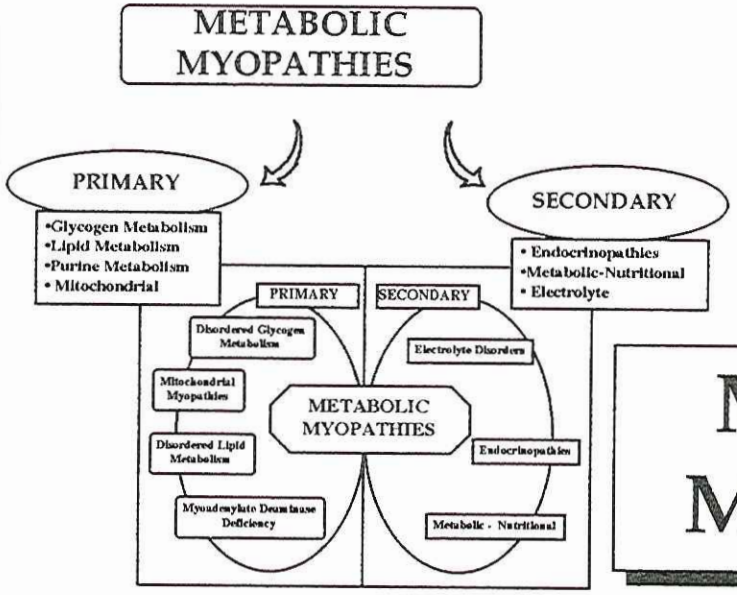
METABOLIC MYOPATHIES

DEFINITIONS:

- **Metabolic Myopathy**
  - Heterogenous group of conditions that have in common abnormalities in muscle energy metabolism that result in skeletal muscle dysfunction.



Glycogen Storage Diseases

McArdle's Disease

- Myophosphorylase deficiency
- Abnormal deposition and accumulation of glycogen in skeletal muscle
- Exercise intolerance: pain, fatigue, weakness
- Elevated CPK
- EMG Findings:
  - increased insertional irritability
  - increased numbers of polyphasic motor unit potentials
  - Fibrillations and positive waves
- Diagnosis:
  - Muscle biopsy: glycogen deposition, fiber necrosis and phagocytosis
  - Forearm ischemic exercise test

Pompe's Disease [Acid Maltase]

- Alpha-1,4 glucosidase [Acid maltase] deficiency
- Abnormal deposition and accumulation of glycogen in skeletal muscle
- Cardiomegaly, hypotonia, Respiratory failure
- Proximal muscle weakness after 20years.
- Elevated CPK
- EMG Findings:
  - intense electrical irritability
  - Myotonic discharges in absence of myotonia
- Diagnosis:
  - Muscle biopsy: glycogen deposition, fiber necrosis and phagocytosis
  - Forearm ischemic exercise test: not useful

Mitochondrial Myopathies

- Morphological abnormalities in number, size, or structure of mitochondria.
- Inherited via maternal transmission
- Associated metabolic derangements
  - Nutrient transport defects - Beta Oxidation
  - Pyruvate & Acetyl-CoA processing - Electron transport
- Clinical spectrum:
  - Progressive proximal myopathy
  - External ophthalmoplegia - Multisystem disease
- Medications: AZT

Myoadenylate Deaminase [MADA] Deficiency

- MADA: Isoenzyme of adenylate deaminase found only in skeletal muscle.
- Primary Form vs Secondary Form
- Common:
  - 21% of muscle biopsies
  - 22% of normal individuals
- MADA: catalyst for deamination of AMP -> IMP
  - in this conversion ammonia is released and stimulates glycolysis by activating phosphofructokinase.
- Diagnosis:
  - Forearm Ischemic Exercise testing:
    - Lactate mildly elevated / Ammonia levels do not increase
  - Muscle bx: Histochemical absence of myoadenylate deaminase

Primary Form:

- **Autosomal Recessive**
- **Exercise intolerance:**
  - pain/cramping
  - decreasing excercise tolerance
  - Asymptomatic at rest
- **CPK & EMG = Normal**
- **Forearm ischemic test:**
  - mildly elevated lactate
  - ammonia does not increase

Arthritis Rheum 1987;30:661-8

Secondary Form:

- Acquired

Associated Neuromuscular Diseases:

- Duchenne's dystrophy
- Becker's dystrophy
- Limb-girdle dystrophy
- Spinal muscular atrophies
- DM
- SLE
- Periodic paralysis
- Polymyositis / Dermatomyositis
- Systemic Sclerosis
- Amyotrophic Lateral Sclerosis
- Hyperthyroidism
- Gout

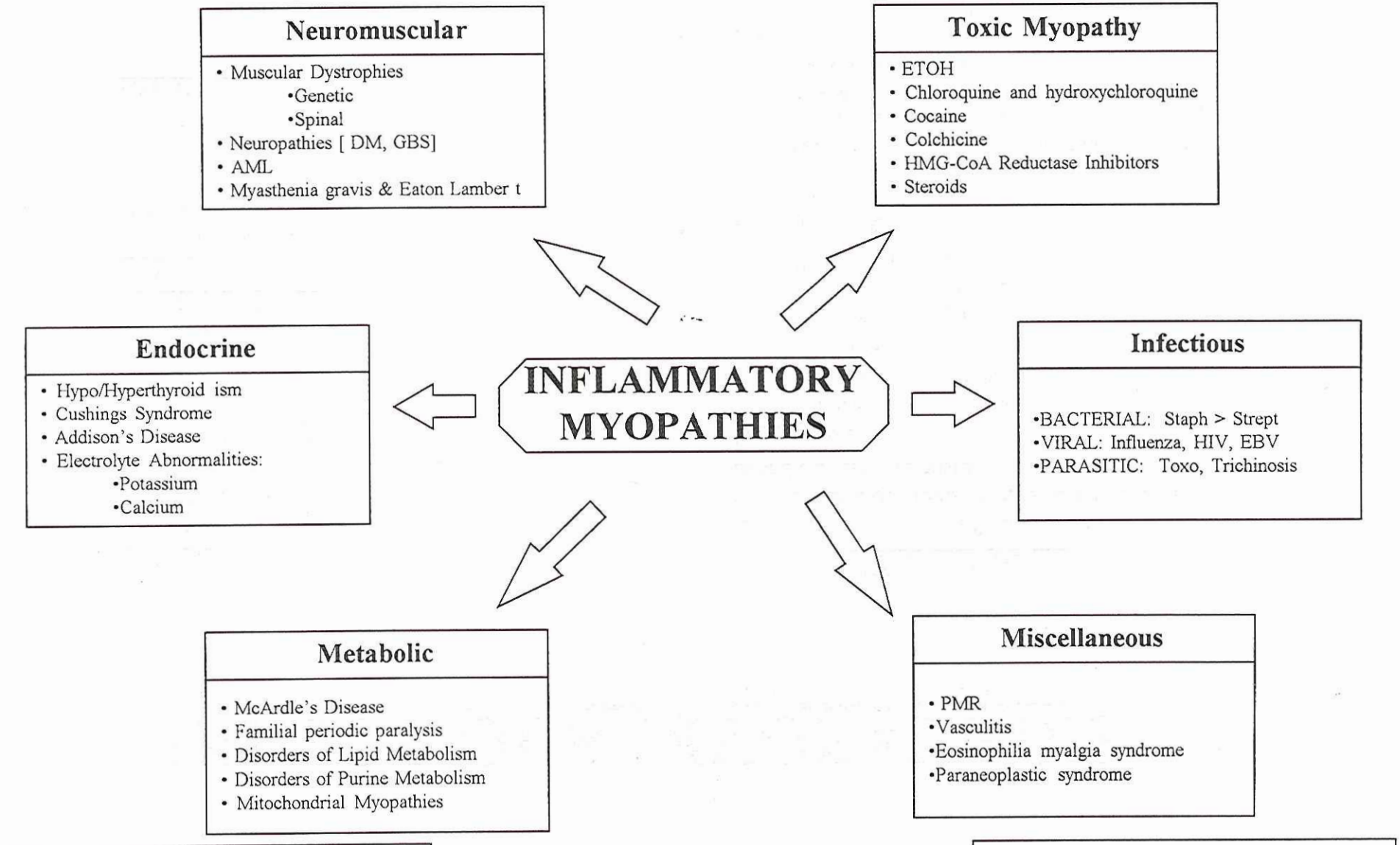
Neurology 1982;32:857-63

Lipid Metabolism Disorders

Carnitine deficiency

- Inherited or Acquired: abnormal lipid deposition
- CPKs elevated
- EMG: polyphasic MUP of small aplitude & short duration.
- Treatment: L-Carnitine/ Steroids /propranolol
- Primary Carnitine deficiencies:
  - A. *Systemic Type*: Early development -> Coma / Hypoglycemia
  - B. *Muscle Type*: Later childhood : facial & pharyngeal muscles
- Acquired Carnitine deficiencies:
  - Pregnancy - Pregnancy - Myexedema
  - ESRD & HD - Adrenal Insufficiency
  - Medications: Valproate / Pirampicim

DIFFERENTIAL DIAGNOSIS OF IDIOPATHIC INFLAMMATORY MYOPATHY



REF: Primer on Rheumatic Diseases. 10th Edition

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INFECTIOUS MYOPATHIES

- **BACTERIAL**
  - Staphylococcus aureus
  - Group A streptococcus
  - Clostridial spp
  - Haemophilus Influenza
  - Escherichia coli
  - Klebsiella spp.
  - Yersinia enterocolitica
  - Pseudomonas spp.
- **MYCOBACTERIA**
- **FUNGI**
  - Aspergillus fumigatus
- **VIRAL**
  - Influenza
  - HIV
  - EBV
- **PARASITIC**
  - Toxoplasmosis
  - Trichenosis
  - Schistosomiasis
  - Cysticercosis

PYOMYOSITIS

- Definition: A bacterial infection of muscle occurring in the absence of a redisposing site of infection. Blood Cultures are positive in less than 5% of cases.
- Pathogenesis:
  - Penetrating wound infection
  - Prolonged vascular insufficiency
  - Contiguous Infection
- Most Common Organism:
  - *Staphylococcus aureus* [95%]
  - Group A Streptococcus [1-5%]
- Clinical Findings:
  - All ages at risk.
  - Onset is often subacute with localized muscle pain, then swelling with induration and tenderness.
  - Most common sites of infection: Lower Limb and Trunk muscles.
  - Leukocytosis and Eosinophilia are common.
- Therapy:
  - Surgical drainage of all Abscesses
  - Identification and speciation of organism
  - Parenteral Antibiotic Therapy

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Antisynthetase Syndrome

I. Autoantibodies

Autoantibodies to nuclear and cytoplasmic antigens are found in a high proportion of patients with polymyositis (PM) and dermatomysoitis (DM), and they are some of the clearest indications of the autoimmune origin of these diseases. Some of these autoantibodies are found exclusively or most commonly in PM-DM or PM-DM overlap syndromes and referred to as myositis-specific autoantibodies (MSA). *Rheum Dis Clin N Amer 1992;18:455-482*

II. Myositis Specific Autoantibodies: (*Clinical Aspects of Autoimmunity 1993;5:1-18*)

TABLE 1

AUTOANTIBODIES IN POLYMYOSITIS AND DERMATOMYOSITIS

Antibody	Antigen	Subunit molecular weight (kD)	PM/DM patients (%)	Subgroup association	Comment
Anti-synthetases					
Anti-Jo-1	Histidyl-tRNA synthetase	50	18	PM/DM,ILD*	35% in PM dimer
Anti-PL-7	Threonyl-tRNA synthetase	80	3	PM/DM,ILD*	
Anti-PL-12	Alanyl-tRNA synthetase	110	3	PM/DM,ILD*	Sera also react with tRNA**
Anti-OJ	Isoleucyl-tRNA synthetase	145	<3	PM/DM,ILD*	Part of multi-enzyme complex
Anti-EJ	Glycyl-tRNA synthetase	75	<3	PM/DM,ILD*	
Antibodies to non-synthetase cytoplasmic antigens					
Anti-SRP	Signal recognition particle	54	4	PM	Particle of six proteins and 7SL RNA
Anti-KJ	Unidentified translation factor	?	<1	PM,ILD*†	
Anti-Fer	Elongation factor 1α	48	<1	Nodular myositis†	
Anti-Mas	Unidentified tRNA	?	1	Alcohol, myositis†	
Antinuclear antibodies:					
Anti-Mi-2	Nuclear protein complex	?	5	DM	15% - 20% in DM, rare in PM
Anti-PM-Scl	Nucleolar protein complex	100 & 70	8	PM/DM/SSc overlap	Complex of 11-16 proteins
Anti-56 kD	Component of ribonucleoprotein	56	87	DM,PM	Common in all subgroups
Found in myositis as part of overlap or other primary association					
Anti-Ku	DNA-binding protein	70,80	<5	PM/SSc overlap	Common in SLE
Anti-U1RNP	U1 small nuclear ribonucleoprotein	70	12	SLE overlap	Common in SLE and overlap
Anti-U2RNP	U2 small nuclear ribonucleoprotein	B" and A'	1	Overlap	Most also have anti-U1RNP
Anti-Ro/SSA	Ribonucleoprotein	60	10	SLE, Sjögren's syndrome	
(w/ anti-La/SSB)		48	<5	SLE, Sjögren's syndrome	
Other autoantibodies					
Anti-cytoskeletal antibodies			86	Not disease-specific	
Anti-myosin			90	Myositis, muscle disease	
Anti-myoglobin			71	Myositis, muscle disease	

*Associated with features of anti-synthetase syndrome
†More patients needed to establish association

III. Antibodies to Aminoacyl-tRNA Synthetases:

The aminoacyl-tRNA synthetases are a family of enzymes, each of which catalyzes the formation of aminoacyl-tRNA from a specific amino acid and its cognate t-RNAs. These autoantibodies occur in the sera of 25-40% of adult patients with idiopathic inflammatory myopathies, polymyositis and dermatomyositis. Anti-Jo-1 (antihistidyl-tRNA synthetase) was the first of these antisynthetases to be discovered and is detectable in approximately 15-30% of myositis patients. Subsequent antibodies discovered include:

- Anti-PL-7 (antithreonyl-tRNA synthetase)
- Anti-PL-12 (antitialanyl-tRNA synthetase)
- Anti-OJ (anti-isoleucyl-tRNA synthetase)
- Anti-EJ (antiglycyl-tRNA synthetase)

Individuals usually produce only one of these autoantibodies, but when present they are usually considered highly specific for an inflammatory myopathy.

Semin Arthritis Rheum 1996;26: 459-467

IV. Antisynthetase Syndrome:

A characteristic group of clinical features occur more frequently in patients with anti-synthetase antibodies.

Clinical Features:

Myositis
Interstitial Lung Disease
Arthritis
Raynauds Phenomenon
Fever
Mechanic's hands
Predominately young patients
Onset early in calender year
HLA DR3,Drw52

Table 2: Clinical Course of Patients With Antisynthetase Antibodies and Interstitial Lung Disease

Patient no.	1	2	3	4	5	6	7	8	9	10
Duration of follow-up (yr)	0.04	3.5	3.5	1.25	13.5	1.5	9.5	0.04	3	1
Treatment(s)	Pred, Ctx	Pred	Pred	Pred, Ctx	Pred, Aza	Pred	Pred, Aza	Pred	Pred, Ctx	None
Outcome										
Current status	Unknown	Living	Living	Living	Living	Deceased	Living	Deceased	Living	Living
ILD	Unknown	Resolved	Stabilized	Improved × 6 months, then progressed	Improved × 11 years, then progressed. Improved after re-starting Pred & Aza	Progressed	Resolved	Progressed	Progressed	Stable

Abbreviations: Pred, predisone; Ctx, oral cyclophosphamide; Aza, azathioprine; ILD, interstitial lung disease.

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IMMUNE THROMBOCYTOPENIC PURPURA

In general, thrombocytopenia is due to one of three mechanisms: reduced production, sequestration, or increased platelet destruction. In outpatients with rheumatological disorders, the most common cause of thrombocytopenia is immune thrombocytopenia. *J Clin Rheumatol 1997;3:95-101*

I. Pathogenesis:

A. Autoantibodies:
- antibody directed against platelet glycoproteins
  Ib, IIb/IIIa, Ia/IIa, IX

B. Attachment of the antibody to the platelet surface results in clearance of the PLT via fragment [Fc] receptor mediated uptake by phagocytic cells in the spleen and liver.

II. Classification of Immune Thrombocytopenia:

A. Acute:
B. Chronic: Greater than 6 months.

Primary
- Idiopathic immune thrombocytopenia purpura

Secondary
- Rheumatic:
  SLE
  RA
  MCTD
  Chronic active hepatitis
- Infectious Diseases
  HIV
  Viral [EBV/CMV/Adenovirus]
  Mycobacterium
- Malignancy
  CLL
  NHL
  HD
- Drugs
  Gold
  Quinidine
  Heparin

III. Diagnosis:

A. Clinical presentation:
- Bleeding /bruising
- splenomegaly

B. Evans Syndrome:
- ITP with AIHA
- Lymphoproliferative disorders / SLE

C. Antiplatelet Antibodies:
- Antiplatelet glycoprotein autoantibodies: Ib, IIb/IIIa, Ia/IIa & IV
- Monoclonal antibody immobilization of platelet antigen [MAIPA] assay
- Antiphospholipid antibodies / aCL

D. Bone Marrow Biopsy [BMBX]
- Not required for diagnosis
- Normal to increase levels of megakarocyte production.

IV. Treatment:

A. Corticosteroids: 1mg/kg/d
- -Complete Remission [CR]: <20%
- Lowest possible dose and QOD dosing when stable

B. Splenectomy
- CR @ 75% inclusive of patients with SLE

C. Intravenous Immunoglobulin [IVIG]
- Drug related Fc receptor blockade --> decreased immune-coated PLT removal.
- 1g/kg q 24-48hrs [infused over 6-8hrs] --> PLT peak in 2wks
- Response rate = 75% [transient]

D. Rh Immunoglobulin Therapy [Anti-D]

E. Transfusion therapy:
- serious or life threatening hemorrhage

F. Danazol [800-1200mg/day]
- Androgenic corticosteroid
- Decreases Fc receptor function / Stimulates PLT production/ Modulation of T cell fxn
- May be more effective in SLE

G. Colchicine

H. Cyclophosphomide [1-2 mg/kg/d]

I. Imuran

V. References:

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RELAPSING POLYCHONDRITIS

Relapsing polychondritis is a rare disease of unknown etiology characterized by episodic attacks of inflammation of hyaline and elastic cartilagenous structures including the ears, nose, laryngotracheal, and articular cartilages, as well as the organs of special sense. It is thought to be an autoimmune disease based upon the observed association with other conditions such as vasculitis, the presence of cartilage inflammation on biopsy, anticollagen antibodies and a therapeutic response to corticosteroid therapy.

I. Associated Conditions

- Systemic vasculitis syndromes
- RA
- Ankylosing spondylitis
- Reiter's Syndrome
- Bechet's syndrome
- Dermatitis herpetiformis
- Panniculitis
- Retroperitoneal fibrosis
- SLE
- Sjogren's Syndrome
- Psoriasis
- Inflammatory bowel disease
- MDS
- Hodgkin's Disease
- Primary biliary cirrhosis
- Thymoma

II. Clinical Manifestations:

Manifestation	Presenting [%]	Cumulative at 6yrs: [%]
Auricular condritis	39	85
Hearing loss	9	26
Vertigo	4	13
Nasal condritis	24	54
Saddle nose	18	29
Laryngealtracheal Sx	26	48
Stricture	15	23
Inflammatory arthritis	36	52
Valvular regurgitation	0	4
Aneurysm	0	4
Microhematuria	15	26
Elevated creatinine	7	13
Cutaneous	7	28

Ann Intern Med 1986;104:74-78

A. Auricular chondritis is most common manifestation: >85%
- Unilateral / Bilateral —-> Simultaneous or sequential pattern
- Inflammation occurs in remitting attacks lasting days to weeks
- Violaceous or erythematous tender swelling of cartilagenous pinna with sparring of soft lobule.
- EAM: narrowing
- Hearing loss: Conductive and sensorineural loss

B. Cardiovascular [10%]
- Valulitis with ruputure
- Complete heart block
- Aortitis
- Aneurysm

II. Clinical Manifestations: [cont.]

C. Ocular Involvement: [50%]
- Episcleritis/ Scleritis -Uveitis
- Proptosis -SICCA

D. Neurological Involvement
- Associated with vasculitis? - CN2/3/6/7/8
- Mononeuritis multiplex - Sensory peripheral neuropathy

E. Dermatological:
- LCV [5%] -Urticaria -Angioedema
-Erythema multiforme - Livedo reticularis - E. nodosum
- Panniculitis -Migratory superficial thrombophlebitis

III. Diagnostic Criteria for Relapsing Polychondritis:

Criterion*	Description
1	Recurrent chondritis of both auricles
2	Nonerosive inflammatory polyarthritis
3	Chondritis of nasal cartilage
4	Inflammation of ocular structures, including conjunctivitis, keratitis, scleritis/episcleritis, and or uveitis.
5	Chondritis of the respiratory tract involving laryngeal a/o tracheal cartilage
6	Coclear and or vestibular damage manifest by neurosensory hearing loss, tinnitus and or vertigo.

* RP can be diagnosed with certainty when three or more of the criteria are present, with biopsy confirmation from a clinically involved site. *McAdam LP et al. Medicine 1976;55:193-215*

IV. Laboratory studies:
A. Nondiagnostic / Nonspecific: ESR elevation, NCNC Anemia, Leukocytosis, Thrombocytosis
B. Antibodies to both native and denatured Type II Collagen [IgG] @ 50%: Disease activity???
C. Radiographs / CT scans: Use in assessment of airway patency
D. Biopsy: Loss of basophilic staining of matrix / fuse PMN/Mono infiltration / collagen matrix destruction and replacement of normal cartilated by fibrous granulation tissue.

V. Prognosis:
A. 5 year prognosis: 74% 10yrs: 55%
B. Decrease survival: Older age / anemia / Laryngotracheal stricture

VI. Treatment:
A. There is no standard medical therapy
B. Prednisone: 0.5 - 1mg/kg/d —> Pulse Solumedrol in Life threatening disease
C. Dapsone: 500-200mg/d
D. Immunosuppressive Rx: Cytoxan / cyclosporine A / MTX have been reported
E. Tracheostomy
F. Valve replacement

References:
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PULMONARY ALVEOLAR HEMORRHAGE

Pulmonary vascular inflammatory disorders may involve all components of the pulmonary vasculature, including capillaries. The principal histological features of pulmonary capillaritis include capillary wall necrosis with infiltration by neutrophils, interstitial erythrocytes, and or hemosiderin, and interalveolar septal capillary occlusion by fibrin thrombi. Immune complex deposition is variably present. Patients with pulmonary capillaritis often present clinically with diffuse alveolar hemorrhage, which is characterized by dyspnea and hemoptysis; diffuse , bilateral alveolar infiltrates on CXR; and anemia. Whereas diffuse alveolar hemorrhage is often diagnosed clinically, capillaritis is strictly a pathologic diagnosis and suggests an underlying systemic vasculitis.

Chest 1996;110:1305-16

I. Etiologies of Diffuse Alveolar Hemorrhage:

Etiologies

Pulmonary Capillaritis described in this setting
- Antiphospholipid Syndrome
- Behcet's Syndrome
- Diphenylyhydantion
- Goodpasture's Syndrome
- HSP
- Wegener's Granulomatosis
- Microscopic PAN
- IgA Nephropathy
- Systemic Lupus Erythematosus

Pulmonary Capillaritis not described in this setting
- Bone marrow transplantation
- Cardiac amyloidosis
- DIC
- Essential Mixed Cryoglobulinemia
- Idiopathic pulmonary hemosiderosis
- Inhalation injury
- Lymphangiography
- Mitral Stenosis
- Penicillamine
- Pulmonary Angiosarcoma
- Tuberous sclerosis

II. Clinical Features of Diffuse Alveolar Hemorrhage

- Dyspnea
- Hemoyptysis
- Diffuse crackles
- Diffuse alveolar filling pattern on CXR
- Hypoxemia
- Anemia
- Extrathoracic evidence of systemic vasculitis
- Adequate platlet number and function
- No evidence of marked coagulation defect

III. Diagnosis:

A. Appropriate Clinical Setting:
- Dyspnea / Hemopytysis / Anemia / Alveolar infiltrates / Hypoxemia

B. Marked Elevated Carbon Monoxide diffusing capacity

C. Bronchoscopy with BAL:
- Aspiration of successive aliquots of increasingly bloody BAL fluid.
- Hemosiderin ladin macrophages

D. Recommended ancillary studies:

- ANA	- ANCA	- dsDNA
- CH50	-RF	- AGBM AB
-Cryoglobins	- ESR	- Urinary sediment
- Renal BX	- Skin BX	

III. Treatment:

A. Pulse dose corticosteroids: [1gm/d x 3d]

B. WG Protocol: Corticosteroids [1-2mg/kg/d + Cyclophosphamide [2-5 mg/kg/d]

C. Plasmaphoresis

D. PCP prophylaxis [Bactrim/Septra DS BID QMWF]

IV. References:

1. Orens JB et al. Pleuropulmonary manifestations of systemic lupus erythematosus Rheum Dis Clin North Am 1994;20:159-93

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GASTROINTESTINAL MANIFESTATIONS OF SJOGREN'S SYNDROME

Sjogren's Syndrome is the result of lymphocyte-mediated destruction of exocrine glands. The gastrointestinal tract consists of a nine meter long canal that is lined almost entirely with mucous membrane. Through this membrane, a great number of small endocrine glands pour their secretions into the lumen. In addition, the major salivary glands, the pancreas, and the liver contribute their secretions to the luminal content. *Rheum Dis Clin North Am 1992;18:617-631*

I. Anatomical Classification of GI manifestations in Sjogren's Syndrome:

A. Salivary Gland:
- Mucosal dryness: xerostomia
- Mucosal Atrophy
- Accelerated dental decay

B. Stomach:
- Nausea
- Epigastric pain
- Dyspepsia
- Chronic atrophic gastritis
  - decreased serum pepsinogen increased gastrin levels
  - Achlordydria

C. Pancreas:
- Sublinical pancreatitis
- Acute Pancreatitis
- Chronic Pancreatitis

D. Esophagus:
- Dysphagia
- Mucosal atrophy

E. Small Bowel:
- Malabsorption
- IBD?

F. Liver:
- Abnormal biochemical function
- Hepatomegaly
- Histological abnormalities:
  - Primary biliary cirrhosis
  - Mild portal tract fibrosis
  - Chronic active hepatitis

G. Colon:
- Inflammatory bowel disease
- Ischemic colitis?

II. Vasculitis in Sjogren's Syndrome:

The incidence of vasculitis in SS ranges from 13-33% and cutaneous vasculitis is the most common. There is a striking association between antibodies to Ro [SSA] and the occurance of vasculitis in SS.

A. Histological categories:

1. Acute necrotizing vasculitis
2. LCV
3. Lymphocytic vasculitis
4. Endarteritis obliterans

B. Isolated Necrotizing and Granulomatous Vasculitis causing Ischemic Bowel Disease in Primary Sjogren's Syndrome. *J Rheumatol 1995;22:2375-2377*

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NEUROSARCOIDOSIS

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Rheumatology and Clinical Immunology
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Sarcoidosis is a multisystem granulomatous disorder of unknown etiology. It is characterized by enhanced cellular immune processes at the sites of disease activity.[1,6] Neurosarcoidosis develops in 5% of patients with sarcoidosis.[2,6] Central nervous system involvement may be the presenting symptom in 50% or more of patients with neurosarcoidosis.[6] Approximately 50-70% of patients who develop neurosarcoidosis do so within the first two years of their systemic illness.[2,6]

I. Clinical Presentation: [4]

Neurological Manifestation	Frequency [%]
Cranial Neuropathy	50-75
Facial paresis [CN VIII/V/II]	25-50
Aseptic Meningitis	10-20
Hydrocephalus	10
Parenchymal Disease	
- Endocrinopathy	10-15
- Mass Lesion	5-10
- Encephalopathy	5-10
- Vasculopathy	5-10
- Seizures	5-10
Peripheral Neuropathy	6-18
Myopathy	9-23

II. Diagnosis:

1. Neuroradiology:
   - Enhanced MRI preferred over CT scan.
   - Most common CT finding is periventricular white matter lesions.
   - Sarcoid lesions are usually well-defined, slightly hyperdense, homogenously enhancing lesions with varying degrees of edema.
2. Cerebrospinal Fluid:
   - Findings may be nonspecific or normal
   - Elevated Protein and mild pleocytosis
   - Elevations in CSF ACE/ Lysozyme [LZM]/ B2Microglobulin > 50%
   - Increased IgG-index and oligoclonal bands
   - Single studies on CSF lymphocytes: elevated T4/T8 ratios
3. Neurophysiology:
   - Visual evoked potential [VEP]
   - Brain stem auditory evoked potential [BAEP]

III. Treatment:

1. Corticosteroid Therapy:
   - Suppress elevated CD4/CD8 lypmphocyte ratio
   - Decrease Interleukin -2 production
   - Inhibit Collagen synthesis
   - Prednisone 40-80mg QD

2. Cyclospoine A
   - Inhibits amplification of T-cell immune mechanism
   - Used as steroid reducing agent in refractory neurosarcoid
   - 4-6mg/kg /day
   - Expensive, avoid in severe HTN or renal disease

3. Azathioprine
   - Moderate cost
   - monitor for neutropenia
   - avoid in liver or hematological disorders

4. Radiation Therapy
   - Use in life-threatening CNS disease or failure of other therapy.

5. MTX
6. Cyclophosphamide
7. Chlorambucil

Avoid in Severe Pulmonary or hematological disease.

References:

1. Johns CJ ed. Tenth Conference on sarcoidosis and other granulomatous disorders. Ann NY Acad Sci.1986;465:1
2. Stern BJ et al. Sarcoidosis and its neurologic manifestations Arch Neurol 1985;42:909-917
3. Scott, TF. Neurosarcoidosis: Progress and clinical aspects. Neurology 1993;43:8-12
4. Oksanen, V. Neurosarcoidosis. Sarcoidosis1994;11:76-79
5. Stern, BJ et al. The treatment of Neurosarcoidosis with Cycloporine. Arch Neurol 1992;49:1065-1072
6. Agbogu, BN et al. Therapeutic Considerations in patients with refractory Neurosarcoidosis. Arch Neurol 1995;52:875-79.
7. Sharma, OP. Sarcoidosis of the Nervous System. A clinical approach. Arch Int Med 1991;151:1317-21

D. DRUGS

"Healing is a matter of time, but it is sometimes also a matter of opportunity"
-Hipprocrates

OUTLINE:

D1. Adrenocorticotropic Hormone (ACTH)
D2. Calcitonin
D3. Cyclophosphamide
D4. Etanercept
D5. Glucocorticoids
D6. Minocycline
D7. Immunosuppressive drugs in Pregnancy

Frederick T. Murphy
NIH ARB Rounds
30 Jan 1998

CORTICOTROPIN
(ADRENOCORTICOTROPIC HORMONE)

Corticotropin (ACTH) is a polypeptide secreted by the basophilic cells of the anterior pituitary gland. The polypeptide which containes 39 amino acides, has a MW of about 4500. Only the first 24 amino acids from the N-terminal end are required for full biologic activity. The sequence of these 24 amino acids is the same in humans,cows,pigs and sheep. For commercial use, ACTH is extracted from the pituitaries of mammals used for food by humans. ACTH via cAMP controls the initial rate limiting step in steroidogenesis from cholesterol and leads to the the synthesis of adrenocortical hormones.

I. Pharmacology.

Exogenous corticotropin elicits all the pharmacologic responses usually produced by endogenous corticotropin. In patients with normal adrenocortical function, ACTH stimulates the adrenal cortex to secrete cortisol (hydrocortisone), corticosterone, several weekly androgenic substances and aldosterone. The primary physiologic and pharmacologic effects of corticotropin result from secretion of cortisol.

A. ACTH 100 IU ----> Cortisol 100mg secretion over 16 hours.

B. Peak plasma concentrations: 1 hour with decline in 2 hours.

II. Effects of Glucocorticoids on Modulators of Inflammation:

A. Inhibited by Glucocorticoids: IL-1,IL-2,IL-3,IL-4,IL-5,I-L6,IL-8, TNF, IFN, GM-CSF
Prostaglandins and Leukotrienes
COX-2
Inducible nitric oxide synthase & Plaminogen activator

B. Stimulated by Glucocorticoids: Lipocortin-1, Neutral endopeptidase, Vasocortin

III. Anti-inflammatory Effects of Glucocorticoids in Cells:

Cell	Effects
Neutrophils	Increased numbers, decreased trafficking, relatively unaltered function
Macrophages & Mononcytes	Decreased numbers, decreased trafficking, decreased phagocytosis and bactericidal effects, inhibited antigen presentation, decreased cytokine release, decreased eicosanoid release.
Lymphocytes	Decreased numbers & trafficking, decreased cytokine production, decreased proliferation and impaired activation; little effect on immunoglobulin synthesis.
Eosinophils	Decreased numbers, increased apoptosis
Basophils	Decreased numbers, decrease release of mediators

IV. Indications for Corticosteroids in Gout (J Clin Rheumatol 1997;3:35-40)

A. Comorbid conditions:

- Cardiac Failure
- Renal Failure
- Peptic Ulcer Disease
- Hepatic Insufficiency
- Bleeding Diathesis
- HTN
- ETOH
- GI Bleeding
- Anticoagulation therapy

B. Advanced Age

C. Postoperative State

D. Severe attacks refractory to NSAIDs & Colchicine

F. NSAID Hypersensitivity

V. Clinical Considerations

A. 1950: Wolfson et al, Gutman & Yu : (Amer J Med 1950;9:24)

- 11 cases of polyarticular, colchine resistant cases responsive to ACTH
- 40 patients treated with concomitant colchine --> 60% complete remission in 24-48hrs.

B. Smith et al. (N Engl J Med 1987;317:1266)

- Demonstrate that peripheral blood mononuclear leukocytes possess ACTH binding sites on their surface membranes.
- Researchers hypothesize that this receptor may modulate ACTH's role in gout.

B. 1988: Axelrod & Preston (Arthritis Rheum 1988;31:803)

- Randomized 100 male patients (ACTH 40 IU x1 vs Indocin 50 QID)
- Faster relief with ACTH than Indomethacin.

C. 1990: Groff et al. (Semin Arthritis Rheum 1990;19:329)

- Compared ACTH with Corticosteroids in polyarticular gout
- Time to resolution: ACTH (5.5days) vs Triamcinolone (7.5days)
- Steroids unlike ACTH have potential for adrenal suppression.

D. 1994: Ritter et al. (J Rheumatol 1994;21:696-9)

- ACTH safe and effectiv in 38 pts with MMP with gout or pseudogout.

VI. Dosage and Administration (GOUT)

A. Injection, repository for IV, IM or SQ. (IM or SQ most common route of administration)

B. 40 -80 IU IM or SQ q 24-72hours

C. Commitant administration of colchicine may decrease relapse rate.

References:

Smith MD et al. The clincial and immunolgicaleffects of pulse methylprednisolone therapy in rheumatoid arthritis. J Rheumatol 1993;20:111

Weusten BLAM et al. Corticosteroid pulse therapy in active rheumatoid arthritis. Semin Arthritis Rheum 1993;23:183-192.

Fauci A S. Alternate - day corticosteroid theapy. Am J Med 1978;64:729

Wilder, RL. Arthritis and Allied Conditions: a textbook of rheumatology. 13th edition. 1997; 731-51. Williams & Wilkins. Baltimore MD

Axelrod D, Preston S. Comparison of parenteral adrenocorticotropic hormone with oral indomethacin in the treatment of acute gout. Arthritis Rheum 1988;31: 803.

Ritter J. Kerr LD, Valerian-Marcet J. ACTH revisited: Effective treatment of acute crystal induced synovitis in patients with multiple medical problems. J Rheumatol 1994;21:696

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CALCITONIN

CALCITIONIN HORMONE

- 32 amino acid hormone
- Parafollicular C cells
- Blunts acute episodes of hyperCA
- BASAL CALCITONIN LEVEL
  - Men > Women
  - decrease with age

CALCITONIN ACTIONS:

- INHIBITS OSTEOCLAST FXN
- Causes:
  - Calciuria
  - Natriuria
  - Phosphaturia
- Modulates nocioception
- Modulates appetite
- Increases renal hydroxylation of vitamin D.

CALCITONIN RECEPTORS

- Mature Osteoclasts
- Renal epithelium
- Gastric and intestinal crypt cells
- Hypothalamus (median eminence)
- Malignant Cells (Breast, Lung)

SIDE EFFECTS

INJECTABLE CALCITONIN

- Anaphylaxis /Allergy (rare)
- Gastrointestinal: (10%)
- Dizziness / flushing (2-5%)
- Injection site reactions (10%)
- Antibody-mediated Calcitonin resistence

NASAL CALCITONIN

- Rhinitis (12%)
- Nasal mucosal irritation(11%)
- Dizziness / flushing (rare)
- Antibody production (unknown)

CALCITONIN INDICATIONS

- FDA Approved
  - Hypercalcemia
  - Paget's disease (mod-severe)
  - Postmenopausal osteoporosis (≥5yrs)
- Additional Indications:
  - Early estrogen intolerant osteoporosis
  - Male osteoporosis
  - Analgesia (compression Fx)
  - Cotherapy in steroid dependence

ADMINISTRATION

- Subcutaneous / Intramuscular
  1. 200IU/cc in 2cc bottles - refrigerate
  2. test dose: 10 IU in ss- observe 20min
  3. SIG: 0.3/0.5cc/d (60/100 IU/d) SQ QHS
- Nasal Spray
  1. 200IU/puff - refrigerate until opened
  2. Single daily spray alternating nares
  3. Cotherapy with calcium & vit D

J CLIN RHEUMATOL 1997;3:S40-S45

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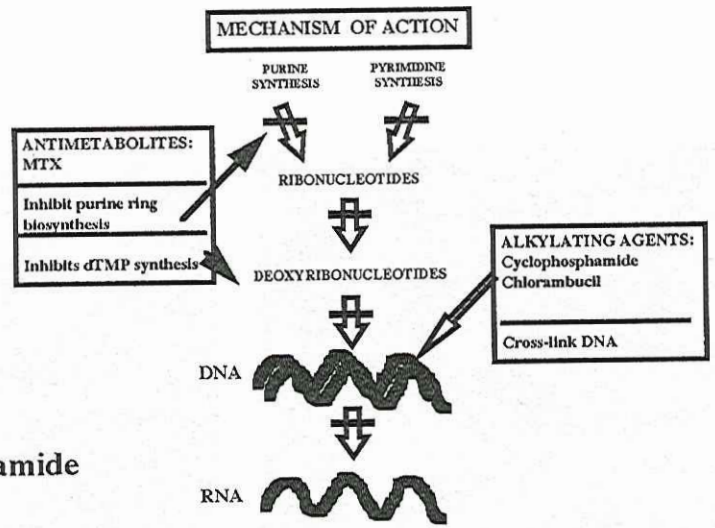
Cyclophosphamide
(Pharmacokinetics)

- T 1/2: 2- 10 (7)hrs
- Protein bound: 12-14%
- Peak Plasma []: 1hr
- Metabolism:

  Inactive -> P450 -> phosphoramide mustard & acrolein



Cyclophosphamide

Administration:

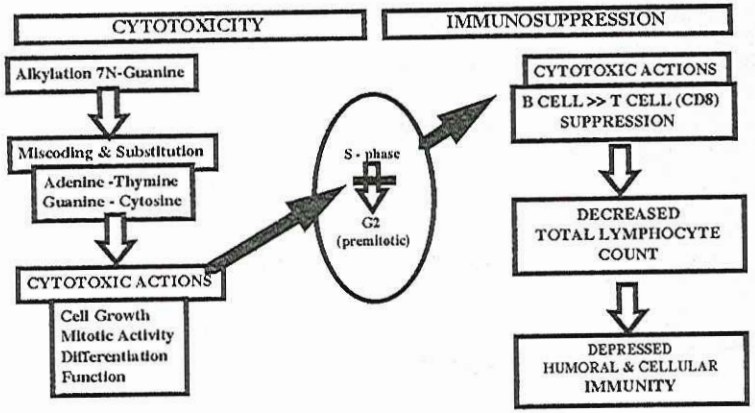
- Parenteral: (750 - 1000)mg/m² IVPB
- Oral: 2 - 4 mg/kg/d

DISEASE STATES:

- –Wegener's Granulomatosis
- –SLE (Renal / CNS /Vasculitis)
- –Systemic Vasculitis
- –PAN
- –RA (Vasculitis)
- –Behcet's Disease

CYCLOPHOSPHAMIDE

Cyclophosphamide



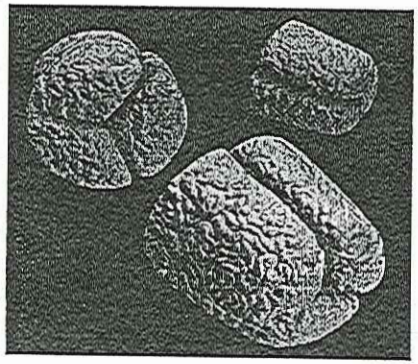
Cyclophosphamide

Toxicities:

- –Nausea & Vomiting/ Alopecia
- –Teratogenicity
- –Bone marrow suppression (WBC>PLT)
  - –Dose related
- –Infections
- –Gonadal failure: ovarian failure / azoospermia
- –Acrolein - acute hemorrhagic cystitis, bladder fibrosis,TCCA
- –Oncogenicity: bladder/skin >> hematopoetic / lymphoreticular

Tumor Necrosis Factor

- Trimer
  - 3 identical subunits
- Actions in RA
  - chemotaxis
  - inhibits cartilage synthesis
  - stimulates
    - osteoclasts
    - fibroblasts
    - cytokine production

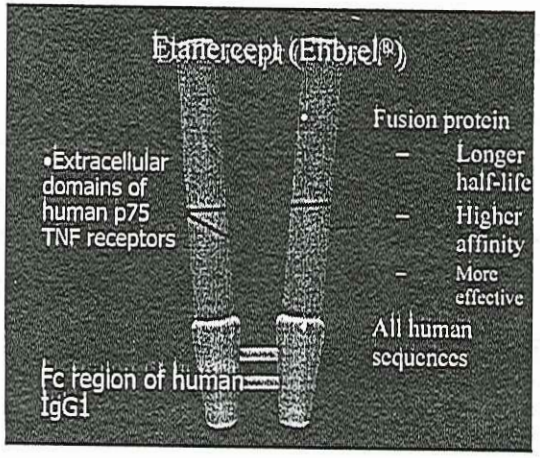


Role Of TNF

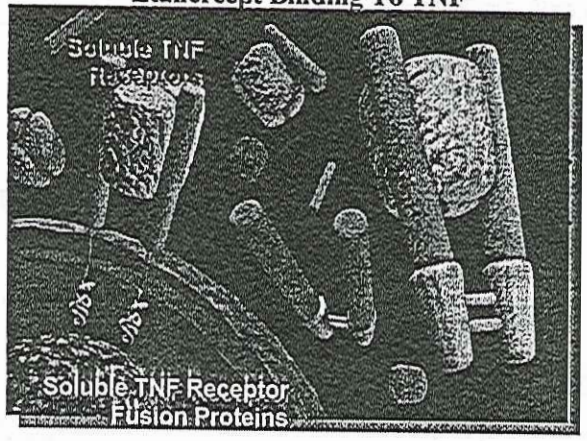
Binding of TNF to cell-surface receptors initiates a variety of inflamm events

- Pain
- Fatigue
- Stiffness
- Impaired function
- Joint destruction

Etanercept (Enbrel®)



Etanercept Binding To TNF



- **INDICATION AND USAGE: Reduction in signs and symptoms of moderately to severly active RA in patients who have had an inadequate response to one or more DMARDS**

DOSAGE: 25MG SQ BIW

PREGNANCY CATEGORY B

Ann Intern Med 1999;130:478-86
N Engl J Med 1999;340:253-9
N Engl J Med 1997;337:141-7

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GLUCOCORTICOIDS

Glucocorticoids are 21-carbon steroid molecules. Corticosteroids with an 11-ketone group [cortisone and prednisone] are biologically inactive until reduced to the 11-hydroxyl compound [hydrocortisone and prednisolone]. Therefore preparations such as cortisone and prednisone that require bioactivation in the liver are used only systemically. In plasma, corticosteroids bind primarily to the alpha-globulin transcortin and, to a smaller extent, albumin. Corticosteroids enter target cells and couple with a specific cytoplasmic receptor before transfer to the nucleus where the corticosteroid receptor complex binds to chromatin and modulates protein synthesis.

I. Comparative Properties of Glucocorticoids.

Drug	Duration of Action	Glucocorticoid Potency	Mineralcorticoid Potency	Equivalen dose [mg]
Short Acting				
Hydrocortisone	8-12 hr	1	1	20
Cortisone	8-12 hr	0.8	0.8	25
Intermediate Acting				
Prednisone	12-36hr	4	0.8	5
Prednisolone	12-36hr	4	0.8	5
Methylprednisolone	12-36hr	5	0.5	4
Triamcinolone	12-36hr	5	0	4
Long Acting				
Betamethasone	36-54hr	20-30	0	0.6
Dexamethasone	36-72hr	25-30	0	0.7

II. Effects of Glucocorticoids on Modulators of Inflammation:

A. Inhibited by Glucocorticoids: IL-1,IL-2,IL-3,IL-4,IL-5,I-L6,IL-8, TNF, IFN, GM-CSF
Prostaglandins and Leukotrienes
COX-2
Inducible nitric oxide synthase & Plaminogen activator

B. Stimulated by Glucocorticoids: Lipocortin-1, Neutral endopeptidase, Vasocortin

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Lymphocytes	Decreased numbers & trafficking, decreased cytokine production, decreased proliferation and impaired activation; little effect on immunoglobulin synthesis.
Eosinophils	Decreased numbers, increased apoptosis
Basophils	Decreased numbers, decrease release of mediators

IV. Adverse effects of Glucocorticoid Therapy:

Immunologic	Increased susceptibility to infection, decreased inflammatory responses, suppressed delayed hypersensitivity, neutrophilia, lymphocytopenia
Musculoskeletal	Osteoporosis c/b fractures, AVN, myopathy
Gastrointestinal	Peptic ulceration in combination with NSAIDs, pancreatitis
Cardiovascular	HTN, fluid retention,accelerated atherosclerosis
Dermatologic	Acne, hiruitism, purple striae, skin fragility,ecchymoses
Neuropsychiatric	Altered mood, emotional lability, euphoria,insomnia, depression, psychosis, pseudotumor cerebri
Ophthomologic	Posterior subcapsulare cataracts, glaucoma
Endocrine & Metabolic	Glucose intolerance, DM, weight gain, hyperlipoproteinemia, negative nitrogen balance, growth suppression, muscle wasting, impaired wound healing, water retention, hypokalemic alkalosis, impotence, Adrenal insufficiency.

V. Clinical Considerations

A. Rheumatoid Arthritis:
   1. Prednisone is not a DMARD; therefore recommend only short courses of tx.
   2. Dose: Prednisone 5- 7.5 mg/day or less

B. SLE/ Vasculitis/Inflammatory Myopathies:
   1. DOSE: 1mg/kg/day---> taper w/in 2wks---> Goal: <20mg/d w/in 3 months.
   2. Initiate immunomodulatory Rx [MTX, Imuran, Cytoxan] if on steriods > 60days.

C. Pulse Glucocorticoid Therapy
   1. Does not suppress HPA axis.
   2. Does not appear to cause osteoporosis.
   3. Clinical response that lasts 4-12 weeks.

D. Alternative - Day Oral Glucocorticoid Therapy.
   1. Fewer adverse effects: suppression of growth,HPA axis and cushinoid features are reduced but osteoporosis is not prevented.

VI. References:

1. Smtih MD et al. The clincial and immunolgicaleffects of pulse methylprednisolone therapy in rheumatoid arthritis. J Rheumatol 1993;20:111

2. Weusten BLAM et al. Corticosteroid pulse therapy in active rheumatoid arthritis. Semin Arthritis Rheum 1993;23:183-192.

3. Fauci A S. Alternate - day corticosteroid theapy. Am J Med 1978;64:729

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MINOCYCLINE IN RHEUMATOID ARTHITIS

On the basis of the theory that persistant Mycoplasma or other infectious agent may cause rheumatoid arthritis, there were a number of case reports and case series reporting a benefit from lengthy courses of tetracyclines. Subsequent evaluation of Minocycline, a semi-synthetic tetracycline, in randomized, double blinded placebo-controlled studies has demonstrated efficacy in the treatment of rheumatoid arthritis. Minocycline has been shown to have anti-inflammatory, immunomodulatory and chondroprotective effects in addition to its antibacterial activity. The efficacy of tetracyclines in arthritis and their limited toxicity, even when long courses are used prompted consideration of these drugs as a possible new treatment for rheumatoid arthritis.

I. Mechanism of Action:

a. Inhibition of Metalloproteinases[collagenase and gelatinase] activity
   - gingival fibroblasts in diabetic rats
   - synovial tissue cultures

b. Inhibition of protein synthesis in rapidly dividing cells.
c. Perturbation of leukocyte function / inhibit leukotaxis & phagocytosis
d. Interference with lymphocyte proliferation
e. Anti-inflammatoyr effects.
f. Antioxidant activity
g. Inhibit production of tumor necrosis factor
g. Suppress induction of experimental arthritis in rats
h. Suppress delayed type hypersensitivity

II. Clinical Trials:

A. Minocycline in Active Rheumatoid Arthritis. Arthritis & Rheumatism 1994;37:629
   - 26 week ,double blinded, placebo-controlled study in 80 patients with active RA treated with DMARDs.
   - Decrease in Acute phase reactants [ESR, IgM-RF]
   - Some clinical parameters improved [Ritchie articular index,# swollen joints]
   - Efficacy parameters: [DHAQ/AM stiffness improved slightly]

B. Minocycline in Rheumatoid Arthritis [MIRA Trial Group] Ann Intern Med 1995;22:81
   - 48 week, double blind, placebo-controlled study in 219 patients with mild to moderate RA treated with DMARDs.
   - Clinical parameters: joint swelling /joint tenderness [$p<0.023$]
   - Laboratory Parameters: [HCT/ESR/PLT/IgM RF $p<0.001$]
   - Reported side effects similar b/w groups. No toxicity.

C. Treatment of Resistant Rheumatoid Arthritis with Minocycline: J Rheumatol 1992;19:1502
   - 48 wk open trial in 18 patients with active RA, resistant to second line agents
   - 12 patients completed study
   - Cytoflurographic analysis revealed significant decrease in Tcell activation antigen
   - Results: Improvement noted [$p<0.05$]
     Patient global assessment score
     Physician global assesment
     Ttender joint count
     Swollen joint count
     Grip strength
     ESR

D. Treatment of Early Rheumatoid Arthritis. Arthritis & Rheumatism 1997;40:842

- Randomized, double blind, placebo-controlled study in 46 patients with SPRA <1yr.
- End point: suuccessful completion of 6 months of therapy without toxicity while demonstrating at least 50% improvement in composite symptoms of arthritis.*
    * Modified Paulus criteria
    * AM stiffness < 30minutes or improved by 50%
    * Joint tenderness improved by 50%
    * Joint swelling improved by 50%
    * ESR < 30 [F] or < 20 [M]
- Additional evaluation criteria:
    Modified Ritchie Articular Index
    Patient global assessment
    Physician global assessment
- Results:
    a. Efficacy: 18 pts had improved 50% at 3 months -> 6months.
        -- 15/23 minocycline [65%] / 3/23 placebo [13%] : p<0.001
    b. Other measures of efficacy:
        -- Patients in minocycline group tended to have clinical improvment in all 7 efficacy measure.
        -- AM stiffness [p<0.03]
        -- Patient global assessment [p<0.006]
        -- Physician global assessment [p< 0.01]

References:

1. Tilley, BC et al. Minocycline in Rheumatoid Arthritis [MIRA]. Ann Intern Med 1995;122:81-89
2. Langevitz P. et al. Treatment of resistant rheumatoid arthritis with minocyline: An open study. J Rheumatol 1992;19:1502-4.
3. Kloppenburg, M. et al. Minocycline in Active Rheumatoid Arthritis. Arthritis & Rheumatism 1994;37:629-636
4. O'Dell J R. et al. Treatment of early rheumatoid arthritis with minocycline or placebo. Arthritis & Rheumatism 1997;40:842-848

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IMMUNOSUPPRESSIVE DRUGS DURING PREGNANCY

Rheumatic diseases commonly affect women of childbearing age. During pregnancy, autoimmune diseases can be active and patients may require aggressive immunosuppressive therapy for disease control. There are a myriad of physiological alterations which occur in pregnancy in patients with rheumatological diseases to include: physiologic anemia, decrease glucose tolerance, decrease biliary excretion and marked increase in hepatic synthesis of proteins, notably those of the coagulation cascade and the complement system. Fibrinogen concentrations are increased resulting in an increased ESR. Additional physiological changes include an increase in total leukocyte count, a decrease in CD4 & IL 2 production. Selection of pharmacological agents in pregnancy requires aunderstanding of informed patient education, gestational age of the fetus and the disease activity of the gravid patient.[1,2]

I. Key to FDA use-in pregnancy Ratings:

Category	Interpretation
A	Controlled studies show no risk. Adequate, well controlled studies in pregnant women have failed to demonstrate risk to the fetus.
B	No evidence of risk to humans. Either animal findings show risk, but human findings do not; or, if no adequate human studies have been done, animal findings are negative.
C	Risk can not be ruled out. Human studies are lacking, and animal studies are either positive for fetal risk or lacking as well. However potential benefits may justify the potential risk.
D	Positive Evidence of Risk. Investigational or post-marketing data show risk to the fetus. Never the less, potential benefits may outweigh the potential risk.
X	Contraindicated in pregnancy. Studies in animals or human, or investigational or post-marketing reports have shown fetal risk which clearly outweighs any possible benefit to the patient.

II. FDA rating for DMARDs

Medication	Category
Sulfasalzine:	B
Gold	C
Cyclosporin A	C
Cyclophosphamide	D
Azathiaprine	D
Chlorambucil	D
6 - mercaptopurine	D
MTX	X

III. General Statements:

A. NSAIDs: Are routinely used in obstetrical practice.

i. Indomethacin - possible constrictive effect on the fetus ductus arteriosus.
- Reduction in fetal urine production and resultant oligohydamnios

ii. Low-dose ASA [80mg/d] is safe and may be used.

B. Corticosteriods: LD prednisone preferable to NSAIDs during pregnancy
- Dosages used comparable to nonpregnant patients.
- Rarity of fetal complications secondary to placental 11-beta-dehyrogenase which oxidizes prenisolone and hydrocortisone to inactive 11-keto forms.

C. Chlorambucil: induces congenital anomalies in early pregnancy including renal hypoplasia.
- 3x more teratogenic [mg/kg] than cyclophosphamide.
- teratogenicity potentiated by caffeine.
- AVOID DURING PREGNANCY

D. Cyclophosphamide: in animal models produces congenital cleft palate & skeletal abnormalities. May be potentiated by caffeine.
- Teratogenicity dose dependent in mice [>20mg/kg]
- Risk factors for future infertility:[age>31, cum dose > 10gm & >15 pulse cycles]
- Brestfeeding is Contraindicated.

E. Azathioprine:
- dosages > 20mg/kg can be embryocidal
- dosages of 2-4mg/kg: normal implantation & fetal development observed
-SLE patients tx withimuran during pregnancy had increased offspring survival.
- In cases where immunosuppression is necessary during pregnancy imuran may be used.

F. 6- mercaptopurine: associated with cleft palate & hydrocephalus
- Clearly teratogenic animals.
-AVOID DURING PREGNANCY.

G. Methotrexate:
- Potential abortifacient
- AVOID DURING PREGNANCY

H. Cyclosporin A:
- Little or no transplacental transfer.
- Teratogenicity is low but premature labor & SGA offspring have been reported.
- Should be avoided in breastfeeding mothers.
- Viable alternative therapy in pregnancy.

I. Gold, SSZ and Plaquenil all cross the placenta but are not considered to be teratogenic.[2]

IV. References:

1. Bermas BL et al. Effects of Immunosuppressive Drugs During pregnancy. Athritis Rheum 1995;38:1722-32.

2. Buyon JP et al. First International Conference on rheumatic diseases in preganancy Arthritis Rheum 1993;36:59-64

3. ACR AD HOC committee. Guidelines for monitoring drug therapy in rheumatoid arthritis. Arthritis Rheum 1996;39:723-731.

4. Petri M. Systemic Lups erythematosus and pregnancy. Rheum Dis Clin North Amer 1994;20:87

5. Ramsey-Goldman R et al. Pregnancy outcome in women with systemic lupus treated with immunosuppressive drugs. J Rheumatol1993;20: 1152.

6. Schaff EA et al. Methotrexate and misoprostol for early abortion Fam Med 1996;28: 198-203.

E. VASCULITIS

"You can not conceive the many without the one"
-Plato

OUTLINE:

E1. Giant Cell Arteritis (Temporal Arteritis)
E2. Wegener's Granulomatosis
    a. Overview
    b. Cutaneous manifestations
    c. Ocular manifestations
    d. Antineutrophil cytoplasmic Antibody (ANCA)
E3. Classification of Cutaneous Vasculitis
    a. Livedoid Vasculitis
E4. Vasculitis of the Central Nervous System
E5. Small Vessel Vasculitis
E6. Takayasu Arteritis
E7. Pseudovasculitis
    a. Fibromuscular dysplasia

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Rheumatology & Clinical Immunology

GIANT CELL ARTERITIS

DEFINITIONS:

GIANT CELL ARTERITIS

- Medium and Large vessel Systemic Vasculits
- Females > Males
- Age ≥ 50 [90% >60yrs]

Clinical Manifestations:

- Fatigue
- Visual Acuity changes
- Fever
- Scalp tenderness
- Headaches
- Jaw claudication
- PMR

J Rheumatol 1992;19:1120-8

Ann Intern Med 1994;120:987-992

Epidemiology

- Most frequent vasculitis in North America
- Incidence: 17 per 100,000
- Prevalence: 23 per 100,000 [>50yrs]
  Scandinavian whites > Hispanic > African American >Asian
- HLA-DR4

Rheum Dis Clin North Amer 1995;21:1013-27

CLASSIFICATION OF GIANT CELL ARTERITIS

1990 ACR [Traditional Format]

Criterion
1. Age at disease onset >50yrs.
2. New Headache
3. Temporal artery abnormality
4. Elevated ESR [≥ 50mm/hr]
5. Abnormal artery biopsy

The presence of 3 or more criteria yields a sensitivity of 93.5% and specificity of 91.2%

CLASSIFICATION OF GIANT CELL ARTERITIS

1990 ACR [Tree Format]

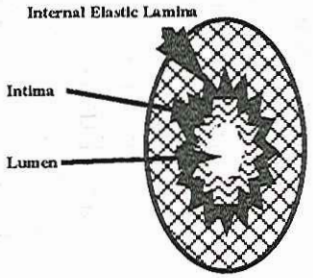
Criterion
1. Age at disease onset >50yrs.
2. New Headache*
3. Claudication of jaw,tongue or on deglutination
4. Temporal artery abnormality
5. Scalp tenderness or nodules*
6. Abnormal artery biopsy

Sensitivity 95.3% and Specificity of 90.7%

Arthritis & Rheum 1990;33:1122-28

DIAGNOSIS

(Temporal Artery Biopsy)



HISTOPATHOLOGIC FINDINGS

- Intima proliferation
- Inflammatory cells (lymphocytes) at the intima-media interface
- Obliteration of internal elastic lamina
- Multinucleated giant cells

TREATMENT

(Giant Cell Arteritis)

WITH Visual Symptoms

- Prednisone 40-80mg/d (8wks)
- Reduce to 20mg/d (4wks)
- Reduce by 5mg Q3-4wks until 10mg/d.

Maintenance dosage:

- Prednisone 5-7mg/d (6-12months)

Final Reduction:

- Taper 1mg every 6-8wks.
- Duration: 2-4yrs

Without Visual Symptoms

- Prednisone 20-40mg/d (8wks)
- Reduce by 5mg Q3-4wks until 10mg/d.

Maintenance dosage:

- Prednisone 5-7mg/d (6-12months)

Final Reduction:

- Taper 1mg every 6-8wks.
- Duration: 2-4yrs

GIANT CELL ARTERITIS

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Giant Cell Arteritis [GCA] is a systemic vasculitis primarily affecting the medium sized extracranial arteries although the aorta and its primary branches may also be involved. The etiology of GCA is unknown. Epidemiologic studies demonstrate an incidence rate of 15-30 cases per year per 100,000 persons over the age of fifty. It affects females more commonly than males. It rarely occurs before the age of 50 and approximately 90% of patients are over the age of 60 at time of diagnosis. Frequent manifestations include fatigue, fever, headaches, jaw claudication, loss of vision, scalp tenderness and polymyalgia rheumatica.

I. Clinical Presentation:

Criterion	Sensitivity [%]	Specificity [%]
Age at onset $\geq$ 50	98.6	63.8
Headache	64.5	81.9
Claudication	40.6	97.6
PMR	52.9	79.3
Diplopia	11.3	93.9
Ischemic optic neuritis	7.5	98.4
Amaurosis fugax	11.2	95.7
Partial unilateral LOV	4.2	99
Complete unilateral LOV	3.3	99.7
Partial bilateral LOV	2.3	99.1
Optic atrophy	4.7	99
Visual abnormality	27.6	88.8
TA abnormality [tender/dec pulse]	57.3	96.8
Scalp tenderness	40.6	97.9
Scalp nodules	43.9	99.5
ESR $\geq$ 50mm/hr	86.5	47.7
Alk phos >1.5 normal	11.2	81.1
AST or ALT > 1.5 normal	17.4	69.7
Artery Biopsy:		
Predominately Mononuclear cell infiltration	90	80.6
Presence of granulomatous inflamation with giant cells	84.3	88.8

Arthritis and Rheumatism 1990;33:1122-28. Comparing 214 patients with GCA and 593 patients with other forms of vasculitis.

II. Diagnosis:

A. 1990 Criteria for classification of giant cell arteritis [Traditional Format]

Criterion	Definition
1. Age at disease onset >50 yrs.	Development of symptoms or findings begining at age 50 or older.
2. New Headache	New onset of or new type of localized pain in the head.
3. Temporal Artery abnormality	Temporal artery tenderness to palpation or decreased pulsation unrelated to arteriosclerosis of cervical arteries.
4. Elevated ESR	ESR greater than or equal to 50mm/hr [Westerngren]
5. Abnormal artery Bx	Biopsy specimen with artery showing vasculitis characterized by a predominance of mononuclear cell infiltration or granulomatous inflammation, usually with multinucleated giant cells.

For purposes of classification, a patient shall be said to have giant cell [temporal] arteritis if at least 3 of 5 criteria are present. The presence of any 3 or more criteria yeilds a a sensitivity of 93.5% and a specificity of 91.2%.

B. Tree Format for the classification of giant cell arteritis

Criterion	Definition
1. Age at disease onset >50 yrs.	Development of symptoms or findings begining at age 50 or older.
2. New Headache*	New onset of or new type of localized pain in the head.
3. Claudication of jaw, tongue or on deglutination	Development or worsening of fatigue or dicomfort in muscles of mastication, tongue, or swallowing muscles while eating.
4. Temporal Artery abnormality	Temporal artery tenderness to palpation or decreased pulsation unrelated to arteriosclerosis of cervical arteries.
5. Scalp tenderness or nodules*	Development of tender areas or nodules over the scalp, away from the temporal artery or other cranial arteries
6. Abnormal artery Bx	Biopsy specimen with artery showing vasculitis characterized by a predominance of mononuclear cell infiltration or granulomatous inflammation, usually with multinucleated giant cells.

* Used as surrogate if artery biopsy is not available [criterion 2] or if temporal artery abdnormality is not present [criterion 5]. The classification tree yields a sensitivity of 95.3% and a specificity of 90.7%.

III. Treatment:

1. Corticosteroid Therapy:
   - Prednisone 40-60mg QD
   - Suppress elevated CD4/CD8 lypmphocyte ratio
   - Decrease Interleukin -2 production
   - Inhibit Collagen synthesis

2. MTX
   - Primarily used as steroid sparing agent
   - MTX 7.5mg - 12.5mg q Wk.

3. Azathioprine
   - Steroid sparing agent
   - Azathioprine 50- 150mg/d

4. Dapsone
   - Steroid sparing agent.

References:

1.Hunder et al. The American College of Rheumatology 1990 Criteria for the classification of Giant Cell Arteritis. Arthritis Rheum 1990;33:1122-8

2. Liu, G T et al. Visual Morbidity in Giant Cell Arteritis Ophthalomology 1994;101:1779-1785

3. Wise, C M etal. Temporal Arteritis with low erythrocyte sedimentation rate: A review of five cases. Arthrtitis Rheum 1991;34: 1571-4

4. Klein RG et al. Larege artery involvement in giant cell[temporal] arteritis. Ann Intern Med 1975;83: 806-812

5. Hunder GG et al. Giant cell arteritis and polymyalgia rheumatica. Clin Rheum Dis 1980;6:389-403

6. Krall PL, et al. Methotrexate for corticosteroid-resistant polymyalgia rheumatica and giant cell arteritis. Cleve Clin J Med 1989;56:253-257

7.DeSilva M et al. Azathioprine in giant cell arteritis/polymyalgia rheumatica: a double blind study. Ann Rheum Dis 1986;45:136-138

8. Doury P et al. The use of dapsone in the treatment of giant cell arteritis and polymyalgia rheumatica. Arthritis Rheum 1983;26:689-690

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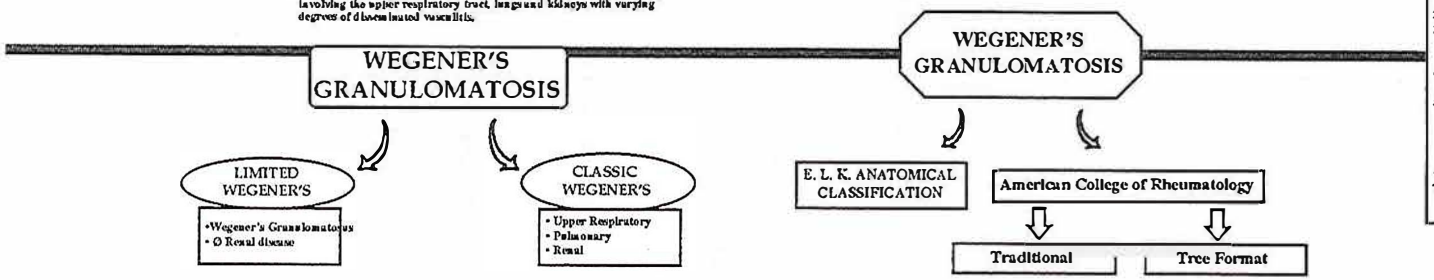
WEGENER'S GRANULOMATOSIS (Overview)

DEFINITIONS:

WEGENER'S GRANULOMATOSIS

- Necrotizing granulomatous small & medium vessel vasculitis classically involving the upper respiratory tract, lung and kidneys with varying degrees of disseminated vasculitis.

CLASSIFICATION OF WEGENER'S GRANULOMATOSIS



CLASSIFICATION OF WEGENER'S GRANULOMATOSIS

1990 ACR [Tree Format]

Criterion	Definition
1. Nasal or Oral Inflammation	Development of painful or painless oral ulcers or purulent or bloody nasal discharge.
2. Hemoptysis *	Hemoptysis during illness
3. Abnormal chest radiograph	Chest radiograph showing the presence of nodules, fixed infiltrates or cavities.
4. Urinary Sediment	Microhematuria [>5 RBC/HPF] or red cell casts in urine sediment
5. Granulomatous Inflammation on biopsy	Histologic changes showing granulomatous inflammation within the wall of an artery or in the perivascular or extravascular area [artery or arteriole]

* Used as a surrogate if biopsy data are not available
The presence of 2 of 4 criteria yields a sensitivity of 87.1% and specificity of 93.6%

Arthritis & Rheum 1990;33:1101 - 1107

CLASSIFICATION OF WEGENER'S GRANULOMATOSIS

1990 ACR [Traditional Format]

Criterion	Definition
1. Nasal or Oral Inflammation	Development of painful or painless oral ulcers or purulent or bloody nasal discharge.
2. Abnormal chest radiograph	Chest radiograph showing the presence of nodules, fixed infiltrates or cavities.
3. Urinary Sediment	Microhematuria [>5 RBC/HPF] or red cell casts in urine sediment
4. Granulomatous Inflammation on biopsy	Histologic changes showing granulomatous inflammation within the wall of an artery or in the perivascular or extravascular area [artery or arteriole]

The presence of 2 of 4 criteria yields a sensitivity of 88.2% and specificity of 92.0%

Arthritis & Rheum 1990;33:1101 - 1107

DIAGNOSIS

- **DEFINITIVE DIAGNOSIS:**
  - Clinical evidence of disease in ≥ 2/3 areas:
    - Upper airways
    - Lung
    - Kidney
  - Biopsy c/w vasculitis & granulomatous involvement in at least one and preferable two of these three organ systems.
- **Tissue Biopsy:**
  - Open Lung Bx: Most sensitive.
  - TBBX
  - Kidney Bx

TREATMENT

Systemic Form:

- Cyclosphosphamide [2mg/kd/day] and adjust dose to keep leukocyte count between 3,000-3,400 per ul.
- *and* Prednisone: start therapy at dose of 1mg/kg/day for two to four weeks, then taper the amount and eventually discontinue
- [Doses may be doubled in patients with rapidly progressive disease and deteriorating renal function]

Clin Exp Immunol 1995;101:S23-6
Ann Intern Med 1983;98:76-85

LIMITED FORM:

- Trimethoprim-sulfamethoxazole [Bactrim, Septra]: 160/800mg BID

or

- Prednisone: 1 mg/kg/day or every other day

Clin Exp Immunol 1995;101:S23-6
Mayo Clin Proc 1985;60:27-32
Ann Intern Med 1983;98:76-85

REMISSION

PARTIAL REMISSION:

1. Clearcut suppression of the progression of disease activity.
2. Stabilization of renal abnormalities, both functional and urinary findings.
3. Pulmonary infiltrates should no longer progress and should begin to resolve.
4. No worsening of other organ system disease activity with progression towards improvement.
5. ESR may still be elevated, BUT, it must be decreasing progressively.

COMPLETE REMISSION:

1. Complete absence of evidence of active disease.
2. Pulmonary infiltrates must clear completely, or show evidence of stable scarring without signs of active inflammation.
3. Renal function must stabilize or improve with no further evidence of active renal sediment.
   A. Proteinuria may persist for months to years in the presence of completely inactive renal disease.
   B. Minor urinary sediment abnormalities may be seen that are c/w previous glomerular damage without evidence of persistent activity [ie. rbc casts].
4. ESR must return to normal.

Ann Intern Med 1983;98:76-85

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WEGENER'S GRANULOMATOSIS
Cutaneous Manifestations

Wegener's granulomatosis {WG} is a systemic vasculitis with distinct clinical and pathological features. The disease is characterized by involvement of the upper airway, the lung and the kidneys. Skin involvement is not unusual [40-50%] and occasionally may be the initial sign at time of presentation. There are no specific skin changes in WG although palpable purpura affecting the lower extremities is most commonly seen. The presence of cutaneous involvement may have prognostic implications in that is frequently associated with active WG disease.

I. Frequency and Localization of Dermatologic Manifestations

Dermatologic manifestation	Number [%] of patients	Localization
Purpura	26[74]	Limbs & trunk
Oral ulcer	15[43]	Lips,tongue,cheeks,tonsils,uvula
Nodules	6[17]	Limbs, trunk
Skin ulcer	5[14]	Face, limbs,back,perineum
Necrotic papule	5[14]	Limbs
Gigival hyperplasia	3[8]	Gingiva
Pustule	2[5.7]	Face, limbs
Xanthoma	2[5.7]	Eyelids
Livedo reticularis	1[2.8]	Limbs
Genital ulcer	1[2.8]	Penis
Digital ulcer	1[2.8]	Fingers

Arch Dermatol 1994;130:861-867

II. Individual Studies:

A. Barksdale et al. *Am J Surg Pathol 1995;19:161-172*

- Clinicopathologic study of 75bx in 46 WG patients with cutaneous disease: 82 WG control patients.

1. LCV[31%]
   a. WG at Earlier Age [30] vs control [45]
   b. Early onset of dz [15 mos]
   c. All lesions associated with active disease
   d. LCV + ESR 2x LCV - ESR
   e. Rapidly progressive + widespread disease
   f. Increased incidence of arthralgias & renal disease
2. Granulomatous Inflammation [19%]
   a. Earlier age [30]
   b. Early onset of dz [12 mos]
   c. 62% of patients with active disease
   d. Slower disease progression

B. Frances et al. *Arch Dermatol 1994;130:861-867*

- Dermatological manifestations in 75 cases with clinicopathologic correlation
- Retrospective review 1973- 1992 -> 75 WG pts -> 35 WG with cutaneous disease
- Palpable purpura most common cutaneous manifestation
    - Necrotic purpura [Less frequent more suggestive of WG]
    - Ecchymotic purpura
- Nongranulomatous vasculitis associated with purpuric lesions [p<0.0001]
- Granulomatous vasculitis associated with nonpurpuric lesions [p<0.0001]
- Cuntaneous dz associated with Articular and Renal involvement
- Except for Xanthoma, Skin dz associated with active WG
- Survival rates similiar +/- cutaneous disease

C. Daoud et al. *J Am Acad Dermatol 1994;31:605-12*

- A retrospective analysis of 244 cases of WG observed b/w 1988-1992
- Clinical, histopathologic and immunopathologic features of 30 WG patients with cutaneous disease. 81% = C-ANCA+
- Cutaneous histologic pattern:
    1. LCV : 24 [80%]
    2. Extravascular granuloma: 3 [10]
    3. Granulomatous Vasculitis
- Palpable purpura most frequent manifestation [47%]
- Prevalence of cutaneous findings was higher in patients with Generalized WG [EK/LK/ELK]
- Skin lesion = initial manifestation of WG in 21 [62%] patients

III. References:

1. Gyselbrect L. et al. Etiological factors underlying conditions in patients with leucocytoclastic vasculitis. Clin Exp Rheumatol 1996;14:665-668.

2. Smith JG. Vasculitis. J Dermatol 1995;22:812-822

3. Barksdale SK. et al. Cutaneous pathology in Wegener's granulomatosis. A clinicopatholgic study of 75 biopsies in 46 patients. Am J Surg Pathol 1995;19:161-172

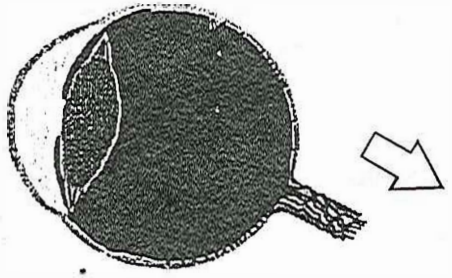
4. Daoud MS et al. Cutaneous Wegener's granulomatosis: clinical, histopathologic, and immunologic features in thirty patients. J Am Acad Dermatol 1994;31:605-612

5. Mangold MC. et al. Cutaneous leukocytoclastic vasculitis associated with active Wegener's granulomatosis. J Am Acad Dermatol 1992;26:579-584

6. DeSpain JD. et al. Collagen vascular disease. Dermatol Clin 1992;10:1-18.

WEGENER'S GRANULOMATOSIS (Ocular Manifestations)

OCULAR MANIFESTATIONS
(Wegener's Granulomatosis)



Contiguous Granulomatous Disease

Focal Vasculitis

CONTIGUOUS DISEASE

- NASOLACRIMAL DUCT OBSTRUCTION
- OCULAR MUSCLE INVOLVEMENT
- OPTIC NERVE INVOLVEMENT
- PROPTOSIS
  - PSEUDOTUMOR
  - ORBITAL CELLULITIS

FOCAL VASCULITIS

- CONJUNCTIVITIS
- EPISCLERITIS
- SCLERITIS
- CORNEOSCLERAL ULCERATION
- UVEITIS
- GRANULOMATOUS VASCULITIS
- RETINAL ARTERY OCCLUSION
- OPTIC NEURITIS

OCULAR MANIFESTATIONS
(Wegener's Granulomatosis)

Ocular Manifestations	Number	Percent (%)
Conjunctivitis	5	16
Episcleritis, scleritis,uveitis	4	12
Optic nerve vasculitis	1	3
Retinal artery occlusion	1	3
Proptosis with orbital pseudotumor	7	22
Nasolacrimal duct obstruction	2	6

Amer J Med 1977;63:131

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Rheumatology & Clinical Immunology

ANCA

[Antineutrophil Cytoplasmic Antibody]

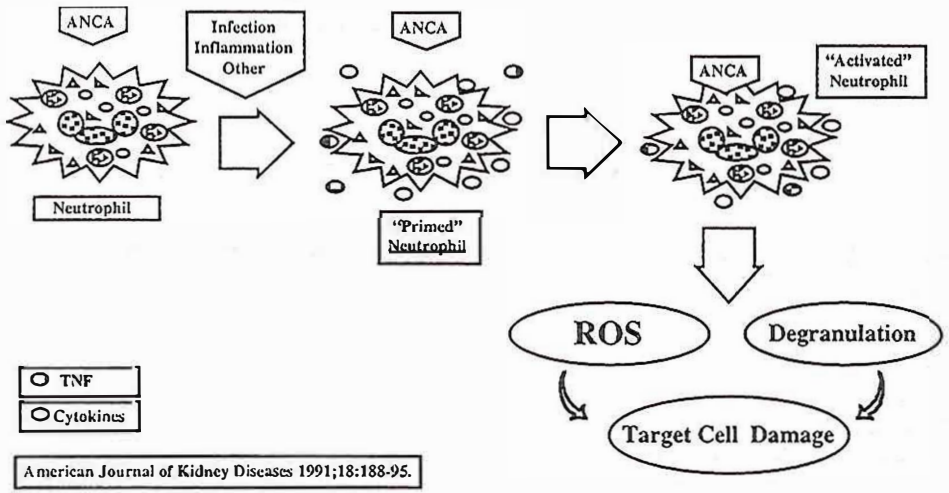
- **Definition:**
  - Autoantibodies specific for antigens located in neutrophil granules and monocyte lysosomes.
- **SUBTYPES:**
  - p -ANCA: [Perinuclear staining pattern]
    - Ag: Myeloperoxidase
  - c -ANCA: [Cytoplasmic staining pattern]
    - Ag: 29-kd serine protease: Proteinase 3
- **ANCA- associated Vasculitis:**
  - -WG
  - -Polyarteritis nodosa
  - -Leukocytoclastic angitis

ANCA

[Antineutrophil Cytoplasmic Antibody]

ANCA

[Pathogenic Role]



American Journal of Kidney Diseases 1991;18:188-95.

c-ANCA

IgG antibodies directed against intracytoplasmic antigens [Proteinase 3] found within the azurophilic granules of neutrophils.

Detection:

- Indirect immunoflourescence [gold standard]
- ELISA
- RIA

Sensitivity: [WG]

- Overall --> 66% [95% CI, 57%-74%]
- Active Disease --> 91% [95% IC, 87%-95%]

Specificity: [WG]

- Overall --> 98% [95% CI, 96% - 99.5%]
- Active Disease --> 99% [95% CI, 97% - 99.9%]

Ann Intern Med 1995;123:925-932
Ann Intern Med 1989;111:28-40

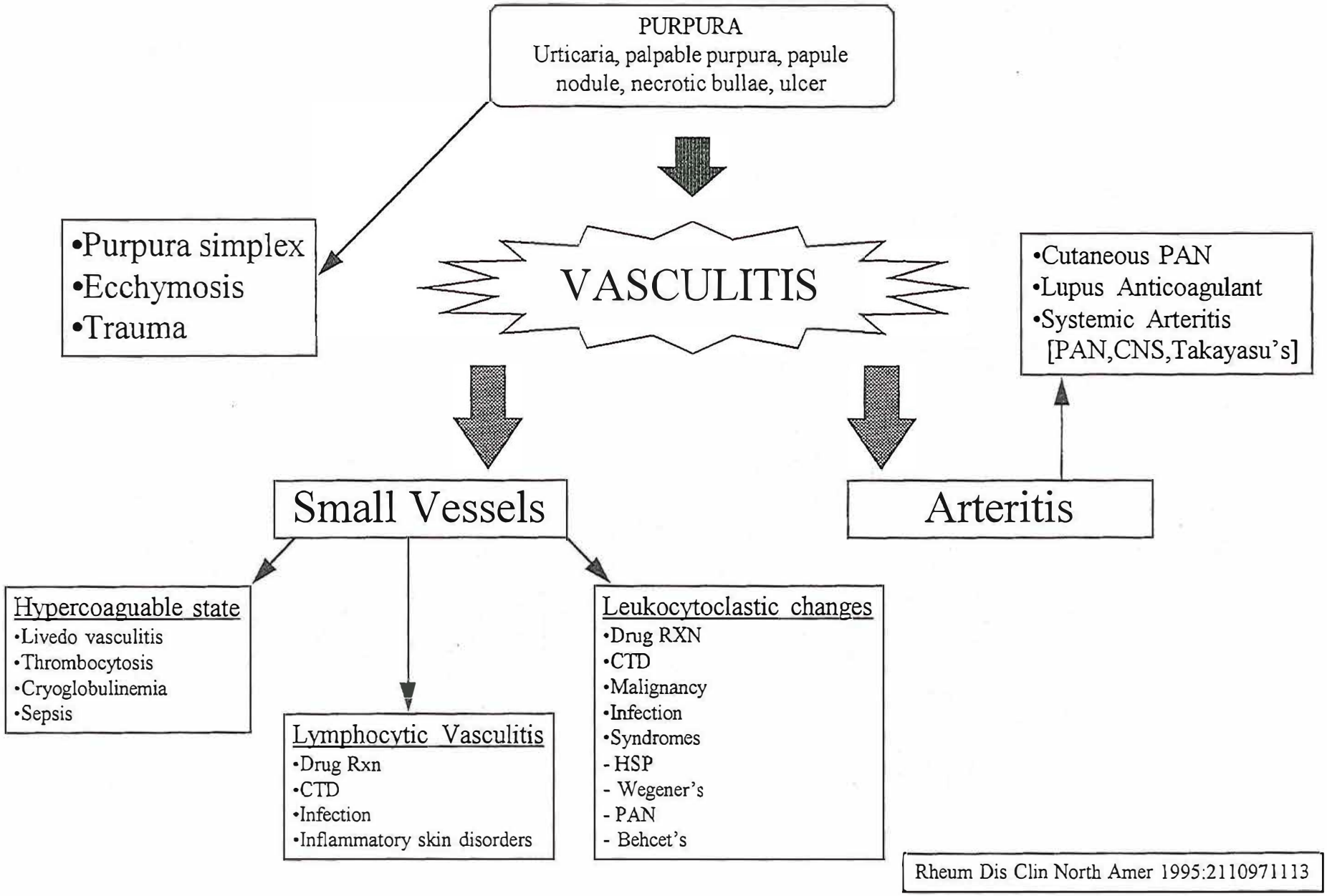
c-ANCA

False Positive Results:

- Tuberculosis
- Hodgkin's Lymphoma
- HIV
- Septal perforation
- Monoclonal Gammopathies
- Drug-induced Wegener's like Disease

Clin Nephrol 1992;37:124-30
Lancet 1990;335:1227-8
Clin Rheumatol 1989;8:512-6.
J Clin Lab Immunol 1990;32:153-9
Pediatr Nephrol 1990;4:299-300

Classification of Cutaneous Vasculitis



Rheum Dis Clin North Amer 1995:2110971113

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LIVEDOID VASCULITIS

Livedoid vasculitis is a form of cutaneous vasculitis which differs from other forms of cutaneous vasculitis in that leucocytoclasis or destruction of the vessel wall is not a characteristic finding. Livedoid vasculitis is known by many other names, including livedo vasculitis, atrophie blanche, livedo reticularis with ulcers, segmental hyalinizing vasculitis. Clinically, livedoid vasculitis is characterized by stellate Ivory white scars with surrounding erythema, telangiectasia, hyperpigmentaion and occasionally angiomatous puncta resembling cayenne pepper. The ulcers characteristically start as small but painful irregular or linear-shaped defects. There is a female predominance with mean age of approximately thirty. Although the lower extremities are the classic areas of involvement, lesions may occur on the forearms, fingers and hand. One extremity can be involved to a much greater extent than the other.

Rheum Dis Clinics North Amer 1995;21:1097-1113

I. Histopathology:

a. Hyalinization and thickening of the small blood vessels of the mid and deep dermis with associated endothelial proliferation.
b. Degeneration of the subintimal portion of vessel.
c. Mild or minimal degree of inflammatory change surrounding vessel
d. DIF: IgM deposition +/- IgG & IgA.
e. Fibrin stain commonly seen in involved areas.

II. Etiology:

a. Unknown
b. Proposed mechanism:
   - Defect in tissue plasminogen activator [t-PA]
   - Abnormal hemostatic factors --> Fibrin clot
c. Medications:
   - Hydroxyurea

III. Associated Disorders:

a. Mononeuritis multiplex
b. Vascular calcification
c. No clear association with venous insufficiency.

IV. Treatment:

a. Bed rest and localized supportive care:
   - Antiseptic wet/dry dressings.
   - Duaderm dressings.
b. Sulfasalazine: 3gms/day
c. Ethylestrenol: used with some success
d. Anticoagulation:
   - heparin
   - coumadin
e. Low dose aspirin
f. Niacinamide
g. Pentoxyifyline
h. Tissue plasminogen activator.

V. References:

1. Gibson, LE. et al. Cutaneous Vasculitis. Rheum Dis Clin North Amer 1995;21:1097-1113
2. Montefusco E. et al. Unusual dermatologic toxicity of long term therapy with hydroxyurea in CML. Tumori 1986;72:317-321

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VASCULITIS OF THE CENTRAL NERVOUS SYSTEM

Vascular inflammatory diseases that affect the CNS may be classified as either primary or secondary. Vasculitis affecting the CNS alone that is unassociated with other systemic diseases is referred to as primary angiitis of the CNS [PACNS]. Secondary vasculitis of the CNS occurs in association with a variety of conditions including infections, lymphoproliferative diseases, drug exposure, connective tissue disease and other forms of systemic vasculitis.

Rheum Dis Clin North Am 1995;21:1059-76.

I. Clinical Manifestations:

Clinical Features of PACNS in 108 patients.

Features	n[%]
Symptoms	
Headache	67[62]
Paresis	60[55.6]
Decreased cognition	55[50.9]
Seizure	23[21.3]
Spinal cord problems	19[17.6]
Cerebral Hemorrhage	4[3.7]
Fever	21[19.4]
Weight loss	13[12]
Duration of Symptoms	
< 4 weeks	43[40.2]
4-12 weeks	22[20.6]
>12 weeks	42[39.9]
Nature of Findings	
Focal only	35[32.4]
Diffuse only	12[11.1]
Focal and diffuse	57[52.8

PACNS may occur at any age but commonly occurs in the forth or fifth decade of life. The onset of disease if variable. There is often a long prodromal period of 6 or more months. The most common symptoms are headache. Nonfocal neurologic defecits are characteristic and include a decrease in cognitive function. Neurological deficits include TIAs, CVA, paraparesis, quadriparesis, cranial neuropathies, ataxia and seizures. Infrequently subarachnoid or intracerebral hemorrhage is the initial event. Importantly, systemic symptoms such as fever, weight loss, arthralgias and myalgias, which are common in systemic necrotizing vasculitis, are uncommon in PACNS.

Rheum Dis Clin North Am 1995;21:1059-76
Cleve Clin J Med 1992;59:293-306

II. Pathology

A. Gross:

- Multiple small foci of infarction or hemorhage; occasional large areas of infarction or hemorrhage.

B. Microscopic:

- Segmental necrotizing and frequently, granulomatous vasculitis of the small & medium sized arteries that predominantly affects the leptomeningeal and cortical blood vessels.
- Veins are affected @ 50%.
- Granulomatous infiltrate [80%], consisting predominatly of lymphocytes with epitheloid cells,plasma cells, macrophages and giant cells.

III. Diagnosis:

A. CSF: sensitivity 53% [95% CI 27-79]
Resembles aseptic meningitis: modest pleocytosis, GLU: WNL, CFP elevated.

B. CT Scan: [sensitivity 33-50%] 65%[95% CI 38-86%]
Results may be WNL in more than 50% PACNS patients.

C. MRI: [sensitivity 50-100%] 75% [95% CI 48-93%]
- multiple bilateral, supratentorial infarcts in cortex, deep white matter a/o leptomeninges.

D. Angiography [sensitivity 75%, specificity 80%]
- Intermittant narrowing and dilatation of small vessels [beading]
- nearly 40% WNL in histologically proven PACNS

E. Biopsy: gold standard.
- Stereotactic biopsy reported to be successful
- Preferred site: temporal tip of nondominant hemisphere, sampling area with a longitudinally-oriented surface area.

F. Combined Modalities: [*J Rheumatol 1994;21:1277-82*]
1. LP + CT scan: sens 92% [95% CI 62-100]
2. LP + MRI: sens 100% [95% CI 74-100]

IV. DIAGNOSTIC CRITERIA FOR PACNS

A. **True PACNS**

1. Definite disease: leptomeningeal, cerebral cortex or spinal cord bx demonstrating primary angiitis [with or without granulomatous features].
and
Exclusion of other conditions

2. Possible Disease: Cerebral angiogram demonstrating the classical findings of arteritis, including segmental narrowing, extasia or beading
and
headaches or multifocal neurologic dysfunction present for more than 3 months
and
CSF analysis demonstrating elevated cell and protein levels
and
exclusion of other conditions.

B. **Benign Angiitis of CNS [BACNS]**
Cerebral angiogram demonstrating the classic findings of arteriitis, including segmental narrowing, ectasia, or beading
and
acute presentation of a focal or multifocal neurological deficit
and
relatively normal CSF cell and protein levels
and
exclusion of other conditions

J Rheumatol 1994;21:1277-82
Cleve Clin J Med 1992;59:293-306

V. Differential Diagnosis

Conditions Resembling PACNS Excluded by Diagnostic Criteria

Systemic Vasculitides
- Polyarteritis nodosa
- Allergic granulomatosis
- Hypersensitivity Vasculitis
- Vasculitis with CTD
- Wegener's Granulomatosis
- Temporal Arteritis
- Takayasau's arteritis
- Behcet's disease
- Lymphomatoid granulomatosis
- Cogan's syndrome

Infection
- Viral, bacterial, fungal, Rickettsial

Neoplasm
- Angioimmunoproliferative disorders
- Carcinomatous meningitis
- Infiltrating glioma
- Malignant angioendotheliomatosis

Drugs
- Amphetamines
- Ephedrine
- Phenylpropanolamine
- Cocaine
- Ergotamine

Vasospastic disorders
- Postpartum angiopathy
- Eclampsia
- Subarachnoid hemorrhage
- Migraine and exertional headache

Other Vasculopathies and mimicking conditions
- Fibromuscular dysplasia
- Moyamoya disease
- TTP
- Sickle cell anemia
- Neurofibromatosis
- Cerebrovascular atherosclerosis
- Demyelinating disease
- Sarcoidosis
- Emboli [SBE, atrial myxoma, paradoxical emboli]
- Acute posterior placoid pigment epitheliopathy and cerebral vasculitis
- Antiphospholipid antibody syndrome

VI. Treatment

A. BACNS: Corticosteroids [60 -80mg/day] + Calcium Channel blocker
   If good clinical response taper over wks - months.
   If clinical response unsatisfactory then go to BX & tx for PACNS .

B. PACNS : Corticosteroids + Oral Cyclophosphamide [1-2mg/kg]
   Treat for 6-12 months following remission.

C. Other Immunosuppressive Therapies:
   1. MTX: 7.5-25mg Q week
   2. Azathioprine:
   3. Chlorambucil
   4. Cyclosporine
   5. Plasmaphoresis
   6. IVIG

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SMALL- VESSEL VASCULITIS

Small-vessel vasculitis is defined as vasculitis that affects vessels smaller than arteries, such as arterioles, venules, and capillaries. It is important to note that small vessel vasculitis sometimes also affects arteries and thus the vascular distribution overlaps with that of the medium-sized vessel and large-vessel vasculitides. *N Engl J Med 1997;337:1512-1523*

I. Major Categories of Noninfectious Vasculitis

TABLE 1. MAJOR CATEGORIES OF NONINFECTIOUS VASCULITIS.*

- **Large-vessel vasculitis**
  - Giant-cell arteritis
  - Takayasu's arteritis
- **Medium-sized-vessel vasculitis**
  - Polyarteritis nodosa
  - Kawasaki's disease
  - Primary granulomatous central nervous system vasculitis
- **Small-vessel vasculitis**
  - ANCA-associated small-vessel vasculitis
    - Microscopic polyangiitis
    - Wegener's granulomatosis
    - Churg–Strauss syndrome
    - Drug-induced ANCA-associated vasculitis
  - Immune-complex small-vessel vasculitis
    - Henoch–Schönlein purpura
    - Cryoglobulinemic vasculitis
    - Lupus vasculitis
    - Rheumatoid vasculitis
    - Sjögren's syndrome vasculitis
    - Hypocomplementemic urticarial vasculitis
    - Behcet's disease
    - Goodpasture's syndrome
    - Serum-sickness vasculitis
    - Drug-induced immune-complex vasculitis
    - Infection-induced immune-complex vasculitis
  - Paraneoplastic small-vessel vasculitis
    - Lymphoproliferative neoplasm-induced vasculitis
    - Myeloproliferative neoplasm-induced vasculitis
    - Carcinoma-induced vasculitis
  - Inflammatory bowel disease vasculitis

*Vascular inflammation is categorized as either infectious vasculitis, which is caused by the direct invasion of vessel walls by pathogens (e.g., rickettsial organisms in Rocky Mountain spotted fever), or noninfectious vasculitis, which is not caused by the direct invasion of vessel walls by pathogens (although infections can indirectly induce noninfectious vasculitis — e.g., by generating pathogenic immune complexes). ANCA denotes antineutrophil cytoplasmic autoantibodies.

II. Chapel Hill Consensus Conference on the Nomenclature of Systemic Vasculitis
(Arthritis Rheum 1994;37:187-92)

TABLE 2. NAMES AND DEFINITIONS OF VASCULITIS ADOPTED BY THE CHAPEL HILL CONSENSUS CONFERENCE ON THE NOMENCLATURE OF SYSTEMIC VASCULITIS.*

LARGE-VESSEL VASCULITIS	
Giant-cell (temporal) arteritis	Granulomatous arteritis of the aorta and its major branches, with a predilection for the extracranial branches of the carotid artery. *Often involves the temporal artery. Usually occurs in patients more than 50 years old and is often associated with polymyalgia rheumatica.*
Takayasu's arteritis	Granulomatous inflammation of the aorta and its major branches. *Usually occurs in patients younger than 50.*
MEDIUM-SIZED-VESSEL VASCULITIS	
Polyarteritis nodosa	Necrotizing inflammation of medium-sized or small arteries without glomerulonephritis or vasculitis in arterioles, capillaries, or venules.
Kawasaki's disease	Arteritis involving large, medium-sized, and small arteries and associated with mucocutaneous lymph node syndrome. *Coronary arteries are often involved. Aorta and veins may be involved. Usually occurs in children.*
SMALL-VESSEL VASCULITIS	
Wegener's granulomatosis†	Granulomatous inflammation involving the respiratory tract and necrotizing vasculitis affecting small-to-medium-sized vessels (e.g., capillaries, venules, arterioles, and arteries). *Necrotizing glomerulonephritis is common.*
Churg–Strauss syndrome†	Eosinophil-rich and granulomatous inflammation involving the respiratory tract and necrotizing vasculitis affecting small-to-medium-sized vessels and associated with asthma and eosinophilia.
Microscopic polyangiitis†	Necrotizing vasculitis with few or no immune deposits affecting small vessels (capillaries, venules, or arterioles). *Necrotizing arteritis involving small and medium-sized arteries may be present. Necrotizing glomerulonephritis is very common. Pulmonary capillaritis often occurs.*
Henoch–Schönlein purpura	Vasculitis with IgA-dominant immune deposits affecting small vessels (capillaries, venules, or arterioles). *Typically involves skin, gut, and glomeruli and is associated with arthralgias or arthritis.*
Essential cryoglobulinemic vasculitis	Vasculitis with cryoglobulin immune deposits affecting small vessels (capillaries, venules, or arterioles) and associated with cryoglobulins in serum. *Skin and glomeruli are often involved.*
Cutaneous leukocytoclastic angiitis	Isolated cutaneous leukocytoclastic angiitis without systemic vasculitis or glo-

III. Characteristics of Some Systemic Vasculitides (Robbins Pathologic Basis of Disease, 4th Edition)

Table 12–4. Characteristics of Some Systemic Vasculitides

ANGIITIS	VESSELS INVOLVED	ORGAN OR TISSUE AFFECTED	PRINCIPAL MORPHOLOGIC FEATURES
Polyarteritis nodosa	Muscular arteries	Gastrointestinal tract, mesentery, liver, gallbladder, kidney, pancreas, muscles, other sites	Lesions of varying ages; all layers of vessels with acute fibrinoid necrosis and extensive periarterial inflammation
Hypersensitivity angiitis	Small venules, capillaries, arterioles	All organs and tissues (skin, muscles, heart, kidneys, lungs)	Acute necrotizing vasculitis with fibrinoid necrosis of entire wall; often thrombosis of lumen
Giant cell arteritis (temporal arteritis)	Muscular arteries	Usually temporal, ophthalmic, and cranial arteries; may be systemic	Disruption of elastic lamina with most intense reaction in intimal medial layers; giant cells engulf elastic fiber fragments; occasionally thrombosis of lumen
Takayasu's arteritis	Aorta, arteries	Extremities; head and neck; viscera	Granulomatous arteritis with fibrosis and marked narrowing
Kawasaki's disease	Arteries	Generalized; coronary arteries	Identical to polyarteritis (infantile polyarteritis)
Wegener's granulomatosis	Small arteries and veins	Lungs, kidneys, upper respiratory tract; occasionally systemic	Acute necrotizing vasculitis with fibrinoid necrosis of vessel wall; often proximate to granulomas in tissues
Buerger's disease	Arteries, veins, nerves	Extremities; viscera uncommonly	Thrombosis with microabscesses; acute inflammation permeates wall artery, but preserves underlying architecture

IV. Diagnosis of small-vessel vasculitis

1. Recognize that small-vessel vasculitis is present
2. Determine specific type of vasculitis
   - A. Diagnostic strategy
     - Laboratory assessment:
       ANCA, ANA,CH50,CRYOGLOBULINS,HCV,HBV,RF,
       Fecal blood,UA.
     - CXR, sinus radiographs
     - Exclude: Psuedovasculitis, Rx, RA, SLE, CA, IBD
   - B. Pathological examination of involved tissue /
     Immunohistologic: IgA (HSP) vs IgM/IgG (CRYO)

V. Treatment

A. Dependent upon type of vasculitis (ie immunosuppressive chemoRX in systemic vasculitis)
B. Elimination of offending agents (ie Rx, Toxins, Infection)
C. Bed rest and localized supportive care:
- Antiseptic wet/dry dressings
- LE stockings
- Duoderm

D. Cutaneous Disease alone:
- NSAIDs
- ECASA
- Corticosteroids
- SSZ 3gm/d
- Colchicine
- Dapsone

VI. References:

N Engl J Med 1997;337:1512-1523
Arthritis Rheum 1994;37:187-92
Rheum Dis Clin North Amer 1995;21:1097-1113
Curr Opin Rheum 1992;4:47-55

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TAKAYASU ARTERITIS

Takayasu Arteritis [TA] is a chronic, idiopathic, inflammatory vasculitis of unclear etiology which primarily affects large vessels, such as the aorta and its major branches. The clinical features usually reflect limb or organ ischemia resulting from gradual stenosis of involved arteries. TA often occurs in females during their reproductive years. The estimated incidence rate is 2.6 cases per million persons per year. *Ann Intern Med 1994;120:919-929*

I. Histopathology:

- Continuous or patchy granulomatous inflammatory reaction with lymphocytes, histiocytes and multinucleated giant cells.

II. Clinical Manifestations:

A. Vascular [100%] :

- Most common finding = bruit. most common site = carotid
- Multiple bruits is common [femoral & renal less common]
- Claudication and deminished or absent pulse :
  Upper extremities [53-62%] > Lower extremities [15-32%]
  Less common in peds [at dz onset] than adults
- Lightheadedness [1/3]
  57% verterbral artery stenosis / 23% without vertebral artery stenosis
- HTN: Renal artery stenosis common [33%]
  Bilateral RAS > Unilateral RAS [R >L]

B. Musculoskeletal [53%]

- Arthralgia > Arthritis
- Myalgia
- Chest wall pain

C. Neurological Findings [57%]

- Visual acuity disturbances: Bilateral > Unilateral
- TIA / CVA

D. Cardiac [38%]

- Aortic Regurgitation
- Angina
- Palpatation
- CHF
- Pericarditis

III. Angiographic Findings:

A. Morphology

- Long, stenotic lesions [98%]
- Irregularity of vessel wall & post-stenotic dilatation [33%]
- Aneurysms [27%]

B. Anatomical location:

-Table

Angiographic findings in 60 patients with Takayasu's Arteritis

Artery	Abnormality [%]	Type of Abnormality [%]		
		Stenosis	Irregularity	Aneurysm
Aorta	65	53	12	23
Aoritic arch a/o root	35	13	5	18
Abdominal aorta	47	38	5	10
Thoracic	17	7	2	8
Subclavian	93	92	8	5
Bilateral		42	0	0
Left only		48	5	2
Right only		2	3	3
Common Carotid	58	58	2	5
Bilateral		28	0	2
Left only		25	2	0
Right only		5	0	2
Renal Artery	38	38	0	2
Bilateral		22	0	0
Left only		0	0	2
Right only		17	0	0
Vertebral Artery	35	28	10	2
Bilateral		5	3	0
Left only		20	3	0
Right only		3	3	2
Innominate Artery	27	20	2	7
Axillary artery	20	18	2	0
Bilateral		5	0	0
Left only		7	2	0
Right only		7	0	0
Coeliac axis	18	18	0	0
Superior mesenteric	18	18	3	5
Common Iliac	17			
Bilateral		10	0	0
Left only		5	0	0
Right only		2	0	2

Ann Intern Med 1994;120:923

IV. Treatment

A. Medical Therapy

1. Glucocorticoids alone [1mg/kg/d]
   - Remission in 60% patients [95% CI, 45%-74%]
   - median time to remission = 22 months
2. Glucocorticoids in combination with Cytotoxic Agents:
   - MTX
   - Cyclosphosphamide

B. Indications for Surgery

1. HTN assoc with critical stenosis of renal vessel
2. Extremity ischemia limiting ADLs
3. Clinical features of cerebrovascular ischemia or critical stenosis [>70%] of 3 or more cerebral vessels.
4. Moderate Grade II NYHA aortic regurgitation
5. Cardiac ischemia in the setting of proven coronary artery stenosis

C. Transluminal Angioplasty

- 56% success rate on first attempt
- 33% success rate on second attempt
- Restenosis common within 3.5 - 13.6 months

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FIBROMUSCULAR DYSPLASIA

Fibromuscular dysplasia[FMD] is a noninflammatory, nonatherosclerotic vascular disease that primarily involves the medium-sized and small arteries especially the renal and internal carotid arteries; in addition to , less commonly, the vertebral, iliac, subclavian and visceral arteries. The etiology of FMD is unknown. Humoral, mechanical, genetic, as well as, mural ischemia have been implicated. The natural history of this disease is usually benign with progression occuring in a minority of patients. The typical clinical manifestations include renovascular HTN, CVA, SAH and claudication. *Mayo Clin Proc 1987;62:931-952*

I. Pathological Classification: Based upon predominant site of arterial wall involvement.

A. Intimal fibroplasia [1-2%] Males = Females
- Angiographically indistinguishable from Medial hyperplasia.
- Histologically: circumferential or eccentric intima fibrosis

B. Medial FMD: most frequent type [95+%]
Focal, multifocal or tubular stenosis +/- aneurysm & dissection.

1. *Medial fibroplasia*: most common [70-95%] Focal concentric subtotal stenosis.
   - Angiography: classic "string of beads" stenosis. Beads exceed the diameter of the proximal unaffected part of the artery.
   - Multifocal thickened fibromuscular ridges alternating with areas of thickness
   - Renal artery: distal 2/3 is typically involved.
   - Cerebrovascular --> Internal Carotid @ C1-C2 vertebrae, sparing origins
2. *Perimedial fibroplasia:*
   - Angiography: beads are less numerous & smaller in diameter than proximal unaffected part of artery.
   - Extensive fibroplasia of outer half of media with effaced external elastic membrane. Commonly: women < 30yrs with Right sided renal HTN.

C. Periadventitial fibroplasia [1-2%]
- Rarest form.
- Fibroplasia & collagenous adventitia

II. Anatomic Distribution:

A. Renovascular [60-70%] >> Cerebrovascular [25-30%] *Internal Carotid most common*
B. Multivessel disease is common [24%]
C. Mesenteric / Subclavian arteries [9%]
D. Iliac arteries [5%]

III. Differential Diagnosis:

- Arteriosclerosis
- Takayasu's arteritis
- Vasculitis
- Neurofibromatosis
- Ehlers-Danlos syndrome
- Pseudovasculitides

IV. Diagnosis:

A. FMD is a histological diagnosis; however the dx can be made with a high degree of accuracy on the basis of angiographic appearance.
B. Angiographic characteristics

	Atherosclerosis	FMD
Location	w/in 1 cm of main artery	middle or distal segments
Morphology	eccentric	concentric [smooth appearance]
Aorta	involved	not involved

V. Natural History:

A. Renovascular disease: Progression of dz [16-38%]
progression more common in older patients with focal or tubular stenosis > Medial fibroplasia

B. Cerebrovascular disease: Less pronounced. Noted in one study in 3/79pts over 5 years.

VI. Pathogenesis: Unknown.

A. Humoral Factors: Females [childbearing age] > Males
- Oral contraceptives: One study of cerebrovascular incidents, 18%, angio c/w FMD
May cause intimal hyperplasia

B. Mechanical Factors: Cyclic stretching of arterial smooth muscle--> incr. collagen,hyaluronate and chondroitin in vitro.

C. Genetic Factors: High incidence of dz in some families[AD] and in caucasians.

D. Ischemia: Decreased blood supply to the vascular wall due to functional or morphologic occlusion of the vasa vasorum might be a primary event in the pathogenesis of FMD.

VII. Clinical Manifestations and Management:

A. Renovascular FMD:

1. Clinical features: Younger /Female /shorter duration of HTN/ Right renal artery / Impaired renal function is uncommon in FMD compared to arteriosclerosis.

2. Management: Cure rates [38-85%]
Percutaneous Transluminal dilation of Renal Artery [PTRA] /
Surgery: Revascularization with autogenous vein/artery/ synthetic graft.
Nephrectomy performed only as last resort.
Antihypertensives: ACEI & Beta blockers are favored pathophysiologically

B. Cerebrovascular FMD:

1. Clinical features: Ischemic/hemorrhagic stroke, TIA,amauosis fugax/syncope/HA
Dissection of arterial wall in internal carotids >>> vertebral artery system
Bilateral lesions are common and often extend to or into carotid canal.
Angiogram: Long, irregular filling defects: "String Sign"
Aneurysms [51%]: Most located in intracranial portion of ICA & MCA.
2/3 pts will have multiple intracranial aneurysms.

2. Management:
- Surgery:
1. Resection & primary anastomosis or interposition of venous graft.
2. Patch angioplasty with graduated intraluminal dilation
3. Intraluminal operative graduated dilation with Garrett/Fogarty catheter
4. Bypass b/w carotid and MCA.
- Percutaneous Transluminal angioplasty:
- Antiplatelet therapy/ Anticoagulation

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F. SOFT TISSUE RHEUMATISM

"Form ever follows function"
-Louis Henry Sullivan

OUTLINE:

F1. Shoulder Pain
- a. Rotator Cuff
- b. Bicipital tendinitis
- c. Adhesive capsulitis
- d. Thoracic outlet syndrome
- e. Subacromial/subdeltoid bursitis
- f. Impingement Syndrome

F2. Hip Pain
- a. Trochanteric bursitis
- b. Piriformis Syndrome
- c. Iliopsoas bursitis
- d. Meralgia paresthetica
- e. Fascia Lata Fasciitis

F3. Knee Pain
- a. Prepatellar bursitis
- b. Infrapatellar bursitis
- c. Anserine bursitis
- d. Patellofemoral syndrome
- e. Medial Collateral ligament bursitis
- f. Semimembranosus bursitis

F4. Fibromyalgia Syndrome

F5. Complex Regional Pain Syndrome/ Reflex sympathetic dystrophy

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THE SHOULDER

BONE

Clavical
Coracoid Process
Acromium
Bicepital Grove
Scapula

ARTICULATIONS

I. Sternoclavicular Joint
II. Acromioclavidular Joint
III. Glenohumeral Joint
IV. Scapulothoracic Articulation

Range of Motion

- ABduction - 180′
- ADduction - 45′
- Flexion - 90′
- Extension - 45′
- Internal Rotation - 55′
- External Rotation - 45′

PALPATION

INSPECTION

SOFT TISSUE

I. Rotator Cuff Zone
- S.I.T.S

II. Bursae
- Subacromial
- Subdeltoid

III. Axilla
IV. Muscles

Differential Diagnosis

Thoracic outlet syndrome
Osteoarthritis
Glenohumeral Instability
Adhesive Capsulitis

Bicipital Tendinitis/ Bursitis
Rotator Cuff Tear/Tendinitis
Acromioclavicular Joint
Calcific Tendinitis
Enthesopathy
Spondylitis of Cervical Spine

Neurological

Muscle Grading:
- Flexion = C5
- Exension = C5, C6
- ABduction = C5, C6
- ADduction = C6, C7, C8
- External Rotation = C5, C6
- Internal Rotation = C5 - T1
- Scapular Elevation = CN X1
- Scapular Retraction = C5

Sensory Testing:
- Lateral Arm = C5 nerve root
- Medial Arm = T1 nerve root
- Axilla = T2 nerve root
- Nipple = T4 nerve root

Special Tests

- Yergason Test
- Apprehension test
- Speed's Test
- Drop Arm Test

- Physical Examination of the Spine & Extremities.
Chapter 1, Hoppenfeld.1976
- Practical Rheumatology
Chapter 7, Klippel. 1995

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HIP PAIN

(Nonarticular)

Differential Diagnosis

- Trochanteric Bursitis
- Piriformis Syndrome
- Ischiogluteal Bursitis
- Meralgia Paresthetica
- Fascia Lata Fasciitis

Physical Examination

Anatomy

Range of Motion:

Flexion: 120 °
Extension: 30°
ABduction: 45° - 50°
ADduction: 20° - 30°
Internal Rotation: 35°
External Rotation: 45°

- FABERE Test
  - Flexion ABduction, External Rotation and Extension
  - Test for intrinsic hip disease
- OBER's Test:
  - Test for contracture of the iliotibial band tract
  - Symptomatic side up with posterior extension then bend knee. Test is positive if after the examiner lets go of the involved thigh the knee does not drop to table.

TROCHANTERIC BURSITIS

(Diagnosis)

CLINICAL CRITERIA FOR DIAGNOSIS:

- Aching pain in lateral aspect of the hip
- Distinct tenderness around the greater trochanter
- Pain at extreme of rotation, ABduction or ADduction or FABERE [Flexion/ABduction/External Rotation/Extension]
- Pain on *Forced* hip ABduction
- Pseudoradiculopathy [Pain extending to lateral thigh]

A diagnosis of trochanteric bursitis necessitates the presence of the first two criteria and one of other three criteria.

(Treatment)

1. Patient education & reassurance & Rest
2. NSAIDs / Analgesics / Zostrix cream
3. Stretching exercises:
   - Gluteus muscle: Hips/knees flexed for 30s [x3] QD
   - Iliotibial band stretch
4. Physical Therapy
5. Life style modification:
   - Sleep with soft pillow beneath affected side
   - Avoidance of squatting, stair climbing
6. Percutaneous glucocorticoid injection
   - 1% xylocaine [4cc] + Celestone 6mg [1cc]
   - Inject area of greatest reproducible tenderness
   - Repeat every 2-3 weeks PRN
7. Surgical Iliotibial Band Release

Mayo Clin Proc 1996;71:565-569
Scand J Rheumatol 1985;14:417-420

Iliopsoas Bursitis

- Anatomy:
  - Largest bursa in body
  - Lateral to Scarpa's triangle (Femoral Triangle)
- History
  - Groin pain /Inguinal pain
  - Misdiagnosis of hernia
- Physical Examination
  - Tenderness over hip below inguinal ligament
  - MRI or CT: Bursae enlargement
- Treatment:
  - NSAIDs
  - Steroid injections

Meralgia Paresthetica

- Anatomy:
  - Compressive neuropathy of lateral cutaneous nerve of thigh
- History:
  - Burning sensation, dicomfort over anterior thigh
- Physical Exam:
  - Sensory deficit over anterior thigh
- Treatment:
  - Weight reduction
  - Clothing adjustments (avoid tight fitting clothing)
  - NSAIDs
  - Steroid injection

Fascia Lata Fasciitis

- Anatomy
  - Fascia Lata
- History:
  - Pain over lateral thigh with rest relieved with activity
- Physical Examination:
  - Symptomatic side up with leg overhanging from table examiner applies 1-2kg of resistance over ankle to hip flexion
- Treatment:
  - Stretching exercises
  - NSAIDS
  - Steroid injection

ROTATOR CUFF
(Tendinitis)

Anatomical Structures:
- Supraspinatus (ABduction)
- Infraspinatus (External rotation)
- Teres minor (External rotation)
- Subscapularis: non palpable (Internal rotation)

S.I.T. → Palpable at insertions into greater tuberosity of humerus

Symptoms:
- Painful arc: 60-120 degrees of ABduction or extension

Physical Examination:
- Drop Arm Test
- Supraspinatus test
- Jobe's Sign

SPECIAL TESTS

Impingment Sign: Pain with immobilization of scapula while forcibly flexing the arm to 130°(Neer test)

Supraspinatus Test: Place arm in 90° of ABduction and internal rotation with 30° forward angulation for isolation of supraspinatus. Muscle strength testing = weakness/pain.

Drop Arm Test: Full ABduction of arm then slowly lower arm to side. Rotator cuff tear --> arm drops spontanously from 90°.

Jobe's Sign: Pt holds arm@ 45°ABduction&flexion with thumb down.Pain reproduced with passive dwnwrd resistance.

ROTATOR CUFF
(IMPINGEMENT SYNDROME)

Anatomical structures: The coracocromial arch protects the humeral head and the rotator cuff from direct trauma. IS results when there is impingement of supraspinatus tendon & subacromial bursa b/w the humerus & overlying coracoacromial arch.

Etiology: (Mechanical stress -> microtrauma)
- Overhead athletics: baseball,football,tennis,swimming
- Degenerative changes

Symptoms:
- Painful arc: 60-120 degrees of ABduction or extension

Physical Examination:
- Impingment sign: (Neer Test)
- Supraspinatus test
- Subluxability test

BICIPITAL TENDINITIS

- Anatomical location
  - Insertion of long head of biceps
- Symptoms: Pain localized to anterior shoulder at insertion of bicipital tendon. Pain worse with use of biceps muscle. Antecedent rotator cuff tendinitis common.
- Signs:
  - Yergason's test: Pain reproduced in the bicipital groove by resisting supination of the forearm with the elbow at side and flexed to 90°.
  - Speed's test: Pain reproduced in the bicipital groove by resisting arm flexion with the pt's elbow extended fully and forearm supinated.

SHOULDER
(Nonarticular Differential Diagnosis)

SUBACROMIAL & SUBDELTOID
(Bursitis)

Anatomical Structures: Subacromial bursae is largest shoulder bursa located b/w rotator cuff inferiorly and deltoid & teres major muscles superiorly. Its lateral extension beneath deltoid muscle is the subdeltoid bursae.

Symptoms:
- Local pain & tenderness ->distal 1/3 arm
- Antecedent supraspinatus tendinitis common

Signs:
- Active & Passive ABduction reduced
- Active ROM >>> disproportionately reduced

ADHESIVE CAPSULITIS
(Frozen shoulder)

- Definition: Decreased passive ROM in all planes in the absence of glenohumeral arthritis.
- Associated conditions:
  - DM - Thyroid DZ - Bronchogenic infections
  - CA - MI - Shoulder trauma
- Mean age = 6th decade; Women > men.
- Stages:
  (1) Painful stage: initial stage (3-8months)
  (2) Adhesive stage: pain with stiffness (4-6 months)
  (3) Resolution stage: Profound stiffness (1-3 yrs)

Thoracic Outlet Syndrome

- Anatomy: Neurovascular bundle compression
  - Supraclavicular (triangular space b/w scalenus muscles)
  - Costoclavicular
  - Infraclavicular (beneath pectoralis minor muscle)
- Clinical Triad:
  - Numbness
  - Weakness
  - Sensation of swelling
- Symptoms:Intermittant pain(100%) & parathesia (50%)
  - Vascular obstruction: venous or arterial (bothersome)
- Etiologies:
  - "droopy shoulder syndrome" (sagging musculature)
  - Cervical rib - Occupation

Roos Test: Pt repeatedly clenches & unclenches fists while keeping arms ABducted & externally rotated.Test + reproduction of symptoms

Adson (O) Test: For neurovascular compression by scalenus anterior muscle.Pt seated, arms at sides. Pulse is palpated at the wrist and is auscultated for bruit in the supraclavicular space. Pt performs valsalva with neck fully extended, arm elevated and chin turned both toward and away from involved side. A positive test results in diminished palention of radial artery and a bruit over axillary vessels pt becomes symptomatic.

Costoclavicular maneuver: Pt seated, shoulder rotated backward and downward; neurovascular features similar to those of Adson's test become evident. Another variation is pt stand with elbows flexed at 90degrees with arms ABducted at 90degrees. Arms are placed at 45/90/135degrees with radial pulse palpated, auscultation at mid clavicle -> pulse diminution and bruit = + test.

TREATMENT

1. Patient Education & Reassurance
   - Informational handouts
   - Avoidance of provacative activities
2. REST: limited use of extremitly during acutely painful period
3. Pain Control:
   - Analgesics / NSAIDs >>> narcotic RX
   - Local injections
4. Physical Therapy
   - Progressive stretching exercises: Codman (pendulum swing)
   - Heat / Cold therapy
   - Ultrasound
5. Treatment of underlying conditions

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PIRIFORMIS SYNDROME

ANATOMY

Piriformis Muscle: [1 of 6 deep lateral rotators of femur]

- Origin: Anterior surface of sacrum between and lateral to anterior sacral foramina, capsule of sacroiliac articulation, margin of the greater sciatic foramen and sacrotuberous ligament.
- Insertion: Superior border of greater trochanter of femur.
- Action: Rotates thigh laterally and ABducts the flexed thigh at the hip.
- Nerve supply: Sacral plexus, L5, S1, S2.
- Synergists:
  - Gemellus Inferior
  - Gemellus Superior
  - Quadratus femoris
  - Obturator Internus*
  - Obturator Externus

Sciatic Nerve:

- Lumbosacral plexus [L-4 ->S-3] ---> Peroneal, Posterior tibial & Sural Nerves
- Route of exit throug Pelvis:
  - [1] Below piriformis
  - [2] divided: through & below / Above & Below

DIFFERENTIAL DIAGNOSIS

- Herniated Nucleus pulposus
- Spinal stenosis
- Spondyloarthropathy
- Trochanteric bursitis
- Ischial bursitis
- Posterior facet syndrome
- Pelvic mass
- Metastatic disease

SIGNS/SYMPTOMS

Symptoms:

- Buttock Pain / "Sciatica"
  - +/- radiation to ipsilateral leg.
- Exacerbated by lower extremity activity
  - Hip ADDuction and Internal Rotation [cross-country skiing]
- Prolonged periods of Sitting [hard surfaces, driving, biking]
  - "hip pocket neuropathy" "Wallet neuritis"
- History of Trauma to gluetal region
- Defication / Dyspareunia / Scrotal pain

Signs:

- Tenderness: Buttock [Sciatic foramen --> Greater Trochanter]
  - Freiberg Sign: pain on forced internal rotation of hip
  - Pace Sign: Weakness in resisted ABduction & External Rotation
  - Morton's Foot: Prominant head of 2nd metatarsal bone:
    Which produces an unstabel foot during the pus-off phase of the gait cycle. Foot pronation -->Internal Rotation of Knee/Hip

CARDINAL FEATURES

1. History of Trauma to sacroiliac & gluteal region
2. Pain in the region of the Sacroiliac joint, greater sciatic notch & piriformis muscle, extending down the leg and associated with walking.
3. Acute pain associated with stooping or lifting , which is moderately relieved by traction.
4. Palpable, tender, sausage-shaped mass over piriformis muscle.
5. Positive Lasegue sign: [Sciatica]
   - Hip flexion + knee extension = PAIN + + +
   - Hip flexion + knee flexion = Ø PAIN
6. Possible gluteal atrophy

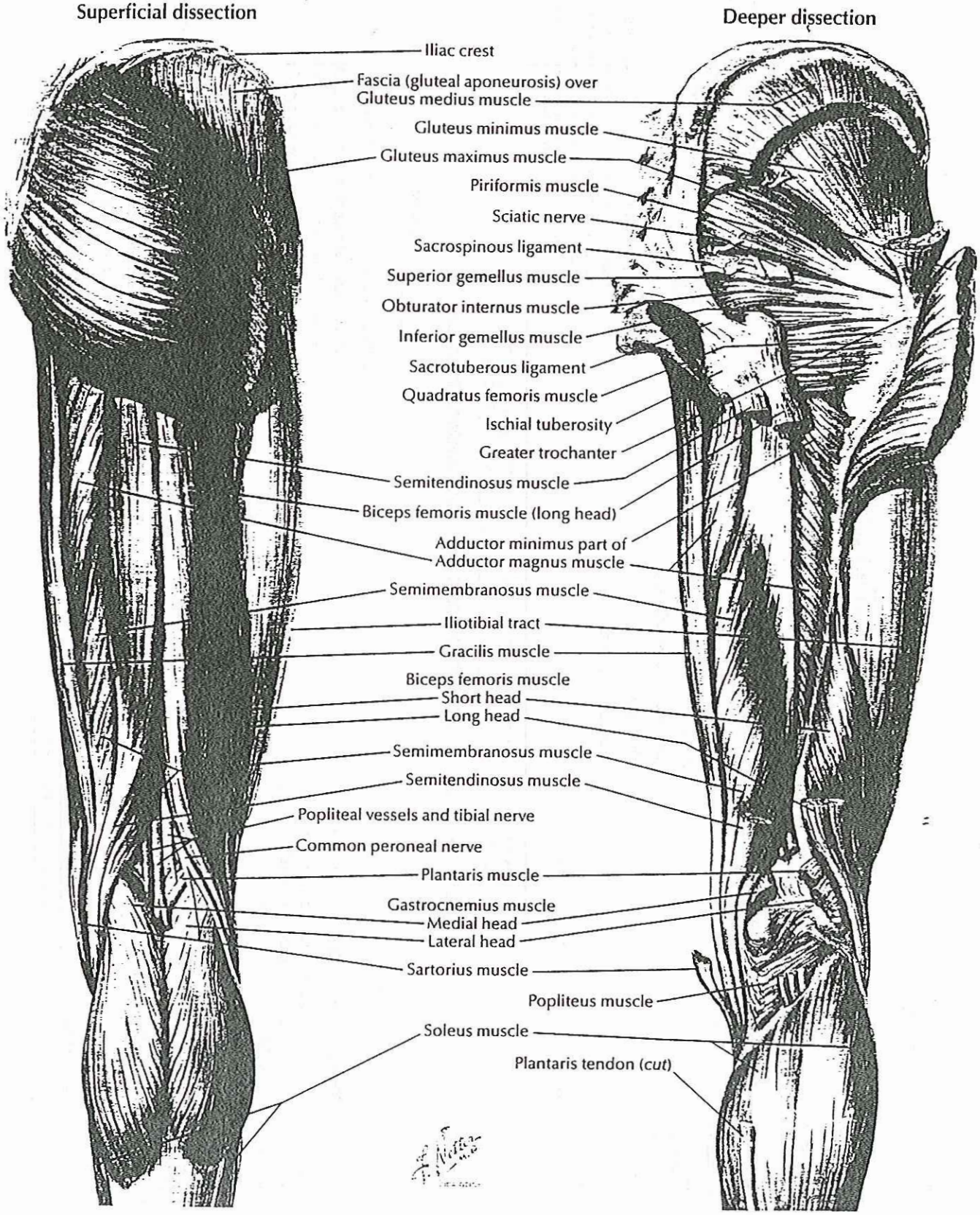
Parziale et al., Amer J Orthopedics 1996

TREATMENT

- Patient education & reassurance
- Correction of abnromal biomechanics:
  - Posture / gait / Leg length discrepancy [heel lift]
- Nonsteroidal Anti-Inflammatory Drugs
- Analgesics
- Muscle Relaxants
- Physical Therapy:
  - Piriformis muscle stretch: Hip Internal Rotation, ADduction/Flexion.
  - Ultrasound Therapy: 2w/cm x 10min
  - Transrectial massage
  - Transcutaneous electrical stimulator [TENS]
- Percutaneous Steroid Injection
- Surgical Release of Piriformis

Muscles of Hip and Thigh: Posterior Views

FOR PIRIFORMIS AND OBTURATOR INTERNUS SEE ALSO PLATES 337, 338; FOR OBTURATOR EXTERNUS SEE PLATE 463



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KNEE PAIN

(Nonarticular)

Differential Diagnosis

- Prepatellar Bursitis
- Infrapatellar Bursitis
- Anserine Bursitis
- Patellofemoral Syndrome
- Medial Collateral Ligament buritis
- Semimembranosus bursitis

Physical Examination

ANATOMY

Range of motion:

Flexion:	135°
Extension:	0°
Internal Rotation	10°
External Rotation	10°

Special Tests:

- **McMurray's Test:**
  - "torn" Meniscus
  - Pt supine: knee flexed, internally & externally rotate tibia (pain)
- **Apley's Compression Test:**
  - "torn "Meniscus
  - Pt prone: knee flexed 90° , internall & externally rotate tibia (pain)
- **Apley's Distraction Test:**
  - Distinguish meniscal vs ligamentous injury
  - Pt. prone: Apley compression test with traction (pain = ligamentous)
- **Apprehension Test:**
  - Patellofemoral pain syndrome / patella hypermobility
  - Pt supine: Medial pressure along patella during quads contrature (pain)

Prepatellar Bursitis

(Housemaid's Knee)

- **Anatomy:**
  - Prepatellar bursae
- **History:**
  - Repetitive kneeling
- **Physical Examination**

  S ++ W +/- T++
- **Treatment:**
  - Rest, Ø KNEELing
  - Protective padding
  - NSAIDs
  - Steroid Injection

PATELLOFEMORAL SYNDROME

(CHRONDROMALACIA PATELLAE)

Clinical Features:

- Anterior Knee pain @ stair climbing, jogging,
- Cinema sign: pain/stiffness after prolonged sitting
- Excessive hip anteversion/ foot valgus or pronation
- Grind sign: crepitus & pain with patella manipulation
- Patellar Aprehension Test:
  - Pain with isometric contraction of quadraceps under resistance at 0 degrees
- Patellar Hypermobility: Quadraceps [Q] angle
  - Normal: <10º [M] <15º [F] at 30º Flexion / >20 = abnormal
  - Patellofemoral Indices: Fulkerson-Schutzer Classification

Therapy:

- NSAIDS / Ice massage / analgesics
- ADL modifications: [high heel shoes, etc.]
- Quadraceps straight leg strengthening exercises [+ weights]
- ROM weight resistant exercises should be avoided
- Physical Therapy / Patellar brace with lateral buttress
- Surgery: Lateral retinaculum release/ vastus medialis obliquus advancement

Infrapatellar Bursitis

(Parson's knee)

- **Anatomy:**
  - Superficial & deep infrapatellar bursae are located on either side of patellar tendon.
- **History:**
  - Trauma
  - Pain with flexion
- **Physical Examination**
  - S++ W+/- T++
- **Treatment:**
  - Rest
  - Protective padding
  - NSAIDs
  - Steroid Injection (caution)

Anserine Bursitis

- **Anatomy:**
  - Pes anserine bursae on medial aspect of knee 3-4 cm beneath joint line under the common tendon of sartorius,gracilis & semitendinosus muscles.
- **History:**
  - Medial pain / tibia pain
- **Physical Examination:**
  - Increase pain: knee flexion
- **Treatment:**
  - Rest
  - NSAIDs
  - Hamstring Stretching exercises
  - Steroid injection

Medial Collateral Ligament Bursitis

(No-Name, No-Fame bursa)

- **Anatomy:**
  - Beneath medial collateral ligament at its insertion onto tibia
- **History:**
  - Injury a/o osteophyte formation
- **Physical Examination:**
  - Pain beneath joint line
  - Pain with full knee extension
- **Treatment:**
  - Rest
  - NSAIDs
  - Steroid Injection

Semimembranosus Bursitis

(Baker's cyst)

- **Anatomy:**
  - semimebranosus & grastronemius bursae in popliteal fossa
- **History**
  - Posterior knee pain
  - Swelling
- **Physical Examination:**
  - Popliteal swelling/ Palpable cyst
  - Homan's sign +/-
- **Treatment:**
  - NSAIDs
  - Aspiration and Drainage
  - Steroid Injection

Fibromyalgia Syndrome Diagnostic Criteria

Fibromyalgia is a distinctive syndrome which can be diagnosed with clinical precision. It may occur in the absence (primary fibromyalgia) or presence of other conditions such as rheumatoid arthritis or systemic lupus erythematosus (concomitant fibromyalgia). It is very rarely secondary to another disease, in the sense that alleviation of the associated disease also cures the fibromyalgia. It may be confidently diagnosed in patients with widespread musculo-skeletal pain (see below) and multiple tender points.

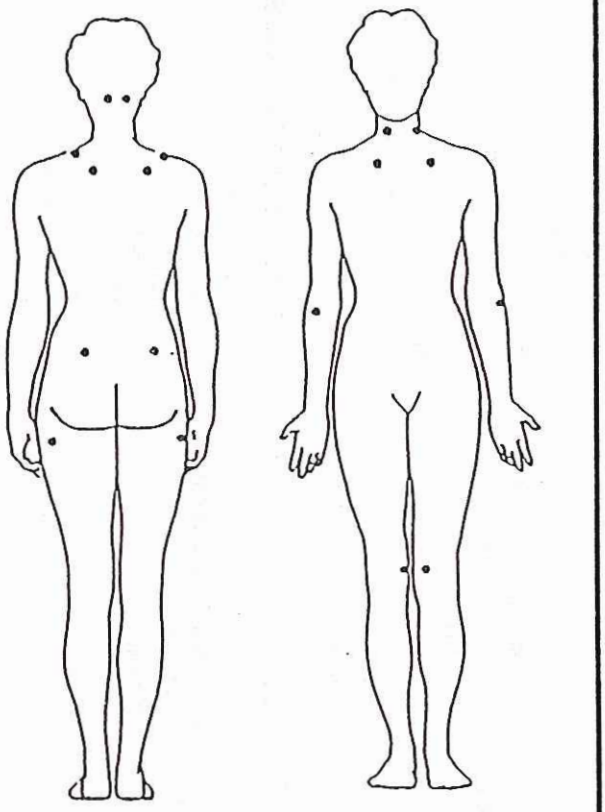

Illustration of Tender Points

The American College of Rheumatology 1990 criteria for the classification of fibromyalgia.

1. **History of widespread pain which has been present for at least a three months.**
   Definition: Pain is considered widespread when all of the following are present: pain in both sides of the body, pain above and below the waist. In addition, axial skeletal pain (cervical spine, anterior chest, thoracic spine or low back pain) must be present. Low back pain is considered lower segment pain.

2. **Pain in 11 of 18 tender point sites on digital palpation.**
   Definition: Pain, on digital palpation, must be present in at least 11 of the following 18 tender point sites:
   Occiput - at the suboccipital muscle insertions.
   Low cervical - at the anterior aspects of the intertransverse spaces at C5-C7.
   Trapezius - at the midpoint of the upper border.
   Supraspinatus - at origins, above the scapula spine near the medial border.
   Second rib - upper lateral to the second costochondral junction.
   Lateral epicondyle - 2 cm distal to the epicondyles.
   Gluteal - in upper outer quadrats of buttocks in anterior fold of muscle.
   Greater trochanter - posterior to the trochanteric prominence.
   Knee - at the medial fat pad proximal to the joint line.

   Digital palpation should be performed with an approximate force of 4 kg. A tender point has to be painful at palpation not just "tender".

Brief Highlights of FMS History

1904 A paper describing fibrositis was written by Sir William Gowers.
1904 Dr. Ralph Stockman published a paper regarding fibrositis nodules.
1915 Llewelly and Jones published a book describing fibrositis.
1953 *"The Management of Pain"* by Dr. John Bonica was published.
1975 Dr. Harvey Moldofsky discovered patients deprived of Stage 4 sleep developed tender points in symmetrical locations in the same areas of the body.
1977 Dr. Hugh Smythe discovered and classified 18 tender points located in specific areas of the body by most of his fibrositis patients. These tender points are in the same body areas as those identified by Dr. Moldofsky in patients with Stage 4 sleep deprivation
1981 Paper regarding Primary Fibromyalgia written by Dr. M. B. Yunus was published in Semin's Arthritis and Rheumatism Journal.
1984 Article entitled "Fibrositis: Does it Exist and Can it be Treated" written by Dr. Robert Bennett, was published in the Journal of Musculoskeletal Medicine.
1987 Dr. Don Goldenberg's article, "Fibromyalgia Syndrome - An Emerging but Controversial Condition: was published in the Journal of American Medicine.
1990 Dr. Frederick Wolfe, Hugh Smythe, Muhammad Yunus, Robert Bennett, et al established the *American College of Rheumatology's Criteria for the Classification of Fibromyalgia.*
1992 MYOPAIN 92 - 2nd World Conference of Pain held in Copenhagen, Denmark. Proceedings published in the Journal of Musculoskeletal Pain, Volume 1, Numbers 3/4, 1993.

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COMPLEX REGIONAL PAIN SYNDROME

The nomenclature, Complex Regional Pain Syndrome (CRPS) Type I and Type II were developed to replace the previous terms RSD and causalgia respectively. As defined by the International Association for the study of Pain (IASP) Task Force, Complex Regional Pain Syndrome (CRPS) Type I or Reflex Sympathetic Dystrophy (RSD) is a syndrome that usually develops following an initiating noxious event, is not limited to the distribution of a single peripheral nerve, and is apparently disproportionate to the inciting event. It is associated at some point with evidence of edema, changes in skin blood flow, abnormal sudomotor activity in the region of the pain , allodynia and/or hyperalgesia. *Hand Clin 1997;13(3):367-72*

I. Diagnostic Criteria:

A. CRPS I: [N.B. Criteria 2 - 4 must be satisfied]

1. Pain syndrome following an initiating noxious event.
2. Spontaneous pain or allodynia/hyperalgesia exists beyond territory of a single peripheral nerve and is disproportionate to inciting event.
3. There is or has been evidence of Edema, skin blood - flow abnormality or abnormal sudomotor activity in the region of the pain since inciting event.
4. The diagnosis is excluded by the existence of conditions that would otherwise account for the degree of pain and dysfunction.

B. CRPS II:

1. Pain occurs following nerve injury;otherwise similar to type I in all other respects. Spontaneous pain or allodynia occurs and is not necessary limited to the territory of the injured nerve.
2. CRPS Type I criteria 3&4.

II. Signs and Symptoms of CRPS

Primary	
Pain	Paramount Feature
Swelling	First Physical Sign
Stiffness	Progressively worsens because of increased fibrosis
Discolorization	Red, Cyanotic, or pale to grayish
Secondary	
Osteoporosis	Marked Radiographic findings
Pseudomotor changes	Initial sweating, then progressive dryness
Temperature changes	Onset of symptoms significant
Vasomotor instability	Vasospasm common
Palmar Fasciitis	Acute nodules and thickening of long bands of palmar fascia
Trophic Changes	Glossy appearance caused by atrophy of skin and SQ tissue

III. Clinical Stages:

1. Stage I: The traumatic stage [sympathetic denervation] (0-3 months).
2. Stage II: The Dsytrophic Stage[sympathetic overativity] (3-12 months)
3. Stage III: The Atrophic Stage (9 months - years)

IV. Imaging in RSD:

1. Plain Radiographs:
   - Soft Tissue Swelling & Patchy Osteopenia are radiographic hallmarks.
   - Periarticular swelling and cortical [reversible] erosions have been described.
   -Bone Resorption [5]
      a. Resorption of cancellous or trabecular bone
      b. Subperiosteal bone resorption
      c. Intracortical bone resorption
      d. Endosteal bone resorption
      e. Subchondral and juxta-articular erosions
2. Dynamic Scintigraphy
   - Asymmetrical blood flow, increased blood flow and uptake in affected area [phase 1]
   - The second phase of bone scan [blood pool] is very reliable in diagnosing RSD
   - Diffusely increased periarticular uptake on delayed bone scan [phase 3].
   - The bone scan appears to lag behind the clinical stage and is not the parameter of choice to monitor response to therapy.
3. MRI

V. Therapy

1. Postganglionic Segmental/ Sympathetic Blockade:
   a. Guanethidine Blockade
      A local anesthetic [lidocaine] is given before intravenous injection of 10-20mg of guanethidine is given after emptying the venous bed of the limb and a proximal tourniquet is left in place for 15 minutes.
   b. Other sympathicolytic drugs:
      - Reserpine / Buflomedil
2 Adjunctive Medical therapy :
   (1) Nifedipine a/o Alpha-sympathetic blocker [Clin Neurol Neurosurg 1997;99:26-30]
   (2) Bisphosphate Therapy [Ann Rheum Dis 1997;56:201-204]
   (3) Calcitonin 100 MRC units QD. [Clin Rheumatol 1986;5:382-388]
   (4) Corticosteroids a/o NSAIDs [Hand Clin 1997;13:339-353]
3. Physical Therapy and Rehabilitation:
   a. Rest and Motion
      - partial rest in a good position with a molded resin support
      - daily activity of affected region
   b. Heat and Cold:
      - heat therapy unproven efficacy especially in phase I
      - Cold/cryotherapy immediately effective against swelling and pain
   c. Massage: not effective in early stages
   d. Electrotherapy: Not indicated in stage I
   e. Radiotherapy: Consider obsolete and dangerous
   f. Acupuncture: No literature to support this modality at this time.

VI. References:

Curr Opin Rheumatol 1994;6:210-216
Hand Clin 1997;13(3) 477-483
Clin Exp Rheumatol 1992;10(4) 401-409
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